

# The Fabius Function and Rvachev's Up-Function

Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics

Human-readable synthesis of the formal development

Vladimir Reshetnikov's ProveIt repository

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## Abstract

This article is the human-readable exposition of the formally verified Fabius-function development in the ProveIt repository. It keeps three related functions separate: the bounded distribution function  $F$ , Rvachev's even compactly supported bump up, and the signed global extension  $\mathcal{F}$ . Their construction and uniqueness, differential and refinement laws, strict shape properties, optimal Lipschitz constants, derivative bounds, exact dyadic-cell derivative value distributions, effective flatness, and precisely formalized analytic and non-elementarity results are presented together with direct pointers to their Lean proofs.

The probabilistic part identifies  $F$  and up with exact probability expressions for an infinite weighted sum of independent uniform coordinates. Finite atomic convolution laws, weak convergence, half-weighted step approximants, and centered uniform splines give proved finite models and quantitative approximation results. At the reusable level, arbitrary absolutely summable vector weights now have probability, head–tail, reflection, continuous-linear naturality, operator-affine recursion, and exact support-as-range laws, while every nontrivial absolutely summable real-weight law is absolutely continuous with respect to Lebesgue measure and has null singletons, and arbitrary signed real weights have exact min/max interval ranges and topological supports. A named geometric-uniform specialization supplies the weights  $(1 - q)q^n$ , their series and law, affine self-similarity and reflection, absolute continuity and null singletons for  $|q| < 1$ , support equal to the series range for  $|q| < 1$ , exact unit-interval support for  $0 \leq q < 1$ , and uniqueness among all probability laws satisfying the same affine independent-copy equation. The uniqueness theorem includes  $q = 0$  and negative  $q$ , and assumes no support, density, or moment condition. Its named CDF is continuous and reflection-symmetric for  $|q| < 1$ , has the exact exterior tails for  $0 \leq q < 1$ , and, for  $0 < q < 1$ , is  $C^\infty$  with an explicit nonnegative, compactly supported  $C^\infty$  Radon–Nikodym density; the half-base CDF and density recover the bounded Fabius function and the affine Rvachev bump exactly. A support-free quantile theorem transports uniform measure through the clamped Fabius inverse and gives Banach-valued quantile substitution. The analytic part develops the formal sinc and cosine products, Fourier inversion and Poisson summation, exact moment recurrences, and the Fourier–Legendre expansion. The arithmetic part gives reciprocal-power values, Thue–Morse formulas for arbitrary dyadic arguments, executable binary reduction, two-adic information, exact half-base values, and denominator-free arbitrary-base  $q$ -binomial and geometric-interpolation identities present in Lean. A compact-rectangle inverse-pair theorem integrates arbitrary absolutely-continuous outer functions without a clock-measurability hypothesis. Finite formal-series filters package arbitrary-node moment exactness and the complete denominator-free Gaussian residual on geometric nodes; their analytic companion gives the same head–tail law for unconditionally summable weighted sampled coefficient series, without a convergence obligation at zero-weight nodes.

Repeated integration is developed once, before its Fabius specialization, as an oriented normalized Volterra calculus whose algebra takes values in an arbitrary real normed space. For

continuous inputs in a complete target, literal iteration and the Cauchy kernel are identified exactly; strict FTC, additive composition of orders, Taylor reconstruction, and finite polynomial commutators then give closed forms for every integer-order primitive of the signed extension and every natural-monomial weight. A separate Gamma-normalized operator supplies the positive-real Riemann–Liouville kernel over arbitrary real normed spaces on ordered endpoints and agrees exactly with the normalized integer operator at every positive natural order. For continuous inputs in a complete target, two positive real orders compose additively on every ordered compact interval. Positive affine covariance, Gamma-normalized order raising from an interior right derivative, and the first exact fractional shifts of the signed, bounded, and folded Fabius functions are also proved on their stated ordered domains.

The endpoint chapters report only the negative-Laplace decompositions, periodic correction, lower-Lambert formulas, saddle estimates, and correction terms that have named proved counterparts in the formal source. Plausible extensions, numerical observations, conditionally convergent boundary filters, sinc-specific generic- $q$  acceleration, complex-order and iterated fractional formulas, and other arguments not yet represented by an exact Lean theorem are kept in the companion volume under docs/semi-formalized-research-frontiers/, not in this primary exposition.

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# 1 Historical note and scope

The probability distribution now attached to the name of Fabius is older than Fabius’s two-page note. It occurs in the 1935 work of Jessen and Wintner on infinite convolutions of independent random variables [1], was rediscovered several times, and entered a different mathematical culture through the compactly supported “finite function” of V. A. Rvachev. Fabius used it in 1966 as a particularly economical probabilistic construction of a nowhere analytic  $C^\infty$  function [2]. Independently, and for entirely different reasons, the Rvachevs introduced the same object in 1971 as the canonical compactly supported solution of a functional–differential equation [3], and V. A. Rvachev later developed it into the theory of *atomic functions* used in approximation theory and numerical analysis [4, 5, 6]. Rvachev’s normalization emphasizes the even compactly supported bump, whereas the Fabius normalization emphasizes its cumulative distribution function. Arias de Reyna’s 1982 paper gave a systematic treatment of the compactly supported function, including its Fourier product, probability interpretation, partitions of unity, derivatives, and dyadic values; an English translation appeared in 2017 [7]. His later arithmetic paper organized the exact rational sequences and denominator questions [8]. Haugland’s note [9] gives a practical algorithm for evaluating  $F$  at dyadic rationals, closely related to the bit-recursion of Section 8.

The present article is not merely a paraphrase of those papers. It follows the proof dependency graph of the Lean development in the `ProveIt` repository [22]. That development first constructs the bounded distribution function, then derives the Rvachev and signed-global forms, and finally builds the arithmetic and asymptotic layers on top. This order has two advantages. It prevents identities valid for the signed extension from being silently applied to the bounded distribution, and it makes the exact dyadic algorithms independent of any numerical approximation.

The formalization also settles several questions that are only implicit, conjectural, or incorrectly normalized in the source material. Most importantly, the sharp endpoint asymptotic contains a nonconstant periodic function of an exact lower-Lambert phase. The periodic term is not a cosmetic improvement: every nonzero Fourier mode is explicit and nonzero, and an elementary formula without it fails at the error scale usually claimed. The development further proves a full inverse-Lambert expansion, establishes an exact bit-recursive evaluator for every dyadic rational, and supplies an absolutely and uniformly convergent Fourier–Legendre expansion of  $\text{up}$ .

Arias de Reyna’s arithmetic paper [8] enumerates the independent occurrences of this function in chronological order, and two of them deserve mention beside Jessen–Wintner, Fabius, and the Rvachevs. In 1981 G. Kh. Kirov and G. A. Totkov arrived at it in a study of the distribution of the zeros of the derivatives of the function they wrote  $\lambda(x)$  (*Godishnik Vissh. Uchebn. Zaved. Prilozhna Mat.* **17** (1981), no. 4, 157–165, in Bulgarian with English and Russian summaries); related dyadic derivative phenomena are represented in the formal Taylor and exact-evaluation APIs used below. In 1985 R. Schnabl gave a further independent construction, “Über eine  $C^\infty$ -Funktion,” in *Zahlentheoretische Analysis*, Lecture Notes in Mathematics 1114, Springer, Berlin (1985), 134–142. Neither paper is used in the proofs below, which is why their data are given here rather than in the bibliography; they are recorded because a reader tracing the function through the literature will meet both names.

Our convention is fixed throughout:

- $F$  is the bounded cumulative distribution function, equal to 0 on  $(-\infty, 0]$  and to 1 on  $[1, \infty)$ ;
- $\text{up}$  is the even compactly supported bump on  $[-1, 1]$ ;
- $\mathcal{F}$  is the signed global extension satisfying  $\mathcal{F}'(x) = 2\mathcal{F}(2x)$  on all of  $\mathbb{R}$ .

All logarithms are natural logarithms. We write  $L = \log 2$  in the asymptotic sections,  $\text{wt}_2(n)$  for the sum of the binary digits of  $n$ , and  $\text{sinc } z = \sin z/z$  with the removable value 1 at the origin.

## 1.1 A guide to the main results

The exposition follows the dependency order of the formal development. The opening chapters separate the bounded function  $F$ , the bump up, and the signed extension  $\mathcal{F}$ ; construct the canonical function; and record the exact uniqueness and shape theorems. The probability chapter then gives the weighted-uniform representation and its three finite approximations: atomic convolution measures, half-weighted step functions, and centered uniform splines. It also isolates the arbitrary-weight distribution laws and the support-free and compact inverse-CDF transport theorems before specializing them back to the Fabius quantile.

The Fourier and global-extension chapters collect the proved product, inversion, Poisson, derivative, normalized repeated-integration, exact dyadic-cell distribution, bound, and flatness APIs. The repeated-integration layer is first proved for arbitrary continuous Banach-valued inputs and polynomial weights, then specialized to the signed and bounded Fabius functions. These chapters lead to the exact analytic locus of  $F$ , the proved nonanalyticity regions for up and  $\mathcal{F}$ , the totalized inverse on  $[0, 1]$ , and the formal non-elementarity results. The moment and arithmetic chapters then derive the rational recurrences, reciprocal-power identities, path and composition formulas, parity and two-adic statements, exact dyadic evaluation, binary reduction, and the specialized half-base  $q$ -binomial formulas.

The later analytic chapters summarize the formal endpoint machinery: the exact negative-Laplace product and logarithmic decomposition, the periodic Fourier correction, the lower-Lambert coordinate, the proved saddle asymptotics and correction coefficients, and the rigorous obstructions to deleting the periodic term. The Fourier–Legendre, summation, and computability chapters finish the proved story. The final module guide names the Lean declarations supporting each retained theorem.

A statement appears here only when its complete mathematical content has an exact proved Lean counterpart. Material removed from earlier drafts because that test is not yet met is preserved in the research-frontier volume. Thus the division between this document and that volume is a proof-status boundary, not a judgment about mathematical plausibility or importance.

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**One formal object, three global functions.** The bounded  $F$  is the cumulative distribution function used for probabilities and dyadic values. The compactly supported up is obtained by folding  $F$ , and its Fourier transform has the proved sinc and cosine product descriptions. The signed  $\mathcal{F}$  is the global Thue–Morse continuation and satisfies the pure dilation equation on all of  $\mathbb{R}$ . Domains and normalizations are stated explicitly whenever a theorem passes from one presentation to another.

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## 2 Three functions that must not be conflated

The literature often uses the same letter  $F$  for three related, but globally different, functions. That economy of notation is harmless in a local calculation and dangerous in a long proof. We therefore begin by separating them.

### 2.1 The bounded Fabius distribution function

**Definition 2.1** (Bounded Fabius function). A function  $F : \mathbb{R} \rightarrow [0, 1]$  is called a *bounded Fabius function* if

$$F(x) = 0 \quad (x \leq 0), \quad (1)$$

$$F(x) = 1 \quad (x \geq 1), \quad (2)$$

$$F(1-x) = 1 - F(x) \quad (0 \leq x \leq 1), \quad (3)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad (4)$$

and  $F$  is infinitely differentiable on  $\mathbb{R}$ .

The derivative equation on the right half follows from symmetry:

$$F'(x) = 2F(2 - 2x), \quad \frac{1}{2} \leq x \leq 1. \quad (5)$$

In particular  $F'(x) > 0$  for  $0 < x < 1$ , once positivity in the interior is known. Thus  $F$  is a genuine cumulative distribution function, not merely a smooth interpolation between 0 and 1.

Because  $F$  is constant on both exterior half-lines and is  $C^\infty$ , every derivative of positive order vanishes at both endpoints:

$$F^{(m)}(0) = F^{(m)}(1) = 0 \quad (m \geq 1). \quad (6)$$

This flatness is the source of both its extraordinary endpoint decay and its failure to be analytic there.

## 2.2 Rvachev's compactly supported bump

**Definition 2.2** (Rvachev up-function). Given  $F$ , define

$$\text{up}(x) = \begin{cases} F(x + 1), & x \leq 0, \\ F(1 - x), & x > 0. \end{cases} \quad (7)$$

Equivalently,

$$\text{up}(x) = F(1 - |x|). \quad (8)$$

Here the right side is interpreted using the globally bounded function  $F$ .

It follows immediately that  $\text{up}$  is even, nonnegative, supported on  $[-1, 1]$ , and  $\text{up}(0) = 1$ . On the unit interval the two functions are simply translates:

$$F(x) = \text{up}(x - 1), \quad 0 \leq x \leq 1. \quad (9)$$

The graph of  $F$  is an infinitely flat sigmoid; the graph of  $\text{up}$  is its symmetric fold.

The primary exposition omits sampled plots: their decimal coordinates are not themselves Lean declarations. Exact point evaluation is developed below through rational dyadic formulas and executable evaluators.

## 2.3 The signed global extension

Let  $\text{wt}_2(n)$  denote the sum of the binary digits of  $n$ , and put

$$\varepsilon_n = (-1)^{\text{wt}_2(n)}. \quad (10)$$

The sequence  $(\varepsilon_n)$  is the  $\{\pm 1\}$  form of the Thue–Morse sequence [14].

**Definition 2.3** (Signed extension). Define

$$\mathcal{F}(x) = \sum_{n \geq 0} \varepsilon_n \text{up}(x - 2n - 1). \quad (11)$$

At a fixed  $x$  only finitely many summands can be nonzero, so this is a locally finite sum. On every block  $[2b, 2b + 2]$  only the  $b$ -th translate can be nonzero (and at the endpoints it too vanishes):

$$\mathcal{F}(x) = \varepsilon_b \text{up}(x - 2b - 1), \quad 2b \leq x \leq 2b + 2. \quad (12)$$

It vanishes for  $x \leq 0$ , and it agrees with the bounded  $F$  for every  $x \leq 1$  (both functions vanish on  $(-\infty, 0]$ ). It does *not* remain in  $[0, 1]$  to the right of that interval. This is the global object that satisfies the cleanest differential equation:

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}). \quad (13)$$

Arias de Reyna's arithmetic paper uses this extension. Statements about its values outside  $[0, 1]$  must therefore not be transferred blindly to the bounded cumulative distribution.

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**Notation used in this article.**  $F$  always means the bounded distribution function;  $\text{up}$  always means the compactly supported even bump; and  $\mathcal{F}$  always means the signed global extension. We have  $F = \mathcal{F}$  throughout  $(-\infty, 1]$  (hence on  $[0, 1]$ ), but globally they are different functions.

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## 2.4 Exact shape: positivity, bijectivity, and unimodality

The three defining equations (1)–(4) already determine the qualitative picture, but the statements that are actually used later are the sharp ones: not  $F \geq 0$  but  $F(x) > 0$  exactly for  $x > 0$ ; not  $\text{up} \leq 1$  but  $\text{up}(x) = 1$  exactly at the origin. We record them here, in the form of equivalences, because the strict versions are what make the convexity and Lipschitz theorems below sharp.

Everything rests on a single formula that merges (4) and (5) into one identity valid on the whole line.

**Lemma 2.4** (Unified derivative formula). *For every real  $x$ ,*

$$F'(x) = 2 \text{up}(2x - 1). \quad (14)$$

*Proof.* By (8),  $\text{up}(2x - 1) = F(1 - |2x - 1|)$ . For  $0 \leq x \leq \frac{1}{2}$  one has  $|2x - 1| = 1 - 2x$ , so  $\text{up}(2x - 1) = F(2x)$  and (14) is (4). For  $\frac{1}{2} \leq x \leq 1$  one has  $|2x - 1| = 2x - 1$ , so  $\text{up}(2x - 1) = F(2 - 2x)$  and (14) is (5). For  $x < 0$  or  $x > 1$  one has  $|2x - 1| > 1$ , hence  $1 - |2x - 1| < 0$  and  $\text{up}(2x - 1) = 0$  by (1), while  $F'(x) = 0$  because  $F$  is constant on each exterior half-line by (1) and (2).  $\square$

Since  $\text{up} \geq 0$ , formula (14) shows at once that  $F$  is nondecreasing on all of  $\mathbb{R}$ . It is also convenient to note that the reflection symmetry (3), stated on the unit interval, in fact holds globally:

$$F(1 - x) = 1 - F(x) \quad (x \in \mathbb{R}). \quad (15)$$

Indeed, for  $x < 0$  both sides equal 1 by (1) and (2), and for  $x > 1$  both sides equal 0.

**Theorem 2.5** (Positivity, strict monotonicity, bijectivity). *The bounded Fabius function satisfies the following.*

1.  $F(x) > 0$  if and only if  $x > 0$ .
2.  $F(x) < 1$  if and only if  $x < 1$ .
3.  $F'(x) > 0$  for  $0 < x < 1$ , and  $F$  is strictly increasing on  $[0, 1]$ .
4.  $F$  maps  $[0, 1]$  bijectively onto  $[0, 1]$ .

*Proof.* The heart of the matter is the following doubling step: if  $0 \leq x \leq \frac{1}{2}$  and  $F(x) = 0$ , then  $F(2x) = 0$ . To see it, observe that  $F \geq 0$  everywhere, so a zero of  $F$  is a global minimum of a function differentiable on all of  $\mathbb{R}$ ; Fermat's rule gives  $F'(x) = 0$ , and (4) turns this into  $2F(2x) = 0$ .

Suppose now that  $F(x) = 0$  for some  $x > 0$ . Since  $2^n x \rightarrow \infty$ , there is a smallest integer  $n \geq 0$  with  $2^n x \geq \frac{1}{2}$ . For  $0 \leq m < n$  we have  $2^m x < \frac{1}{2}$ , so the doubling step applies at  $2^m x$  and propagates the vanishing: from  $F(x) = 0$  we obtain  $F(2^m x) = 0$  for every  $m \leq n$ , and in particular  $F(2^n x) = 0$ . But  $F$  is nondecreasing and  $2^n x \geq \frac{1}{2}$ , so  $F(2^n x) \geq F(\frac{1}{2}) = \frac{1}{2}$ , where the midpoint value comes from (3) at  $x = \frac{1}{2}$ . This contradiction proves that  $F$  has no zero on the positive half-line; combined with (1) it proves (i).

Assertion (ii) follows from (i) and (15):  $F(x) < 1$  if and only if  $F(1 - x) > 0$ , that is, if and only if  $1 - x > 0$ .

For (iii), let  $0 < x < 1$ . Then  $2x - 1 \in (-1, 1)$ , so  $1 - |2x - 1| \in (0, 1]$ , and (i) gives  $\text{up}(2x - 1) = F(1 - |2x - 1|) > 0$ . Now (14) yields  $F'(x) > 0$ . A continuous function on  $[0, 1]$  whose derivative is positive throughout the interior is strictly increasing on  $[0, 1]$ .



For (iv),  $F$  is continuous with  $F(0) = 0$  and  $F(1) = 1$ , so the intermediate value theorem makes  $F([0, 1])$  contain  $[0, 1]$ ; since  $F$  takes values in  $[0, 1]$ , the image is exactly  $[0, 1]$ . Strict monotonicity from (iii) gives injectivity.  $\square$

The fold (8) now transfers the whole picture to Rvachev’s bump, and it does so in strict form.

**Theorem 2.6** (Strict unimodality of  $\text{up}$ ). *Rvachev’s function satisfies the following.*

1.  $\text{up}$  is strictly increasing on  $[-1, 0]$  and strictly decreasing on  $[0, 1]$ .
2.  $\text{up}(x) = 1$  if and only if  $x = 0$ ; the maximum value 1 is attained at the origin and nowhere else.
3.  $\text{up}(x) = 0$  if and only if  $x \notin (-1, 1)$ . Thus  $\text{up}$  is nonzero exactly on  $(-1, 1)$ , and  $\text{supp } \text{up} = [-1, 1]$ .
4.  $\text{up}'(x) > 0$  for  $-1 < x < 0$  and  $\text{up}'(x) < 0$  for  $0 < x < 1$ .

*Proof.* By (8),  $\text{up}(x) = F(1 - x)$  for  $x \geq 0$ . As  $x$  increases through  $[0, 1]$  the argument  $1 - x$  decreases through  $[0, 1]$ , on which  $F$  is strictly increasing by theorem 2.5(iii); hence  $\text{up}$  is strictly decreasing on  $[0, 1]$ , and evenness gives (i) on  $[-1, 0]$ .

For (ii),  $\text{up}(x) = F(1 - |x|)$  and theorem 2.5(ii) say that  $\text{up}(x) < 1$  unless  $1 - |x| \geq 1$ , that is, unless  $x = 0$ ; and  $\text{up}(0) = F(1) = 1$ . For (iii),  $\text{up}(x) > 0$  if and only if  $1 - |x| > 0$  by theorem 2.5(i), and  $\text{up} \geq 0$ , so the zero set is exactly the complement of  $(-1, 1)$ ; the closed support is the closure  $[-1, 1]$ .

For (iv), the identity  $\text{up}(x) = F(1 - x)$  holds on the whole open half-line  $x > 0$ , hence may be differentiated there:  $\text{up}'(x) = -F'(1 - x)$ , which is negative for  $0 < x < 1$  by theorem 2.5(iii). Evenness gives  $\text{up}'(x) = -\text{up}'(-x)$ , so  $\text{up}' > 0$  on  $(-1, 0)$ .  $\square$

Parts (i) and (iii) are the classical description of the bump [6, 7]; the strict forms, and the equivalences in theorem 2.5, are exactly what the formal development records.

## 2.5 Convexity and the maximal slope

The derivative formula transfers the monotonicity of  $\text{up}$  to the derivative of  $F$ .

**Theorem 2.7** (Convexity, concavity, and the maximal slope). *The derivative  $F'$  is nondecreasing on  $(-\infty, \frac{1}{2}]$  and nonincreasing on  $[\frac{1}{2}, \infty)$ . It is strictly increasing on  $[0, \frac{1}{2}]$  and strictly decreasing on  $[\frac{1}{2}, 1]$ . Consequently  $F$  is convex on  $(-\infty, \frac{1}{2}]$ , concave on  $[\frac{1}{2}, \infty)$ , strictly convex on  $[0, \frac{1}{2}]$ , and strictly concave on  $[\frac{1}{2}, 1]$ . Moreover*

$$F'(\tfrac{1}{2}) = 2 = \max_{x \in \mathbb{R}} F'(x), \quad (16)$$

and equality  $F'(x) = 2$  holds exactly at  $x = \frac{1}{2}$ .

*Formal proof.* The derivative monotonicity statements are `monotoneOn_deriv_fabiusReal`, `antitoneOn_deriv_fabiusReal`, `strictMonoOn_deriv_fabiusReal_Icc`, and `strictAntitoneOn_deriv_fabiusReal_Icc`. The four convexity statements are `convexOn_fabiusReal_Iic_half`, `concaveOn_fabiusReal_Ici_half`, `strictConvexOn_fabiusReal_firstHalf`, and `strictConcaveOn_fabiusReal_secondHalf`. Finally `deriv_fabiusReal_half`, `deriv_fabiusReal_eq_two_iff`, and `isGreatest_deriv_fabiusReal` prove the exact maximal-slope statement. All are in `Convexity.lean`.  $\square$

The same module records the complete pointwise curvature profile. For every  $x \in \mathbb{R}$ ,

$$F''(x) = 8(\text{up}(4x - 1) - \text{up}(4x - 3)), \quad (17)$$

and therefore

$$\begin{aligned} F''(x) > 0 &\iff 0 < x < \tfrac{1}{2}, \\ F''(x) < 0 &\iff \tfrac{1}{2} < x < 1, \\ F''(x) = 0 &\iff x \notin (0, 1) \text{ or } x = \tfrac{1}{2}. \end{aligned} \tag{18}$$

The derivative-level statement and its pointwise corollary are `deriv_fabiusReal_hasDerivAt` and `deriv_deriv_fabiusReal`; the three equivalences are `deriv_deriv_fabiusReal_pos_iff`, `deriv_deriv_fabiusReal_neg_iff`, and `deriv_deriv_fabiusReal_eq_zero_iff`, with `deriv_deriv_fabiusReal_half` naming the midpoint specialization.

## 2.6 The optimal Lipschitz constant

The bound  $F' \leq 2$  of (16) is a Lipschitz estimate, and because the bound is attained the estimate cannot be improved.

**Theorem 2.8** (Two is the least Lipschitz constant). *Both  $F$  and  $\text{up}$  are 2-Lipschitz on  $\mathbb{R}$ :*

$$|F(x) - F(y)| \leq 2|x - y|, \quad |\text{up}(x) - \text{up}(y)| \leq 2|x - y| \quad (x, y \in \mathbb{R}). \tag{19}$$

Moreover 2 is the least admissible constant for each: if  $K \geq 0$  is such that  $|F(x) - F(y)| \leq K|x - y|$  for all  $x, y$ , or such that  $|\text{up}(x) - \text{up}(y)| \leq K|x - y|$  for all  $x, y$ , then  $K \geq 2$ .

*Proof.* By (14) and  $0 \leq \text{up} \leq 1$  we have  $0 \leq F' \leq 2$  on all of  $\mathbb{R}$ , so the mean value theorem gives the first estimate in (19). The map  $x \mapsto 1 - |x|$  is 1-Lipschitz, so the second estimate follows from the first through the fold (8).

For optimality, let  $K$  be admissible for  $F$ . Dividing  $|F(x) - F(\frac{1}{2})| \leq K|x - \frac{1}{2}|$  by  $|x - \frac{1}{2}|$  and letting  $x \rightarrow \frac{1}{2}$  gives  $K \geq |F'(\frac{1}{2})| = 2$  by (16). For  $\text{up}$  the same limit at the same point works: the proof of theorem 2.6(iv) gives  $\text{up}'(\frac{1}{2}) = -F'(\frac{1}{2}) = -2$ , so  $K \geq 2$ .  $\square$

The two extremal points are the same point in the two pictures:  $\frac{1}{2}$  is the centre of the sigmoid and the midpoint of the decreasing half of the bump, and the fold sends one to the other.

**Corollary 2.9** (Linear majorants at the endpoints). *For all admissible arguments,*

$$F(x) \leq 2x \quad (x \geq 0), \tag{20}$$

$$1 - F(x) \leq 2(1 - x) \quad (x \leq 1), \tag{21}$$

$$\text{up}(x) \leq 2(1 - |x|) \quad (|x| \leq 1). \tag{22}$$

*Proof.* Applying (19) with  $y = 0$  and using  $F(0) = 0$  gives (20). Replacing  $x$  by  $1 - x$  there and using (15) gives (21). Substituting  $1 - |x| \geq 0$  into (20) and using (8) gives (22).  $\square$

The content of (20) lies on  $[0, \frac{1}{2}]$ ; for  $x \geq \frac{1}{2}$  it is weaker than  $F \leq 1$ . Its value is that it holds with no hypothesis at all on the size of  $x$ , so it can be used as a linear majorant in any estimate near the left endpoint, and (22) says that the bump vanishes at least linearly at both ends of its support. These majorants and the optimality of the constant 2 are contributions of the formal development; the plain 2-Lipschitz bound is used implicitly throughout [7, 6].

## 3 Existence and uniqueness by a contraction

The most economical construction of  $F$  is a Banach fixed-point argument. It has an additional advantage: the contraction estimate later gives a clean uniqueness theorem, without presupposing Fourier analysis or probability.

### 3.1 The admissible function space

Let  $I = [0, 1]$  and let  $C(I)$  carry the supremum norm. Define  $\mathcal{A} \subset C(I)$  by

$$\mathcal{A} = \{f \in C(I) : 0 \leq f \leq 1, f(1-x) = 1 - f(x)\}. \quad (23)$$

The set  $\mathcal{A}$  is closed in the Banach space  $C(I)$ , hence complete. It is nonempty; for example  $f(x) = x$  belongs to it.

Extend  $f \in \mathcal{A}$  to a continuous function  $\tilde{f}$  on  $\mathbb{R}$  by clamping the argument to  $I$ :

$$\tilde{f}(t) = \begin{cases} f(0), & t \leq 0, \\ f(t), & 0 \leq t \leq 1, \\ f(1), & t \geq 1. \end{cases} \quad (24)$$

Symmetry gives

$$\int_0^1 f(t) dt = \frac{1}{2}. \quad (25)$$

Indeed, replacing  $t$  by  $1 - t$  and adding the two integrals gives  $2 \int_0^1 f = 1$ .

### 3.2 The integral transform

For  $f \in \mathcal{A}$ , define  $Tf$  by

$$(Tf)(x) = \begin{cases} \int_0^{2x} \tilde{f}(t) dt, & 0 \leq x \leq \frac{1}{2}, \\ 1 - \int_0^{2-2x} \tilde{f}(t) dt, & \frac{1}{2} < x \leq 1. \end{cases} \quad (26)$$

The two definitions agree at  $x = 1/2$  by (25). Positivity and the upper bound  $f \leq 1$  show that  $0 \leq Tf \leq 1$ , and a direct substitution shows  $(Tf)(1-x) = 1 - (Tf)(x)$ . Hence  $T : \mathcal{A} \rightarrow \mathcal{A}$ .

The central estimate is stronger than the naive integral bound.

**Lemma 3.1** (Contraction estimate). *For  $f, g \in \mathcal{A}$ ,*

$$\|Tf - Tg\|_\infty \leq \frac{1}{2} \|f - g\|_\infty. \quad (27)$$

*Proof.* Write  $h = f - g$ . Since both  $f$  and  $g$  have integral  $1/2$ ,  $\int_0^1 h = 0$ . For  $0 \leq y \leq 1$  we may therefore estimate the primitive in two ways:

$$\begin{aligned} \left| \int_0^y h(t) dt \right| &\leq y \|h\|_\infty, \\ \left| \int_0^y h(t) dt \right| &= \left| \int_y^1 h(t) dt \right| \leq (1-y) \|h\|_\infty. \end{aligned}$$

Consequently

$$\left| \int_0^y h(t) dt \right| \leq \min(y, 1-y) \|h\|_\infty \leq \frac{1}{2} \|h\|_\infty.$$

Every value of  $Tf - Tg$  is one of these primitives, possibly with a minus sign. Taking the supremum proves (27).  $\square$

**Theorem 3.2** (Existence of the bounded Fabius function). *The transform  $T$  has a unique fixed point  $f_* \in \mathcal{A}$ . Extending  $f_*$  by 0 on  $(-\infty, 0]$  and by 1 on  $[1, \infty)$  produces the bounded Fabius function  $F$ .*

*Proof.* The space  $\mathcal{A}$  is complete and  $T$  is a strict contraction with constant  $1/2$ , so Banach’s fixed-point theorem gives a unique fixed point. On  $[0, 1/2]$  its fixed-point equation is

$$F(x) = \int_0^{2x} F(t) dt. \quad (28)$$

Differentiating gives  $F'(x) = 2F(2x)$ . The second half of (26) supplies symmetry and the matching right-hand formula. The endpoint values are immediate.  $\square$

### 3.3 The differential equation for up and smoothness

Folding (4) and (5) yields the refinement-type differential equation

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)) \quad (x \in \mathbb{R}). \quad (29)$$

At  $x = 0$  the two one-sided derivatives glue because  $\text{up}(1) = \text{up}(-1) = 0$ . Once  $\text{up}$  is continuously differentiable, (29) bootstraps regularity: if  $\text{up} \in C^r$ , then the right side is  $C^r$ , so  $\text{up} \in C^{r+1}$ . Induction gives  $\text{up} \in C^\infty$  and then  $F \in C^\infty$ .

**Proposition 3.3** (Basic shape). *The functions  $F$  and  $\text{up}$  have the following properties.*

1.  $F$  is strictly increasing on  $(0, 1)$  and satisfies  $0 < F(x) < 1$  there.
2.  $\text{up}$  is even, strictly positive on  $(-1, 1)$ , and zero outside  $[-1, 1]$ .
3.  $\int_{-1}^1 \text{up}(x) dx = 1$ .
4. Every derivative of  $\text{up}$  vanishes at  $x = \pm 1$ .

*Proof.* For  $0 < x \leq 1/2$ , (4) gives  $F'(x) = 2F(2x)$ . Positivity follows first near  $1/2$  from  $F(1/2) = 1/2$ , and then propagates toward zero by repeatedly doubling the argument; symmetry handles the right half. This proves strict monotonicity and, by (8), positivity of  $\text{up}$  in the interior of its support. Finally,

$$\int_{-1}^1 \text{up}(x) dx = 2 \int_0^1 F(t) dt = 1,$$

where the middle identity uses the fold and the last uses symmetry. Flatness at the support endpoints follows from smoothness and the fact that  $\text{up}$  is identically zero just outside the support.  $\square$

### 3.4 The original Rvachev characterization

The formal predicate `IsOriginalFabius` packages the source characterization exactly. For a function  $\varphi : \mathbb{R} \rightarrow \mathbb{R}$  and a scale  $k$ , it requires  $C^\infty$ -smoothness, topological support  $[-1, 1]$ , strict positivity on  $(-1, 1)$ , the normalization  $\varphi(0) = 1$ , positivity of  $k$ , and

$$\varphi'(x) = k(\varphi(2x + 1) - \varphi(2x - 1)) \quad (x \in \mathbb{R}). \quad (30)$$

The positivity of the scale is redundant: the smart constructor `IsOriginalFabius.mk_of_derivative_law` derives it from the other hypotheses.

**Theorem 3.4** (Original characterization). *If `IsOriginalFabius`  $\varphi$   $k$ , then*

$$k = 2, \quad \varphi = \text{up}. \quad (31)$$

*Conversely, the canonical  $\text{up}$  satisfies this predicate at scale 2; in particular there is exactly one original solution.*

*Formal proof.* The scale is fixed by `originalFabius_scale_eq_two`; the canonical witness is `canonical_isOriginalFabius`. Uniqueness and the displayed characterization are `originalFabius_eq_canonical`, `isOriginalFabius_iff_eq_canonical`, and `existsUnique_originalFabius` in `OriginalUniqueness.lean`.  $\square$

The fold loses only the unused right tail of a bounded candidate. Precisely,

$$\text{up}_G = \text{up}_H \iff G(t) = H(t) \text{ for every } t \leq 1. \quad (32)$$

Thus a bounded candidate is canonical exactly when its fold satisfies the original characterization at scale 2 and its right tail is constantly 1. These are the exact declarations `rvachevUp_eq_iff_eqOn_Iic_one`, `isFabius_iff_isOriginalFabius_rvachevUp_and_rightTail`, and `isOriginalFabius_iff_existsUnique_isFabius` in `OriginalUniqueness.lean`.

## 4 The probability law behind the functions

The fixed-point construction is analytic. A probabilistic construction explains the same self-similarity almost without calculation and makes monotonicity and positivity transparent.

### 4.1 An infinite weighted sum of uniforms

Let  $U_0, U_1, \dots$  be independent random variables, each uniformly distributed on  $[0, 1]$ , and define

$$X = \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}}. \quad (33)$$

The series converges absolutely for every outcome and satisfies  $0 \leq X \leq 1$ . Removing the first coordinate gives an independent copy  $X'$  of  $X$ , and

$$X = \frac{U_0 + X'}{2}. \quad (34)$$

Reflecting every coordinate,  $U_n \mapsto 1 - U_n$ , preserves the product measure and sends  $X$  to  $1 - X$ .

Let  $H(x) = \mathbb{P}(X \leq x)$  be the cumulative distribution function. Conditioning on  $U_0$  in (34) gives

$$H(y) = \int_0^1 H(2y - u) \, du. \quad (35)$$

For  $0 \leq y \leq 1/2$ , the integrand vanishes when  $u > 2y$ , so

$$H(y) = \int_0^{2y} H(t) \, dt. \quad (36)$$

Reflection gives  $H(1 - y) = 1 - H(y)$ . Thus the restriction of  $H$  to  $[0, 1]$  is a fixed point of the contraction  $T$  from [section 3](#).

**Theorem 4.1** (Global probabilistic representation). *For every  $x, y \in \mathbb{R}$ ,*

$$F(x) = \mathbb{P} \left( \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \leq x \right), \quad (37)$$

$$\text{up}(y) = \mathbb{P} \left( \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \leq 1 - |y| \right). \quad (38)$$

*The second identity includes both tails: if  $|y| \geq 1$ , both sides vanish.*

*Formal proof.* The product-space construction proves continuity and measurability of the uniformly convergent coordinate sum, identifies its pushforward distribution, establishes the head–tail self-similarity and the coordinate-reflection law, and invokes uniqueness of the bounded Fabius fixed point. The exact canonical specializations are `fabiusReal_eq_weightedSum_probability` and `rvachevUp_eq_weightedSum_probability_global` in `ProbabilityRepresentation.lean`. The same module makes the transition to the generic law literal: `ProbabilityRepresentation.weightedSumDistribution_eq_weightedUniformDistribution` identifies the dyadic `ProbabilityRepresentation.weightedSumDistribution` with the generic weighted-uniform law for  $w_n = 1/2/2^n = 2^{-n-1}$ . The additional bridges `ProbabilityRepresentation.weightedCoordinateSum_eq_geometricUniformSeries_one_half` and `ProbabilityRepresentation.weightedSumDistribution_eq_geometricUniformDistribution_one_half` identify the dyadic series and law with the named geometric specialization at  $q = 1/2$ . Its exact topological support is recorded by `ProbabilityRepresentation.weightedSumDistribution_support_eq_Icc`. The same `ProbabilityRepresentation.lean` ownership now includes the almost-everywhere carried support and restriction identity `ProbabilityRepresentation.ae_weightedSumDistribution_mem_Icc` and `ProbabilityRepresentation.weightedSumDistribution_restrict_Icc`.  $\square$

## 4.2 Generic weighted laws and support-free quantile transport

The probability mechanism does not depend on the dyadic weights. Let  $\Omega = [0, 1]^{\mathbb{N}}$ , let  $\mathbf{P}$  be product Lebesgue measure, and let  $\tau\omega = (\omega_1, \omega_2, \dots)$ . For a real normed vector space  $E$  equipped with a measurable structure and weights  $w : \mathbb{N} \rightarrow E$ , put

$$S_w(\omega) := \sum_{n=0}^{\infty} \omega_n w_n, \quad \mu_w := (S_w)_* \mathbf{P}, \quad w_n^+ := w_{n+1}. \quad (39)$$

The series  $S_w$  and the pushforward  $\mu_w$  are totalized Lean definitions: they exist syntactically even without summability or measurability. The law statements below add precisely the hypotheses that make those definitions analytically meaningful.

**Theorem 4.2** (Head–tail product law and arbitrary weighted distributions). *For every measurable map  $g : \Omega \rightarrow A$  into an arbitrary measurable space,*

$$(\omega \mapsto (\omega_0, g(\tau\omega)))_* \mathbf{P} = \lambda_{[0,1]} \otimes g_* \mathbf{P}, \quad (40)$$

where  $\lambda_{[0,1]}$  is Lebesgue measure restricted to the unit interval.

Now let  $E$  be a real Banach space with its Borel measurable structure. If  $\sum_n \|w_n\| < \infty$ , then  $\mu_w$  is a probability measure. The weighted joint law is more general and independent of any parent sequence: for every separately chosen  $v : \mathbb{N} \rightarrow E$  with  $\sum_n \|v_n\| < \infty$ ,

$$(\omega \mapsto (\omega_0, S_v(\tau\omega)))_* \mathbf{P} = \lambda_{[0,1]} \otimes \mu_v. \quad (41)$$

Under full summability, with  $W = \sum_n w_n$  and  $A_w(u, z) = u w_0 + z$ ,

$$\mu_w = (A_w)_* (\lambda_{[0,1]} \otimes \mu_{w^+}), \quad (42)$$

$$(x \mapsto W - x)_* \mu_w = \mu_w. \quad (43)$$

Thus every absolutely summable vector-weight law is centrally symmetric about  $W/2$ . If  $E = \mathbb{R}$  and at least one coefficient  $w_n$  is nonzero, then  $\mu_w$  is absolutely continuous with respect to Lebesgue measure and every singleton is  $\mu_w$ -null. No sign condition is imposed on the weights, and the nonzero coefficient may occur at an arbitrary index rather than at the head. More generally, its topological support is the exact range of the continuous series map:

$$\text{supp}(\mu_w) = S_w(\Omega). \quad (44)$$

If  $E = \mathbb{R}$ , with no sign restriction, put

$$L_w = \sum_n \min\{w_n, 0\}, \quad U_w = \sum_n \max\{w_n, 0\}.$$

Then cancellation and sign changes create no gaps:

$$S_w(\Omega) = \text{supp}(\mu_w) = [L_w, U_w]. \quad (45)$$

If every  $w_n \geq 0$ , this becomes the full interval  $[0, W]$ , including the degenerate case  $W = 0$ . Consequently

$$S_w(\Omega) = \text{supp}(\mu_w) = [0, W], \quad \mu_w([0, W]) = 1, \quad \mu_w(\mathbb{R} \setminus [0, W]) = 0. \quad (46)$$

Equation (40) is `ProbabilityRepresentation.uniformProduct_map_head_tail_comp`. Definition (39) uses the existing `ProbabilityRepresentation.weightedUniformSeries` and the new `ProbabilityRepresentation.weightedUniformDistribution`; its totalization boundary is part of the latter definition's statement. The probability and weighted head–tail results are `ProbabilityRepresentation.isProbabilityMeasure_weightedUniformDistribution` and `ProbabilityRepresentation.uniformProduct_map_head_tailWeightedUniformSeries`. The latter is the arbitrary- $v$  statement (41); the shifted choice  $v = w^+$  is made only in the subsequent split of  $\mu_w$ . Equations (42)–(43) are `ProbabilityRepresentation.weightedUniformDistribution_split` and `ProbabilityRepresentation.weightedUniformDistribution_reflection`; the two exact mass statements in (46) are `ProbabilityRepresentation.weightedUniformDistribution_Icc` and `ProbabilityRepresentation.weightedUniformDistribution_comp1_Icc`. Together these declarations give the probability, splitting, reflection, and carried-set layer of `WeightedUniformSeries.lean`.

The exact-support upgrade is isolated in `WeightedUniformSupport.lean`. Its ten public declarations are `ProbabilityRepresentation.isOpenPosMeasure_infinitePi`, `ProbabilityRepresentation.uniformProduct_isOpenPosMeasure`, `ProbabilityRepresentation.support_map_eq_closure_range_of_continuous`, `ProbabilityRepresentation.weightedUniformSeries_constCoordinates`, `ProbabilityRepresentation.weightedUniformDistribution_support_eq_range`, `ProbabilityRepresentation.range_weightedUniformSeries_eq_Icc_min_max`, `ProbabilityRepresentation.weightedUniformDistribution_support_eq_Icc_min_max`, `ProbabilityRepresentation.range_weightedUniformSeries_eq_Icc`, `ProbabilityRepresentation.weightedUniformDistribution_support_eq_Icc`, and `ProbabilityRepresentation.weightedUniformDistribution_support_eq_unitInterval`. The first three establish open positivity for arbitrary products and identify the support of a continuous pushforward with the closure of its range. Compactness of the coordinate cube then removes the closure in (44). In the real case, coordinates selecting each negative or positive coefficient attain the two endpoints, and connectedness fills the interval in (45); constant coordinate paths give the especially transparent nonnegative specialization.

The same module proves a more structural layer before any choice of geometric weights.

**Theorem 4.3** (Naturality and operator-affine recursion). *Let  $G$  and  $H$  be arbitrary measurable spaces, let  $g : \Omega \rightarrow G$  be measurable, and let  $\Phi : [0, 1] \times G \rightarrow H$  be measurable. If*

$$f(\omega) = \Phi(\omega_0, g(\tau\omega)) \quad \text{for } \mathbf{P}\text{-almost every } \omega,$$

*then no separate measurability hypothesis on  $f$  is needed and*

$$f_*\mathbf{P} = \Phi_*(\lambda_{[0,1]} \otimes g_*\mathbf{P}). \quad (47)$$

*There is also the immediate pointwise-recursion specialization.*



Let  $E$  be a real Banach space, let  $F$  be a real normed space, and let  $L : E \rightarrow F$  be continuous linear. The target  $F$  need not be complete. For  $\sum_n \|w_n\| < \infty$ , writing  $(Lw)_n = L(w_n)$ , one has

$$L(S_w(\omega)) = S_{Lw}(\omega). \quad (48)$$

If both spaces carry their Borel measurable structures, then

$$L_*\mu_w = \mu_{Lw}. \quad (49)$$

For scalar multiplication the pointwise identity

$$S_{cw}(\omega) = cS_w(\omega) \quad (50)$$

holds in every real normed space, with no completeness or summability assumption; the law-level identity  $(x \mapsto cx)_*\mu_w = \mu_{cw}$  retains completeness, Borel, and norm-summability hypotheses so that the totalized measure maps are genuine pushforwards.

Finally, let  $A : E \rightarrow E$  be continuous linear and suppose  $w_{n+1} = A(w_n)$  for every  $n$ . Then

$$S_w(\omega) = \omega_0 w_0 + A(S_w(\tau\omega)). \quad (51)$$

If  $E$  carries its Borel measurable structure, then also

$$A_*\mu_w = \mu_{w+}, \quad (52)$$

$$\mu_w = (B_{A,w})_*(\lambda_{[0,1]} \otimes \mu_w), \quad B_{A,w}(u, z) = uw_0 + A(z). \quad (53)$$

The scalar case  $w_{n+1} = cw_n$  is a corollary of the operator statement with  $A = c \text{ id}$ . Here  $c \in \mathbb{R}$  is arbitrary: positivity and contraction are not built into either theorem; norm summability of the displayed weights is the analytic hypothesis.

Equation (47) and its pointwise specialization are `ProbabilityRepresentation.uniformProduct_map_of_ae_head_tail_recursion` and `ProbabilityRepresentation.uniformProduct_map_of_head_tail_recursion`. Equations (48)–(50) are `ProbabilityRepresentation.weightedUniformSeries_map` and `ProbabilityRepresentation.weightedUniformSeries_smul`; their law-level counterparts are `ProbabilityRepresentation.weightedUniformDistribution_map` and `ProbabilityRepresentation.weightedUniformDistribution_smul`. Equation (52) is `ProbabilityRepresentation.weightedUniformDistribution_map_eq_shift`. The pointwise and law recursions (51) and (53) are `ProbabilityRepresentation.weightedUniformSeries_split_of_shift_eq_map` and `ProbabilityRepresentation.weightedUniformDistribution_split_of_shift_eq_map`; their scalar corollaries are `ProbabilityRepresentation.weightedUniformSeries_geometric_split` and `ProbabilityRepresentation.weightedUniformDistribution_geometric_split`. These are exactly the eleven public declarations in the generic naturality tranche.

The word “geometric” in the final two declaration names denotes only the conditional recursion  $w_{n+1} = cw_n$ . The named specialization is supplied separately, rather than being hidden in these generic theorems.

The refactored weighted-law surface also exposes seven compatibility declarations: `ProbabilityRepresentation.weightedUniformSeries_smul_weights` in `WeightedUniformSeries.lean`, together with `ProbabilityRepresentation.uniformProduct_map_head_tail_function`, `ProbabilityRepresentation.weightedUniformDistribution_isProbabilityMeasure`, `ProbabilityRepresentation.weightedUniformDistribution_smul_weights`, `ProbabilityRepresentation.uniformProduct_map_head_tail_weightedUniformSeries`, `ProbabilityRepresentation.weightedUniformDistribution_unitInterval`, and `ProbabilityRepresentation.weightedUniformDistribution_compl_unitInterval`.



in `WeightedUniformDistribution.lean`. The two scalar compatibility identities use the same totalized pointwise equality as (50); at law level the compatibility theorem is oriented as  $\mu_{cw} = (x \mapsto cx) * \mu_w$ . Those two unit-interval compatibility declarations are carried-set statements; the separate exact support equalities are supplied by `WeightedUniformSupport.lean`.

The same compatibility module contains a four-declaration real absolute-continuity tranche. For an arbitrary s-finite measure  $\nu$  on  $\mathbb{R}$  and  $a \neq 0$ , `ProbabilityRepresentation.uniformScaledAdd_absolutelyContinuous` proves that the image of  $\lambda_{[0,1]} \otimes \nu$  under  $(u, x) \mapsto ua + x$  is absolutely continuous with respect to Lebesgue measure. The weighted head-coordinate workhorse is `ProbabilityRepresentation.weightedUniformDistribution_absolutelyContinuous_of_head_ne_zero`. The sharp theorem `ProbabilityRepresentation.weightedUniformDistribution_absolutelyContinuous` assumes only norm summability and  $\exists n, w_n \neq 0$ , and `ProbabilityRepresentation.weightedUniformDistribution_nullSingletonClass` packages the resulting zero mass of every singleton as Mathlib’s `NullSingletonClass`.

The same measure language gives a clean inverse-CDF theorem without differentiating an inverse.

**Theorem 4.4** (Support-free and compact inverse-CDF transport). *Let  $\mu$  be an arbitrary probability measure on  $\mathbb{R}$ , with no support restriction, and let  $Q : \mathbb{R} \rightarrow \mathbb{R}$  be almost-everywhere measurable for  $\lambda_{[0,1]}$ . If, for every  $x \in \mathbb{R}$ ,*

$$Q(y) \leq x \iff y \leq \text{cdf}_\mu(x) \quad \text{for } \lambda_{[0,1]} \text{-almost every } y, \quad (54)$$

then

$$Q_* \lambda_{[0,1]} = \mu. \quad (55)$$

Thus neither  $Q([0, 1]) \subseteq [0, 1]$  nor compact support of  $\mu$  belongs to the foundational theorem; null-set choices of quantile convention are immaterial.

There is also a pointwise compact specialization. Suppose now that  $\mu$  is carried almost everywhere by  $[0, 1]$ , and:

1.  $Q([0, 1]) \subseteq [0, 1]$ ;
2. for every  $x, y \in [0, 1]$ ,

$$Q(y) \leq x \iff y \leq \text{cdf}_\mu(x).$$

Together with the same source-a.e. measurability, these hypotheses again imply (55). Both versions permit atoms.

The support-free theorem is exactly `map_quantile_eq`; its compact specialization is `map_inverseCDF_volume_restrict_Icc`, both in `QuantileTransport.lean`. In both cases only the source measure is restricted to  $[0, 1]$ . The first theorem neither controls nor needs the values of  $Q$  outside a source-full-measure set.

**Corollary 4.5** (Fabius quantile substitution). *Let  $I = \text{fabiusInv}$  be the Fabius inverse obtained by first clamping its argument to  $[0, 1]$ , and let  $\mu_X$  be the dyadic weighted-sum distribution in (37). Then*

$$I_* \lambda_{[0,1]} = \mu_X. \quad (56)$$

Consequently, for every strongly measurable map  $\Phi : \mathbb{R} \rightarrow E$  into a real Banach space,

$$\int_{[0,1]} \Phi(I(y)) \, dy = \int_{\mathbb{R}} \Phi(x) \, d\mu_X(x). \quad (57)$$

No separate integrability hypothesis is imposed. Lean’s Bochner integral is totalized to zero in the non-integrable case, and the measure-map theorem identifies the two totalized integrals there as well.

The measure identity and Banach-valued integral identity are respectively `map_fabiusInv_restrict_Icc_eq_weightedSumDistribution` and `integral_comp_fabiusInv_restrict_Icc_eq_weightedSumDistribution` in `QuantileTransport.lean`. Together with the support-free and compact statements above, they are exactly the four public declarations of that module. The clamped definition of  $I$  is global, but both displayed transport identities deliberately use uniform measure restricted to  $[0, 1]$ . They do not by themselves prove the two-function integration identity; the separate absolutely-continuous theorem below supplies that full-interval result. The transport declarations assert neither complex Mellin continuation nor fractional-power substitutions.

A separate subgraph-Fubini argument does provide the exact first-order weighted inverse identity, without differentiating either clock.

**Theorem 4.6** (Weighted Fabius inverse layer cake). *Let  $\phi$  be interval-integrable on  $[0, 1]$ , with real or complex values, and let  $\psi$  be interval-integrable there with values in a compatible Banach space. Then*

$$\int_0^1 \phi(u) \left( \int_0^{I(u)} \psi(x) dx \right) du = \int_0^1 \left( \int_{F(x)}^1 \phi(u) du \right) \psi(x) dx. \quad (58)$$

*If  $\Psi(0) = 0$ ,  $\psi$  is continuous on  $[0, 1]$ , and  $\Psi'(x) = \psi(x)$  pointwise there, the inner primitive on the left may be replaced by  $\Psi(I(u))$ . If instead  $\Psi : \mathbb{R} \rightarrow \mathbb{R}$  is merely absolutely continuous on  $[0, 1]$ , the same replacement is valid with  $\psi = \Psi'$  almost everywhere, for every real or complex interval-integrable weight  $\phi$ ; no separate integrability hypothesis on the right-hand side is needed.*

The arbitrary-absolutely-continuous Fabius form is `intervalIntegral_mul_comp_fabiusInv_of_absolutelyContinuousOnInterval`. The explicit-primitive and pointwise- $C^1$  forms are respectively `intervalIntegral_smul_intervalIntegral_fabiusInv` and `intervalIntegral_smul_comp_fabiusInv` in `InverseLayerCake.lean`.

The generic theorem does not require the primitive to vanish at the left endpoint.

**Theorem 4.7** (Absolutely-continuous inverse layer cake on a rectangle). *Let  $a \leq b$  and  $c \leq d$ . Suppose  $C, Q : \mathbb{R} \rightarrow \mathbb{R}$  map  $[a, b]$  into  $[c, d]$  and  $[c, d]$  into  $[a, b]$ , respectively, and*

$$C(x) < u \iff x < Q(u) \quad (x \in [a, b], u \in [c, d]).$$

*For  $\mathbb{K} = \mathbb{R}$  or  $\mathbb{C}$ , let  $\phi : [c, d] \rightarrow \mathbb{K}$  be integrable and let  $\Psi : \mathbb{R} \rightarrow \mathbb{R}$  be absolutely continuous on  $[a, b]$ . Then*

$$\int_c^d \phi(u) (\Psi(Q(u)) - \Psi(a)) du = \int_a^b \Psi'(x) \left( \int_{C(x)}^d \phi(u) du \right) dx. \quad (59)$$

*When  $\Psi(a) = 0$ , the subtraction on the left disappears. No measurability hypothesis is imposed on either clock, and equality is allowed in either endpoint pair. Values of the clocks outside the two displayed intervals play no role.*

The order-theoretic engine is `galoisConnection_Icc_restrict_of_lt_iff_lt`: the strict order equivalence makes the two interval restrictions a Galois connection and hence monotone. A canonical clamped extension of the restricted inverse is therefore measurable. Consequently, the generic explicit-primitive and pointwise- $C^1$  theorems `intervalIntegral_smul_intervalIntegral_of_lt_iff_lt` and `intervalIntegral_smul_comp_of_lt_iff_lt` no longer assume clock measurability. The subtraction and normalized forms of (59) are `intervalIntegral_mul_comp_sub_of_lt_iff_lt_of_absolutelyContinuousOnInterval` and `intervalIntegral_mul_comp_of_lt_iff_lt_of_absolutelyContinuousOnInterval`. The reusable arbitrary-s-finite-measure engine is `integral_smul_setIntegral_subgraph` in `SubgraphFubini.lean`. Finally, `intervalIntegral_mul_fabiusInv_eq` is the real weighted-area specialization; `intervalIntegral_fabiusInv_eq_intervalIntegral_rvachevUp` and

`intervalIntegral_fabiusInv_eq_one_half` identify the inverse area with the unit-interval up area and with  $1/2$ , respectively.

The inverse-order relation also supports a genuinely two-function absolutely-continuous calculus. Here the important reusable point is that measurability of the clocks is a conclusion of the order structure on the rectangle, not an extra hypothesis.

**Theorem 4.8** (Absolutely-continuous calculus for an inverse pair). *Let  $a \leq b$  and  $c \leq d$ . Suppose  $C, Q : \mathbb{R} \rightarrow \mathbb{R}$  map  $[a, b]$  into  $[c, d]$  and  $[c, d]$  into  $[a, b]$ , respectively, and*

$$C(x) < u \iff x < Q(u) \quad (x \in [a, b], u \in [c, d]). \quad (60)$$

*If  $A, B : \mathbb{R} \rightarrow \mathbb{R}$ , with  $A$  absolutely continuous on  $[a, b]$  and  $B$  absolutely continuous on  $[c, d]$ , then*

$$\int_a^b A'(x)B(C(x)) \, dx + \int_c^d A(Q(u))B'(u) \, du = A(b)B(d) - A(a)B(c). \quad (61)$$

*Moreover, every  $y \in [a, b]$  cuts out the smaller inverse rectangle  $[a, y] \times [c, C(y)]$ , and*

$$\int_a^y A'(x)B(C(x)) \, dx + \int_c^{C(y)} A(Q(u))B'(u) \, du = A(y)B(C(y)) - A(a)B(c). \quad (62)$$

*No measurability, continuity, or differentiability assumption is imposed on  $C$  or  $Q$ . The strict-order equivalence forces both restrictions to be monotone; canonical clamped extensions are then measurable. Values of the two clocks outside their displayed compact intervals play no role. Equality of either endpoint pair is allowed. The stated orientation is essential: reversed endpoints require swapping the corresponding interval and its order data first.*

The complete and variable-upper statements are, respectively, `intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_of_lt_iff_lt` and `intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_to_of_lt_iff_lt` in `InversePairIntegral.lean`. Their proof combines `intervalIntegral_smul_intervalIntegral_of_lt_iff_lt` with the fundamental theorem for absolutely-continuous functions; no derivative or Jacobian formula for the clocks enters.

**Corollary 4.9** (Fabius inverse-pair calculus and area sum). *For real functions  $A, B : \mathbb{R} \rightarrow \mathbb{R}$  that are absolutely continuous on  $[0, 1]$ ,*

$$\int_0^1 A'(x)B(F(x)) \, dx + \int_0^1 A(I(u))B'(u) \, du = A(1)B(1) - A(0)B(0). \quad (63)$$

*For every  $y \in [0, 1]$ , the corresponding variable-upper identity is*

$$\int_0^y A'(x)B(F(x)) \, dx + \int_0^{F(y)} A(I(u))B'(u) \, du = A(y)B(F(y)) - A(0)B(0). \quad (64)$$

*In particular, taking both outer functions to be the identity gives the inverse-graph area identity*

$$\int_0^1 F(x) \, dx + \int_0^1 I(u) \, du = 1. \quad (65)$$

*At a variable inverse endpoint  $u \in [0, 1]$ , the same specialization gives*

$$\int_0^u I(v) \, dv = uI(u) - \int_0^{I(u)} F(x) \, dx. \quad (66)$$

The complete and variable-upper two-function statements are exactly `intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv` and `intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv_to`; the two area statements are `intervalIntegral_fabiusReal_add_fabiusInv_eq_one` and `intervalIntegral_fabiusInv_to_eq_mul_sub_intervalIntegral_fabiusReal` in `InversePairIntegral.lean`. These are proper-compact first-order identities. Reversed-endpoint, higher-order, improper-integral, fractional-power, complex-valued arbitrary-AC, and complex-Mellin extensions remain outside these declarations.

### 4.3 The normalized geometric family

The same product probability space supports a more general family without changing the measure-theoretic argument. For  $q \in \mathbb{R}$  with  $|q| < 1$ , put

$$w_n(q) = (1 - q)q^n, \quad X_q = \sum_{n=0}^{\infty} w_n(q)U_n, \quad \mu_q = \text{law}(X_q). \quad (67)$$

For every real  $q$ ,  $w_0(q) = 1 - q$  and  $w_{n+1}(q) = q w_n(q)$ . Under  $|q| < 1$ , the weights are absolutely summable and have total mass one. The series is continuous and measurable on the full product space, and its law satisfies the exact affine refinement

$$\mu_q = (\text{Unif}[0, 1] \otimes \mu_q) \text{ map } ((u, x) \mapsto (1 - q)u + qx). \quad (68)$$

Equivalently,  $X_q \stackrel{d}{=} (1 - q)U_0 + qX'_q$ , where  $X'_q$  is an independent copy. The head coordinate and the shifted full series have product law. Coordinate reflection gives  $X_q(R\omega) = 1 - X_q(\omega)$  and  $\mu_q \text{ map } (x \mapsto 1 - x) = \mu_q$ , where  $(R\omega)_n = 1 - \omega_n$ . For every  $|q| < 1$ , the law  $\mu_q$  is absolutely continuous with respect to Lebesgue measure and has null singletons; no positivity hypothesis on  $q$  is needed. For every  $|q| < 1$ , the topological support of  $\mu_q$  is exactly the range  $X_q(\Omega)$ . When  $0 \leq q < 1$ , every weight is nonnegative,  $X_q(\omega) \in [0, 1]$ ,  $\mu_q([0, 1]) = 1$ ,  $\mu_q(\mathbb{R} \setminus [0, 1]) = 0$ , and  $\text{supp}(\mu_q) = [0, 1]$ ; the endpoint  $q = 0$  is included. At  $q = 1/2$ , both  $X_q$  and  $\mu_q$  are exactly the dyadic random series and law above.

The same affine equation admits a reusable uniqueness theorem far beyond this real uniform-digit example. Let  $D$  be a measurable space, let  $E$  be a complete, second-countable real inner-product space with its Borel measurable structure, let  $\rho$  be a probability measure on  $D$ , and let  $d : D \rightarrow E$  be measurable. For a probability measure  $\lambda$  on  $E$ , define

$$\mathcal{A}_{\rho, d, q}(\lambda) = (d_*\rho) * ((x \mapsto qx)_*\lambda) = ((u, x) \mapsto d(u) + qx)_*(\rho \otimes \lambda). \quad (69)$$

The second equality is the product-map form of convolution. Thus  $\mathcal{A}_{\rho, d, q}$  is precisely the law transformer for  $d(U) + qX$  when  $U \sim \rho$ ,  $X \sim \lambda$ , and the two inputs are independent.

**Theorem 4.10** (Contractive affine independent-copy law). *The transformer  $\mathcal{A}_{\rho, d, q}$  preserves probability measures, and its characteristic function factors as*

$$\widehat{\mathcal{A}_{\rho, d, q}(\lambda)}(t) = \widehat{d_*\rho}(t) \widehat{\lambda}(qt). \quad (70)$$

If  $\lambda$  is a fixed point, then for every  $n \in \mathbb{N}$ ,

$$\widehat{\lambda}(t) = \left( \prod_{k=0}^{n-1} \widehat{d_*\rho}(q^k t) \right) \widehat{\lambda}(q^n t). \quad (71)$$

If  $|q| < 1$ , the transformer has at most one probability fixed point. Equivalently, two probability measures satisfying the product-map equation in (69) are equal.

*Proof.* The factorization is the convolution product rule followed by the characteristic-function rule for real scalar multiplication. For two fixed points, their common digit factor has modulus at most one, so one recurrence step gives

$$|\widehat{\lambda}(t) - \widehat{\nu}(t)| \leq |\widehat{\lambda}(qt) - \widehat{\nu}(qt)|.$$

Iteration moves the argument to  $q^n t$ . Since  $|q| < 1$ , this tends to zero; continuity of characteristic functions and their common value 1 at the origin force equality at every  $t$ , and characteristic-function extensionality gives  $\lambda = \nu$ . The proof uses no support, density, absolute-continuity, or moment hypothesis on either fixed point or on the arbitrary measurable digit law. It includes  $q = 0$  and negative  $q$ .  $\square$

**Corollary 4.11** (Characterization of the geometric uniform law). *For every real  $q$  with  $|q| < 1$ , if a probability measure  $\lambda$  on  $\mathbb{R}$  satisfies*

$$\lambda = ((u, x) \mapsto (1 - q)u + qx)_* (\text{Unif}[0, 1] \otimes \lambda),$$

*then  $\lambda = \mu_q$ . Hence the refinement equation (68) characterizes the named law even at  $q = 0$  and for negative  $q$ , without any support, density, or moment assumption on  $\lambda$ .*

The CDF and its positive-ratio density are now named as well. Put

$$F_q(x) = \text{cdf}(\mu_q)(x), \quad f_q(x) = \frac{F_q(x/q) - F_q((x - (1 - q))/q)}{1 - q}. \quad (72)$$

Both definitions are total in the real parameter. For every real  $q$ ,  $F_q$  is measurable and monotone with values in  $[0, 1]$ . If  $|q| < 1$ , then

$$F_q \in C(\mathbb{R}), \quad F_q(1 - x) = 1 - F_q(x), \quad F_q(1/2) = 1/2. \quad (73)$$

If  $0 \leq q < 1$ , including  $q = 0$ , its exterior values are

$$F_q(x) = 0 \quad (x \leq 0), \quad F_q(x) = 1 \quad (x \geq 1). \quad (74)$$

For  $0 < q < 1$ , conditioning on the head coordinate gives, for every real  $x$ ,

$$F_q(x) = \int_0^1 F_q\left(\frac{x - (1 - q)u}{q}\right) du, \quad (75)$$

$$= \frac{q}{1 - q} \int_{(x - (1 - q))/q}^{x/q} F_q(t) dt. \quad (76)$$

The second expression in (72) is then the everywhere pointwise derivative of  $F_q$ . It is continuous, nonnegative, reflection-symmetric, vanishes for  $x \leq 0$  and for  $x \geq 1$ , has ordinary support contained in  $(0, 1)$  and topological support contained in  $[0, 1]$ , and is compactly supported. More precisely,

$$\mu_q = (\text{Leb} : \text{Measure}(\mathbb{R})) \text{ withDensity } (x \mapsto \text{ofReal}(f_q(x))), \quad F_q, f_q \in C^\infty(\mathbb{R}). \quad (77)$$

Thus the smoothness proof and the explicit Radon–Nikodym identification are formal in real space for the full positive-ratio range, without passing through Fourier inversion.

At the half base the named objects recover the classical functions globally:

$$F_{1/2} = \text{weightedSumCDF} = F, \quad f_{1/2}(x) = 2 \text{ up}(2x - 1) \quad (x \in \mathbb{R}). \quad (78)$$

*Formal API crosswalk.* The twenty-three public declarations in `GeometricUniformLaw.lean` are `ProbabilityRepresentation.geometricUniformWeight`, `ProbabilityRepresentation.geometricUniformWeight_zero`, `ProbabilityRepresentation.geometricUniformWeight_succ`, `ProbabilityRepresentation.hasSum_geometricUniformWeight`, `ProbabilityRepresentation.summable_norm_geometricUniformWeight`, `ProbabilityRepresentation.geometricUniformWeight_nonneg`, `ProbabilityRepresentation.geometricUniformSeries`, `ProbabilityRepresentation.continuous_geometricUniformSeries`, `ProbabilityRepresentation.measurable_geometricUniformSeries`, `ProbabilityRepresentation.geometricUniformSeries_split`, `ProbabilityRepresentation.geometricUniformSeries_reflect`, `ProbabilityRepresentation.geometricUniformSeries_mem_Icc`, `ProbabilityRepresentation.geometricUniformDistribution`, `ProbabilityRepresentation.geometricUniformDistribution_isProbabilityMeasure`, `ProbabilityRepresentation.geometricUniformDistribution_absolutelyContinuous`, `ProbabilityRepresentation.geometricUniformDistribution_nullSingletonClass`,

ProbabilityRepresentation.uniformProduct\_map\_head\_tail\_geometricUniformSeries, ProbabilityRepresentation.geometricUniformDistribution\_selfSimilar, ProbabilityRepresentation.geometricUniformDistribution\_reflection, ProbabilityRepresentation.geometricUniformDistribution\_Icc, ProbabilityRepresentation.geometricUniformDistribution\_compl\_Icc, ProbabilityRepresentation.geometricUniformDistribution\_support\_eq\_range, and ProbabilityRepresentation.geometricUniformDistribution\_support\_eq\_Icc.

At compiler-validated checkpoint d312c0603, the generic affine interface in AffineIndependentCopy.lean consists of the definition ProbabilityTheory.affineIndependentCopyLaw and the eight theorems ProbabilityTheory.affineIndependentCopyLaw\_isProbabilityMeasure, ProbabilityTheory.affineIndependentCopyLaw\_eq\_map\_prod, ProbabilityTheory.charFun\_affineIndependentCopyLaw, ProbabilityTheory.charFun\_eq\_mul\_charFun\_of\_affineIndependentCopy\_fixedPoint, ProbabilityTheory.charFun\_iterate\_of\_affineIndependentCopy\_fixedPoint, ProbabilityTheory.eq\_of\_charFun\_affine\_recurrence, ProbabilityTheory.affineIndependentCopyLaw\_fixedPoint\_unique, and ProbabilityTheory.affineIndependentCopy\_map\_fixedPoint\_unique. The named specialization is ProbabilityRepresentation.eq\_geometricUniformDistribution\_of\_selfSimilar in GeometricUniformUniqueness.lean. The generic theorem is formulated for a complete second-countable real inner-product target and an arbitrary measurable digit map from an arbitrary probability space; the named theorem specializes it to uniform digits.

The predecessor GeometricUniformDictionary.lean already supplies finite measure, characteristic-function, MGF, and CGF tail factorizations. The six public theorems in GeometricSincFactorization.lean are charFun\_geometricUniformDigit, charFun\_geometricUniformDistribution\_prefix, charFun\_geometricUniformDistribution\_prefix\_sinc, tendsto\_prefix\_sinc\_charFun, charFun\_weightedSumDistribution\_prefix\_sinc, and tendsto\_prefix\_sinc\_charFun\_weightedSumDistribution. The digit formula is unconditional in real  $q$ . For every  $|q| < 1$ , depth  $m$ , and real frequency  $t$ , the phase-collapsed residual identity is

$$\phi_q(t) = e^{i(1-q^m)t/2} \left( \prod_{k=0}^{m-1} \operatorname{sinc} \left( \frac{(1-q)q^k t}{2} \right) \right) \phi_q(q^m t),$$

and the phase-bearing prefix without the residual converges pointwise to  $\phi_q(t)$ . This includes  $q = 0$  and negative  $q$ ; the final two declarations are exact  $q = 1/2$  wrappers for weightedSumDistribution. The existing geometricSincProduct already packages the same sinc factors, after frequency rescaling, as a tprod. What is missing is a named characteristic-function-to-geometricSincProduct bridge (or explicit HasProd theorem) and compact-uniform convergence of the characteristic prefixes.

The generic CDF mechanisms are continuous\_cdf\_of\_nullSingleton, cdf\_reflection\_sub, and measure\_eq\_withDensity\_of\_cdf\_hasDerivAt in ContinuousCDF.lean.

The exact named interface in GeometricUniformCDF.lean consists of ProbabilityRepresentation.geometricUniformCDF, ProbabilityRepresentation.monotone\_geometricUniformCDF, ProbabilityRepresentation.geometricUniformCDF\_nonneg, ProbabilityRepresentation.geometricUniformCDF\_le\_one, ProbabilityRepresentation.measurable\_geometricUniformCDF, ProbabilityRepresentation.continuous\_geometricUniformCDF, ProbabilityRepresentation.geometricUniformCDF\_reflection, ProbabilityRepresentation.geometricUniformCDF\_one\_half, ProbabilityRepresentation.geometricUniformCDF\_zero\_of\_nonpos, ProbabilityRepresentation.geometricUniformCDF\_one\_of\_one\_le, ProbabilityRepresentation.geometricUniformCDF\_eq\_integral, ProbabilityRepresentation.geometricUniformCDF\_eq\_intervalIntegral, ProbabilityRepresentation.geometricUniformDensity, ProbabilityRepresentation.geometricUniformCDF\_hasDerivAt, ProbabilityRep

resentation.deriv\_geometricUniformCDF, ProbabilityRepresentation.continuous\_geometricUniformDensity, ProbabilityRepresentation.geometricUniformDensity\_nonneg, ProbabilityRepresentation.geometricUniformDensity\_zero\_of\_nonpos, ProbabilityRepresentation.geometricUniformDensity\_zero\_of\_one\_le, ProbabilityRepresentation.support\_geometricUniformDensity\_subset\_Ioo, ProbabilityRepresentation.support\_geometricUniformDensity\_subset\_Icc, ProbabilityRepresentation.tsupport\_geometricUniformDensity\_subset\_Icc, ProbabilityRepresentation.geometricUniformDensity\_hasCompactSupport, ProbabilityRepresentation.geometricUniformDensity\_reflection, ProbabilityRepresentation.geometricUniformDistribution\_eq\_withDensity, ProbabilityRepresentation.contDiff\_geometricUniformCDF, and ProbabilityRepresentation.contDiff\_geometricUniformDensity.

Exact half-base series and law recovery is weightedCoordinateSum\_eq\_geometricUniformSeries\_one\_half and weightedSumDistribution\_eq\_geometricUniformDistribution\_one\_half in ProbabilityRepresentation.lean. The CDF and density bridges there are ProbabilityRepresentation.weightedSumCDF\_eq\_geometricUniformCDF\_one\_half, ProbabilityRepresentation.geometricUniformCDF\_one\_half\_eq\_fabijsReal, and ProbabilityRepresentation.geometricUniformDensity\_one\_half\_eq\_rvachevUp; the exact dyadic support corollary is weightedSumDistribution\_support\_eq\_Icc.  $\square$

The remaining boundary is parameter-sensitive. Although  $\mu_q$  is absolutely continuous throughout  $|q| < 1$ , the displayed total function  $f_q$  is identified as its Radon–Nikodym density only for  $0 < q < 1$ ; no such interpretation is claimed here at  $q = 0$  or for negative  $q$ . The positive-ratio  $C^\infty$  theorem is proved by the real-space derivative refinement. The phase-bearing general- $q$  sinc prefixes and their pointwise characteristic-function limit are formal, while geometricSincProduct already packages the same rescaled sinc factors as a tprod. A named characteristic-function-to-geometricSincProduct bridge (or explicit HasProd theorem), compact-uniform convergence, rapid-decay estimates, and a Fourier-inversion construction remain outside the formal API. So do a named centered  $q$ -family, a closed all-order derivative hierarchy and explicit endpoint-flatness wrappers, shape theorems, further transform formulas, and explicit Bernoulli-cumulant/Bell-moment formulas and asymptotics. The probability-law fixed point itself is uniquely characterized by [Corollary 4.11](#). The oriented Cauchy transform and its adjacent-power hierarchy are also formal for every nonzero  $|q| < 1$ , including negative ratios, in GeometricUniformCauchy.lean; these are distinct from the still-open characteristic-function-to-infinite-product and Fourier-inversion formulas just listed.

#### 4.4 Finite atomic laws and weak convergence

There is also a completely discrete approximation theorem. For  $k \geq 0$ , let  $\beta_k$  be the probability measure assigning mass  $1/2$  to each of  $\pm 2^{-k-1}$ . Starting from  $\nu_0 = \delta_0$ , define

$$\nu_{n+1} = \nu_n * \beta_{n+1}^{*(n+1)}. \quad (79)$$

Thus  $\nu_n$  is a finite atomic probability measure. Its characteristic function is the finite weighted cosine product

$$\widehat{\nu}_n(t) = \prod_{k=0}^{n-1} \left( \cos \frac{t}{2^{k+2}} \right)^{k+1}. \quad (80)$$

The indexing in (79) is worth noticing: the new scale at stage  $n + 1$  is  $2^{-n-2}$ , and it occurs with multiplicity  $n + 1$ .

Let  $\mu_{\text{up}}$  denote the formal measure rvachevMeasure; by definition it is Lebesgue measure with density ofRealo up. It is absolutely continuous with respect to Lebesgue measure; in particular, every Lebesgue-null set is also  $\mu_{\text{up}}$ -null. These are rvachevMeasure\_absolutelyContinuous



and `rvachevMeasure_eq_zero_of_volume_eq_zero` in `StepMeasureBridge.lean`. Its characteristic function uses the positive-frequency normalization

$$\text{charFun}(\mu_{\text{up}}, t) = \widehat{\text{up}}\left(\frac{t}{2\pi}\right) \quad (t \in \mathbb{R}), \quad (81)$$

the exact identity `rvachevMeasure_charFun_pos` in `WeakConvergence.lean`. The finite laws converge weakly to  $\mu_{\text{up}}$ :

$$\nu_n \implies \mu_{\text{up}}. \quad (82)$$

Equivalently, for every bounded continuous real-valued map  $g$ ,

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} g(x) \, d\nu_n(x) = \int_{\mathbb{R}} g(x) \, d\mu_{\text{up}}(x). \quad (83)$$

The same test-function theorem is formalized for complex-valued, and more generally real-or-complex-like, bounded continuous functions.

*Formal proof.* The definitions are `centeredBernoulliMeasure` and `finiteConvolutionMeasure`. Their probability-measure instances and (80) are proved in `EarlyApproximants.lean`. Pointwise convergence of the characteristic functions, weak convergence (82), and the test-function formulations (83) are respectively `finiteConvolutionMeasure_charFun_tendsto`, `finiteConvolutionProbability_tendsto_fabius`, and `integral_finiteConvolutionMeasure_tendsto_fabius` in `WeakConvergence.lean`.  $\square$

#### 4.5 Combinatorial step functions and the half-weight convention

A second finite model replaces each atom by a narrow interval. Define polynomials with nonnegative integer coefficients by

$$\mathcal{P}_0(X) = 1, \quad \mathcal{P}_m(X) = \mathcal{P}_{m-1}(X^2)(1 + X)^m \quad (m \geq 1). \quad (84)$$

They also have the finite product form

$$\mathcal{P}_m(X) = \prod_{j=1}^m (1 + X + \cdots + X^{2^j-1}). \quad (85)$$

Writing  $\mathcal{P}_m(X) = \sum_r a_{m,r} X^r$ , the coefficient  $a_{m,r}$  counts the tuples

$$(s_1, \dots, s_m), \quad 0 \leq s_j \leq 2^j - 1, \quad s_1 + \cdots + s_m = r. \quad (86)$$

Moreover,

$$g_m := \deg \mathcal{P}_m = 2^{m+1} - m - 2, \quad \mathcal{P}_m(1) = 2^{\binom{m+1}{2}}. \quad (87)$$

For  $\alpha < \beta$ , let  $\mathbf{1}_{[\alpha, \beta]}^*$  be 1 in the interior, 1/2 at either endpoint, and 0 elsewhere. Put

$$I_{m,r} = \left[ \frac{2r - 1 - g_m}{2^{m+1}}, \frac{2r + 1 - g_m}{2^{m+1}} \right] \quad (88)$$

and

$$\mathcal{W}_m(x) = \frac{2^m}{2^{\binom{m+1}{2}}} \sum_{r=0}^{g_m} a_{m,r} \mathbf{1}_{I_{m,r}}^*(x). \quad (89)$$



**Theorem 4.12** (Corrected step approximation). *For every  $m \geq 0$  and  $x \in \mathbb{R}$ ,*

$$0 \leq \mathcal{W}_m(x) \leq 1, \quad \mathcal{W}_m(0) = 1, \quad \int_{\mathbb{R}} \mathcal{W}_m(x) dx = 1.$$

*The function  $\mathcal{W}_m$  is nondecreasing on  $(-\infty, 0)$  and nonincreasing on  $(0, \infty)$ . For every real  $x$ ,*

$$\mathcal{W}_m(x) \longrightarrow \text{up}(x). \quad (90)$$

The proof of pointwise convergence passes through a useful exact comparison. Set  $h_m = 2^{-m-1}$ , and use the finite atomic law  $\nu_m$  from (79). Whenever  $a \leq b$ ,

$$\nu_m((a + h_m, b - h_m]) \leq \int_a^b \mathcal{W}_m(x) dx \leq \nu_m((a - h_m, b + h_m]). \quad (91)$$

Consequently,

$$\lim_{m \rightarrow \infty} \int_a^b \mathcal{W}_m(x) dx = \int_a^b \text{up}(x) dx. \quad (92)$$

Monotonicity on the two open half-lines, the exact value at the origin, and continuity of  $\text{up}$  then upgrade these interval limits to (90).

*Formal proof.* The polynomial recursion, product, degree, total coefficient mass, and bounded-partition interpretation are respectively proved by `approximationPolynomial_succ`, the definition `approximationPolynomial`, `approximationPolynomial_natDegree`, `approximationDegree_eq_sub`, `approximationPolynomial_eval_one`, and `approximationPolynomial_coeff_eq_card` in `EarlyApproximants.lean`. The finite atomic description is identified with `finiteConvolutionMeasure` by `polynomialMeasure_eq_finiteConvolutionMeasure` in `EarlyMeasureBridge.lean`.

For nonnegativity, see `stepApproximant_nonneg` in `EarlyApproximants.lean`. The remaining analytic assertions are `stepApproximant_le_one`, `stepApproximant_monotoneOn_Iio`, `stepApproximant_antitoneOn_Ioi`, `stepApproximant_apply_zero`, and `stepApproximant_tendsto_fabius` in `StepApproximationLimit.lean`. The normalization, the two sides of (91), and (92) are `integral_stepApproximant`, `finiteConvolutionMeasure_inner_le_intervalIntegral_stepApproximant`, `intervalIntegral_stepApproximant_le_finiteConvolutionMeasure_outer`, and `intervalIntegral_stepApproximant_tendsto_of_le` in `StepMeasureBridge.lean`.  $\square$

*Warning 4.13* (Why the endpoint convention is explicit). The half-weighted cell representative agrees almost everywhere with the ordinary interval indicator, so it has the same integral. Its point values are nevertheless part of the formal statement: adjacent cells share endpoints, and each contributes exactly half of its coefficient there. In particular, the corrected representative satisfies  $\mathcal{W}_m(0) = 1$  at every level. These facts are formalized by `halfEndpointIntervalIndicator_ae_eq_indicator`, `intervalIntegral_halfEndpointIntervalIndicator_of_le`, and `stepApproximant_apply_zero`.

## 4.6 Centered finite splines and an explicit error bound

The preceding finite random sums also give a completely explicit polynomial approximation to  $F$ . Put

$$T_p = \sum_{i=0}^{p-1} \frac{U_i}{2^{i+1}}, \quad C_p(x) = \mathbb{P}(T_p \leq x - 2^{-p-1}), \quad (93)$$

where  $p \geq 1$ . The shift by half of the omitted tail is important: it centers the one-sided approximation and improves its error by a factor of two.

Let

$$N_p(x) = \max \left( 0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor \right), \quad \varepsilon_r = (-1)^{\text{wt}_2(r)}, \quad (94)$$

and interpret a sum with  $N_p(x) = 0$  as empty. Define

$$S_p(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r \left( r - 2^p x + \frac{1}{2} \right)^p. \quad (95)$$

Equivalently,

$$S_p(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r \geq 0} \varepsilon_r \left( 2^p x - \frac{1}{2} - r \right)_+^p, \quad (96)$$

where only the terms  $r < N_p(x)$  can be nonzero and  $t_+ = \max(t, 0)$ . The cutoff in (94) is a safe natural-number form of the inclusive upper limit

$$0 \leq r \leq \left\lfloor 2^p x - \frac{1}{2} \right\rfloor.$$

It handles the empty case without assigning a negative upper endpoint to a finite sum.

**Theorem 4.14** (Centered spline as a finite distribution function). *For  $p \geq 1$  and  $0 \leq x \leq 1$ ,*

$$S_p(x) = C_p(x). \quad (97)$$

*Consequently  $0 \leq S_p(x) \leq 1$  on the unit interval.*

*Proof.* For  $p = 1$ ,  $T_1 = U_0/2$ , and direct evaluation gives

$$C_1(x) = \max \left( 0, \min \left( 1, 2x - \frac{1}{2} \right) \right).$$

Formula (96) gives exactly the same broken-linear function: the term  $r = 0$  starts at  $x = 1/4$ , and the term  $r = 1$  subtracts the excess after  $x = 3/4$ .

For the induction step, split off the first uniform coordinate. The finite sums satisfy

$$T_{p+1} \stackrel{\text{d}}{=} \frac{U_0 + T'_p}{2},$$

where  $T'_p$  is an independent copy of  $T_p$ . Conditioning on  $U_0$  and accounting for the midpoint shifts gives the exact smoothing recurrence

$$C_{p+1}(x) = \int_0^1 C_p(2x - u) \, du. \quad (98)$$

On the spline side, integrate each truncated power in (96). The two binary children  $2r$  and  $2r + 1$  have opposite Thue–Morse signs, because  $\varepsilon_{2r} = \varepsilon_r$  and  $\varepsilon_{2r+1} = -\varepsilon_r$ . After the change of scale this yields

$$S_{p+1}(x) = \int_0^1 S_p(2x - u) \, du. \quad (99)$$

The normalization is exactly the one required by

$$\sum_{r=0}^{2^p-1} \varepsilon_r r^p = (-1)^p 2^{\binom{p}{2}} p!,$$

while every lower power sum vanishes. Thus the two recurrences and their common base case prove (97).  $\square$

**Theorem 4.15** (Quantitative centered approximation). *For  $p \geq 1$  and  $0 \leq x \leq 1$ ,*

$$F(x - 2^{-p-1}) \leq S_p(x) \leq F(x + 2^{-p-1}), \quad (100)$$

and hence

$$|S_p(x) - F(x)| \leq 2^{-p}. \quad (101)$$

*Proof.* Write  $X = T_p + R_p$ , where the omitted tail satisfies

$$0 \leq R_p = \sum_{i=p}^{\infty} \frac{U_i}{2^{i+1}} \leq 2^{-p}.$$

If  $X \leq x - 2^{-p-1}$ , then certainly  $T_p \leq x - 2^{-p-1}$ . Conversely, if  $T_p \leq x - 2^{-p-1}$ , then  $X \leq x + 2^{-p-1}$ . Taking probabilities and using (37) and (97) proves (100).

The bounded Fabius function is globally 2-Lipschitz. Indeed, on the open unit interval its derivative is a rescaled value of up (or of the signed extension), whose absolute value is at most 1, and outside the unit interval  $F$  is constant. Thus

$$|F(x \pm 2^{-p-1}) - F(x)| \leq 2 \cdot 2^{-p-1} = 2^{-p}.$$

Combining these two inequalities with the sandwich gives (101).  $\square$

The bound is uniform on the closed interval and includes both endpoints. It is stronger than mere weak convergence of the finite distributions, and it is the quantitative input for the computability proof in section 15.

The same finite formula also has the correct signed global behavior. For  $x \leq 0$  its safe cutoff is empty, so  $S_p(x) = \mathcal{F}(x) = 0$ . On the nonnegative half-line, binary block translation and midpoint reflection extend the local identification. Consequently, for every  $p \geq 1$  and every  $x \in \mathbb{R}$ ,

$$|S_p(x) - \mathcal{F}(x)| \leq 2^{-p}. \quad (102)$$

Indeed, every nonnegative two-unit block is a common signed copy of the first block, and on the second unit cell both functions use the same complement. Neither operation changes the absolute error in (101); the nonpositive half-line is exact. This pointwise all-real bound is `abs_fabiusUniformSpline_sub_extendedFabius_1e`; it implies the separately proved uniform convergence `fabiusUniformSpline_tendstoUniformly_extendedFabius`. This global form strengthens and packages the analytic input used by the complex-shift and discrete-limit results below.

## 5 Fourier products, inversion, and Poisson summation

Use the convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i x z} dx. \quad (103)$$

The formal Fourier transform of up is entire and has two exact infinite-product descriptions.

**Theorem 5.1** (Sinc and cosine products). *For every  $z \in \mathbb{C}$ ,*

$$\widehat{\text{up}}(z) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^n}\right), \quad (104)$$

$$= \prod_{m=1}^{\infty} \left(\cos \frac{\pi z}{2^m}\right)^m. \quad (105)$$

The transform also satisfies

$$\widehat{\text{up}}(z) = \text{sinc}(\pi z) \widehat{\text{up}}(z/2). \quad (106)$$

*Formal proof.* The finite-product refinement and its infinite limit are `rvachevFourier_eq_finite_product` and `rvachevFourier_eq_product` in `FourierProduct.lean`. The refinement equation is `rvachevFourier_scaling` in the same module. Writing  $\Phi$  for the first product in (104), the standalone one-step and  $n$ -fold renormalizations

$$\Phi(z) = \text{sinc}(\pi z)\Phi(z/2), \quad \Phi(z) = \left( \prod_{j < n} \text{sinc}(\pi z/2^j) \right) \Phi(z/2^n)$$

are `rvachevFourierProduct_scaling` and `rvachevFourierProduct_shell` in `RenormalizationIdentity.lean`. The whole-plane analyticity theorem is `rvachevFourier_analyticOnNhd` in `FourierAnalytic.lean`. The weighted cosine product is `rvachevFourierProduct_eq_cosineProduct` in `OriginalPaperSupplement.lean`; its finite telescoping identity is proved internally in the same module.  $\square$

### 5.1 A finite Thue–Morse product-to-sum bridge

The finite binary block has a reflection law that matches the parity of a quarter-turn phase. Recall that  $\varepsilon_n = (-1)^{\text{wt}_2(n)}$ .

**Proposition 5.2** (Binary complement and quarter-turn parity). *For  $m, n \in \mathbb{N}$  with  $n < 2^m$ ,*

$$\text{wt}_2(2^m - 1 - n) + \text{wt}_2(n) = m, \quad (107)$$

$$\varepsilon_{2^m-1-n} = (-1)^m \varepsilon_n. \quad (108)$$

*For every  $m \in \mathbb{N}$  and  $x \in \mathbb{R}$ ,*

$$(-1)^m \cos\left(\frac{m\pi}{2} - x\right) = \cos\left(\frac{m\pi}{2} + x\right). \quad (109)$$

*Formal proof.* The three identities are respectively `binaryWeight_compl`, `thueMorseSign_compl`, and `neg_one_pow_mul_cos_sub` in `LacunaryProductToSum.lean`. The first hypothesis makes the natural-number subtractions in (107)–(108) ordinary binary complementation rather than truncated subtraction.  $\square$

**Theorem 5.3** (Affine Thue–Morse sine polynomial). *For every  $m \in \mathbb{N}$  and  $x, \phi \in \mathbb{R}$ ,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n \sin(\phi + nx) = 2^m \left( \prod_{j=0}^{m-1} \sin \frac{2^j x}{2} \right) \sin\left(\phi + \frac{(2^m-1)x}{2} - \frac{m\pi}{2}\right). \quad (110)$$

*The product is empty when  $m = 0$ .*

*Formal proof.* This is exactly `sum_thueMorseSign_mul_sin_affine` in `ThueMorseSineProduct.lean`. Its proof rotates the complex Thue–Morse exponential product through the arbitrary phase  $\phi$ , rewrites  $(-2i)^m$  as a positive magnitude and a quarter-turn phase, and takes imaginary parts.  $\square$

**Theorem 5.4** (Lacunary sine product as a Thue–Morse sum). *For every  $m \in \mathbb{N}$  and  $w \in \mathbb{R}$ ,*

$$2^m \prod_{j=0}^m \sin(2^j w) = \sum_{n=0}^{2^m-1} \varepsilon_n \sin\left(\frac{m\pi}{2} + (2n+1)w\right). \quad (111)$$

*Formal proof.* This is exactly `prod_sin_two_pow_eq_thueMorse_sum` in `LacunaryProductT oSum.lean`. It is the specialization of (110) at

$$x = 2w, \quad \phi = \frac{m\pi}{2} + w.$$

The final sine becomes  $\sin(2^m w)$ , which appends the last factor to the product over  $j < m$ . The formal statement is over real  $w$ ; no complex continuation or neighboring cosine-product identity is asserted.  $\square$

**Theorem 5.5** (Dyadic shell factorization). *For every  $k \in \mathbb{N}$  and  $z \in \mathbb{C}$ , dilation by  $2^k$  extracts exactly the first  $k$  factors of the sinc product:*

$$\widehat{\text{up}}(2^k z) = \left( \prod_{j=0}^{k-1} \text{sinc}(\pi 2^{j+1} z) \right) \widehat{\text{up}}(z). \quad (112)$$

*Formal proof.* This is `rvachevFourier_two_pow_mul` in `RvachevFourierDecay.lean`. Its standalone-product form is `rvachevFourierProduct_two_pow_mul` in `SincProductShells.lean`. On the real axis the removable singularity can be cleared without excluding the zero ray: for every  $k \in \mathbb{N}$  and  $y \in \mathbb{R}$ ,

$$\begin{aligned} & 2^{k(k+1)/2} (\pi|y|)^k \left| \widehat{\text{up}}(2^k y) \right| \\ &= \left( \prod_{j=0}^{k-1} \left| \sin(2^{j+1} \pi y) \right| \right) \left| \widehat{\text{up}}(y) \right|. \end{aligned} \quad (113)$$

This total form includes  $y = 0$ , and at  $k = 0$  both sides are the same Fourier modulus. It is `norm_rvachevFourier_two_pow_mul_cross` in `RvachevFourierDecay.lean`; its product-level source is `norm_rvachevFourierProduct_two_pow_mul_cross` in `SincProductShells.lean`.  $\square$

**Theorem 5.6** (Total arbitrary-prefix identity and envelope). *Write  $\Phi$  for the standalone sinc product in (104). For every  $K \in \mathbb{N}$  and  $x \in \mathbb{R}$ , clearing the first  $K$  sinc denominators gives the total identity*

$$(\pi|x|)^K |\Phi(x)| = 2^{\binom{K}{2}} \left( \prod_{j=0}^{K-1} \left| \sin\left(\frac{\pi x}{2^j}\right) \right| \right) \left| \Phi\left(\frac{x}{2^K}\right) \right|. \quad (114)$$

*In particular,*

$$(\pi|x|)^K |\Phi(x)| \leq 2^{\binom{K}{2}}. \quad (115)$$

*Both formulas include  $x = 0$  and the empty prefix  $K = 0$ . Only after dividing do we require  $x \neq 0$ :*

$$|\Phi(x)| \leq \frac{2^{\binom{K}{2}}}{(\pi|x|)^K} \quad (x \neq 0). \quad (116)$$

*Thus the usual selected-factor constant is  $2^{K(K-1)/2}/\pi^K$ , with no unnecessary assumption  $K \geq 1$ .*

*Formal proof.* The total identity, its total bound, and the divided corollary are respectively `norm_rvachevFourierProduct_prefix_cross`, `norm_rvachevFourierProduct_prefix_cross_le`, and `norm_rvachevFourierProduct_le_prefix_div` in `SincPrefixBound.lean`. The proof combines the finite shell `rvachevFourierProduct_shell`, the removable real-sinc identity `abs_mul_norm_complexSinc_ofReal`, and the global tail bound `norm_rvachevFourierProduct_le_one`; no convergence argument is repeated.  $\square$

The mean-value input behind shell estimates is more general than the dyadic case. Let  $m_0, m_1, \dots$  be positive integer frequencies satisfying  $2m_j \leq m_{j+1}$ . Then, for every  $n \in \mathbb{N}$ ,

$$\int_0^1 \prod_{j=0}^{n-1} \sin^2(\pi m_j t) \, dt = \left(\frac{1}{2}\right)^n, \quad (117)$$

$$\int_0^1 \prod_{j=0}^{n-1} |\sin(\pi m_j t)| \, dt \leq \left(\frac{\sqrt{2}}{2}\right)^n. \quad (118)$$

The exact square mean is `integral_prod_sin_sq` in `LacunaryRieszIntegral.lean`. Applying Cauchy–Schwarz to it gives `integral_prod_abs_sin_le_sqrt_two` in `LacunaryMeanSharp.lean`. The geometric-base and dyadic specializations are `integral_prod_abs_sin_pow_le_sqrt_two` and `integral_prod_abs_sin_two_pow_le_sqrt_two`.

The logarithm of the normalized dyadic sine shell has an exact square mean as well. For every  $n \in \mathbb{N}$ , including the empty product  $n = 0$ ,

$$\int_0^1 \log^2 \left| \prod_{j=0}^{n-1} 2 \sin(\pi 2^j t) \right| \, dt = \frac{\pi^2}{4} n - \frac{\pi^2}{3} (1 - 2^{-n}). \quad (119)$$

The product statement is `integral_sq_log_prod_two_sin_all` in `BirkhoffProductBridge.lean`. Its doubling-cocycle form is `integral_sq_birkhoff_sum_all` in `BirkhoffVariance.lean`, and the finite covariance summation is `variance_closed_form_all` in `VarianceBookkeeping.lean`. The all-index declarations make the zero-length boundary an identity rather than a side condition.

For a quantitative global form, set

$$\kappa_\infty = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{\log 2}, \quad E_\kappa(x) = \exp\left(-\frac{(\log x)^2}{2 \log 2}\right) x^{-\kappa} \quad (x > 0). \quad (120)$$

There is a constant  $C > 0$  such that the single two-sided estimate

$$|\widehat{\text{up}}(x)| \leq C E_{\kappa_\infty}(|x|) \quad (|x| \geq 1) \quad (121)$$

holds on both real tails. This is `norm_rvachevFourier_le_decayGauge_abs` in `RvachevFourierDecay.lean`. On the distinguished dyadic ray the same gauge occurs in an exact identity:

$$|\widehat{\text{up}}(2^k 2/3)| = \frac{|\widehat{\text{up}}(2/3)|}{E_{\kappa_\infty}(2/3)} E_{\kappa_\infty}(2^k 2/3), \quad (122)$$

namely `norm_rvachevFourier_peak_ray_envelope` in the same module.

The exponent  $\kappa_\infty$  is sharp within this family of gauges. Write  $\Phi$  for the standalone sinc product, which equals `up` by `rvachevFourier_eq_product`. For every  $\kappa > \kappa_\infty$ ,

$$\frac{|\Phi(2^k 2/3)|}{E_\kappa(2^k 2/3)} \longrightarrow +\infty \quad (k \longrightarrow \infty). \quad (123)$$

More explicitly, for every real  $C$  there is a  $k \in \mathbb{N}$  such that  $x_k = 2^k 2/3 \geq 1$  and

$$C E_\kappa(x_k) < |\Phi(x_k)|. \quad (124)$$

Consequently no  $C > 0$  satisfies

$$|\Phi(x)| \leq C E_\kappa(x) \quad \text{for every real } x \geq 1.$$

The limit, explicit peak-ray obstruction, and positive-ray nonexistence theorem are `norm_rvachevFourierProduct_peak_ray_ratio_tendsto_atTop`, `exists_peak_ray_norm_gt_mul_decayGauge`, and `not_exists_norm_rvachevFourierProduct_le_decayGauge_of_kappaInf_lt` in `GlobalDecayEnvelope.lean`.

The transform also has the exact conjugation, parity, and real-axis identities

$$\overline{\widehat{\text{up}}(z)} = \widehat{\text{up}}(-\bar{z}), \quad \widehat{\text{up}}(-z) = \widehat{\text{up}}(z), \quad \text{Im } \widehat{\text{up}}(t) = 0 \quad (t \in \mathbb{R}). \quad (125)$$

These are `rvachevFourier_conj` in `FourierAnalytic.lean` and `rvachevFourier_neg`, `rvachevFourier_ofReal_im_eq_zero` in `FourierProduct.lean`.

**Theorem 5.7** (Exact Fourier zero set, positive-real zero-order limits, and canonical product). *The complex zeros of the Fourier transform are precisely the nonzero integers:*

$$\widehat{\text{up}}(z) = 0 \iff \exists m \in \mathbb{Z} \setminus \{0\} \text{ such that } z = m, \quad z \in \mathbb{C}. \quad (126)$$

At every positive integer  $m$ , its real-axis zero order is exactly  $v_2(m) + 1$ , in the following quantitative sense: there is a constant  $C_m > 0$  such that

$$\lim_{\substack{x \rightarrow m \\ x \neq m}} \frac{|\widehat{\text{up}}(x)|}{|x - m|^{v_2(m)+1}} = C_m. \quad (127)$$

The corresponding canonical product records all of these orders at once:

$$\widehat{\text{up}}(z) = \prod_{r=0}^{\infty} \left(1 - \frac{z^2}{(r+1)^2}\right)^{1+v_2(r+1)}. \quad (128)$$

*Formal proof.* The complete complex zero-set equivalence is `rvachevFourier_eq_zero_iff` in `FourierProduct.lean`. The positive-integer limit is `tendsto_norm_rvachevFourier_div_pow_zero_order` in `RvachevFourierDecay.lean`; its limiting constant is exhibited there as an exponential, and is therefore strictly positive. The canonical product is `rvachevFourier_eq_canonical` in the same facade module, built from `rvachevFourierProduct_eq_canonical` in `SincCanonicalProduct.lean`.  $\square$

## 5.2 Geometric reciprocal-Gamma products

The Gamma factorization is most naturally formalized from the entire reciprocal factor

$$R(z) := \frac{1}{\Gamma(1+z)} \quad (129)$$

**Theorem 5.8** (Geometric reciprocal-Gamma products and the dyadic factor). *For a complex contraction  $q$  satisfying  $\|q\| < 1$ , define the locally uniformly convergent product*

$$G_q(z) := \prod_{n=0}^{\infty} R(q^n z) = \prod_{n=0}^{\infty} \frac{1}{\Gamma(1+q^n z)}. \quad (130)$$

*The hypothesis includes  $q = 0$ ; no positivity or reality assumption is made. The reciprocal factor and the geometric product satisfy*

$$R(z)R(-z) = \text{sinc}(\pi z), \quad (131)$$

$$G_q(0) = 1, \quad G_q(z) = R(z)G_q(qz), \quad (132)$$

$$G_q(z) = 0 \iff \exists n, m \in \mathbb{N} : 1 + q^n z = -m. \quad (133)$$



In particular there are no hidden zeros in the infinite product. Define the Gamma-side total meromorphic representative and the matching sinc product by

$$\Gamma_{\langle q \rangle}(z) := G_q(z)^{-1}, \quad \Phi_q(z) := \prod_{n=0}^{\infty} \text{sinc}(\pi q^n z). \quad (134)$$

Then

$$\Gamma_{\langle q \rangle}(z) = \Gamma(1+z)\Gamma_{\langle q \rangle}(qz), \quad (135)$$

$$G_q(z)G_q(-z) = \Phi_q(z). \quad (136)$$

At  $q = 1/2$ , write  $G_{\text{dy}} = G_{1/2}$  and  $\Gamma_{\text{dy}} = G_{\text{dy}}^{-1}$ . The geometric sinc product is exactly the standalone Rvachev product, so for every  $z \in \mathbb{C}$

$$\Phi(z) = G_{\text{dy}}(z)G_{\text{dy}}(-z) = \frac{1}{\Gamma_{\text{dy}}(z)\Gamma_{\text{dy}}(-z)}. \quad (137)$$

Moreover  $G_{\text{dy}}$  vanishes precisely at the negative integers. For every  $n \geq 1$ ,

$$\text{ord}_{z=-n} G_{\text{dy}}(z) = 1 + v_2(n), \quad (138)$$

$$\text{mord}_{z=-n} \Gamma_{\text{dy}}(z) = -(1 + v_2(n)). \quad (139)$$

Thus the positive pole order of  $\Gamma_{\text{dy}}$  is  $1 + v_2(n)$ ; Mathlib records it as the displayed negative meromorphic order.

*Formal proof and exact API.* At formal source checkpoint 71ab6f6728fceb753c88d8b0573077a59acf2682, the reusable convergence engine is `ScaledInfiniteProducts.lean`. Its ten public declarations are `summable_norm_scaled_sub_one`, `hasProdUniformlyOn_scaled`, `multipliableUniformlyOn_scaled`, `hasProdLocallyUniformly_scaled`, `multipliableLocallyUniformly_scaled`, `continuous_tprod_scaled`, `differentiable_tprod_scaled`, `differentiable_tprod_scaled_of_eq_one`, `tprod_scaled_ne_zero`, and `tprod_scaled_eq_zero_iff`. The pointwise summability theorem assumes a normed scalar field, a normed space, a normed-ring target, summability of  $n \mapsto \|a_n\|$ , and  $f(z) - 1 = O(z)$  at zero. The uniform statements take values in a complete commutative normed ring with a normed unit and assume continuity of  $f$ ; the locally uniform statements additionally assume a locally compact domain. The holomorphic statements take values in a complete commutative normed ring with a normed complex-algebra structure and assume complex differentiability, with `differentiable_tprod_scaled_of_eq_one` replacing the explicit big- $O$  hypothesis by  $f(0) = 1$ . The no-hidden-zero statements use a complete commutative normed ring with a normed unit and a multiplicative norm, but need neither continuity of  $f$  nor local compactness of the domain.

The specialization is `GeometricReciprocalGamma.lean`. Its twenty-five public declarations are `shiftedReciprocalGamma`, `shiftedReciprocalGamma_zero`, `shiftedReciprocalGamma_differentiable`, `shiftedReciprocalGamma_sub_one_isBigO`, `shiftedReciprocalGamma_eq_zero_iff`, `shiftedReciprocalGamma_mul_neg`, `summable_norm_qpow`, `geometricReciprocalGamma`, `geometricReciprocalGammaFactors_multipliable`, `geometricReciprocalGamma_differentiable`, `geometricReciprocalGamma_zero`, `geometricReciprocalGamma_mahler`, `geometricReciprocalGamma_eq_zero_iff`, `geometricGamma`, `geometricGamma_meromorphic`, `geometricGamma_mahler`, `geometricSincProduct`, `geometricReciprocalGamma_mul_neg`, `dyadicReciprocalGamma`, `dyadicGamma`, `dyadicReciprocalGamma_differentiable`, `dyadicReciprocalGamma_zero`, `geometricSincProduct_inv_two`, `dyadicReciprocalGamma_mul_neg`, and `rvachevFourierProduct_eq_one_div_dyadicGamma_mul`. Here  $q \in \mathbb{C}$  and  $\|q\| < 1$  are the

exact hypotheses on every generic convergence, holomorphy, Mahler, zero-orbit, meromorphy, and reflection statement; normalization at zero itself needs no contraction hypothesis.

At every reciprocal integer scale, `ReciprocalIntegerGammaZeros.lean` exposes four public theorems: `norm_inv_natCast_lt_one`, `geometricReciprocalGamma_inv_natCast_eq_zero_iff`, `geometricReciprocalGamma_inv_natCast_ne_zero_of_nonneg`, and `geometricGamma_inv_natCast_meromorphic`. For  $b \geq 2$ , the entire product  $G_{1/b}$  vanishes exactly at the embedded strictly negative natural numbers, independently of  $b$ , is nonzero at every embedded nonnegative natural number, and its pointwise inverse is meromorphic. This closes the integer-base zero set, but not the multiplicity of those zeros.

Finally, `DyadicGammaOrder.lean` exposes eight public declarations: `dyadicReciprocalGamma_eq_zero_iff`, `dyadicReciprocalGamma_int_ne_zero_of_nonneg`, `dyadicReciprocalGamma_nat_ne_zero`, `dyadicGamma_meromorphic`, `analyticOrderAt_dyadicReciprocalGamma_int_of_neg`, `analyticOrderAt_dyadicReciprocalGamma_neg_nat`, `meromorphicOrderAt_dyadicGamma_int_of_neg`, and `meromorphicOrderAt_dyadicGamma_neg_nat`. The integer forms assume a strictly negative integer center; the natural-number wrappers assume  $n \neq 0$ .

There is an important representation boundary. `Mathlib` assigns `Complex.Gamma` the value zero at a classical Gamma pole, and its inverse also has the totalized value zero there. Accordingly `geometricGamma` and `dyadicGamma` are defined as pointwise inverses of the entire reciprocal products, and a pole is carried by negative `meromorphicOrderAt`, not by an infinite point value. At that checkpoint, the declarations above do not identify these functions with the raw products  $\prod_n \Gamma(1 + q^n z)$  away from their poles, and they assert no arithmetic Weierstrass regrouping, logarithmic or digamma series, Malmsten or heat-trace integral, D-finiteness result, reciprocal-Gamma derivative coefficient for the Thue–Morse tower, or general integer-base collision order.

The post-checkpoint source commit 0ba35abd4 closes the reciprocal-Gamma first-jet and Thue–Morse Gamma-tower tranche in two further modules. `ReciprocalGammaJets.lean` has exactly five public declarations: `deriv_Gamma_inv_neg_nat`, `hasDerivAt_Gamma_inv_neg_nat`, `hasDerivAt_Gamma_inv_zero`, `analyticOrderAt_Gamma_inv_neg_nat`, and `tendsto_Gamma_inv_div_add_nat`. For every  $r \in \mathbb{N}$ , they identify

$$(\Gamma^{-1})'(-r) = (-1)^r r!, \quad \text{ord}_{-r} \Gamma^{-1} = 1, \quad \frac{\Gamma(s)^{-1}}{s+r} \longrightarrow (-1)^r r! \quad (s \rightarrow -r, s \neq -r),$$

with `hasDerivAt_Gamma_inv_zero` retaining the origin compatibility name. These are statements about the entire reciprocal function  $s \mapsto \Gamma(s)^{-1}$ : they do not assign a value or derivative to the raw Gamma function at one of its poles.

At the same commit, `ThueMorseGammaTower.lean` has exactly eight public declarations: `hasDerivAt_dirichletMellinContinuation_neg_nat`, `deriv_dirichletMellinContinuation_neg_nat`, `thueMorseGammaLog`, `thueMorseGammaTower`, `thueMorseGammaLog_eq_mellin`, `thueMorseGammaLog_eq_integral`, `thueMorseGammaLog_dyadic`, and `thueMorseGammaTower_dyadic`. The first two give, for  $a > 0$  and  $r \in \mathbb{N}$ ,

$$\tilde{D}'_a(-r) = (-1)^r r! \text{ Mellin}(t \mapsto e^{-at} \mathcal{E}(t))(-r).$$

The two definitions are total for every real  $a$ ; their Mellin, integral, and dyadic identification theorems assume exactly  $a > 0$ . In particular `thueMorseGammaLog` is the chosen coordinate  $L_r(a) := \tilde{D}'_a(-r)$ , not a proved identity with `Complex.log` of `thueMorseGammaTower`; the latter is defined as  $T_r(a) := \exp L_r(a)$ . The formal laws are

$$L_r(a) = (-1)^r r! \int_0^\infty t^{-r-1} e^{-at} \mathcal{E}(t) dt, \quad L_r(a) = 2^r (L_r(a/2) - L_r((a+1)/2)),$$

and

$$T_r(a) = \left( \frac{T_r(a/2)}{T_r((a+1)/2)} \right)^{2^r}.$$

The current integrated tree adds the ninth public tower declaration of `Real_exp_mplimit_eq_gammaTower_div`: for  $a, b > 0$ , the complex embedding of `Real.exp` of `mplimit` at  $(a, b)$  is  $T_0(b)/T_0(a)$ . Within this tower tranche, only the differential and iterated differential ladder in the real parameter  $a$  remains frontier; it requires a parameter-under-the-Mellin-integral theorem. The raw Gamma product, canonical regrouping, logarithmic/digamma expansions, Gamma-product Malmsten and heat-trace representations, D-finiteness, and general-base collision order listed above remain separate frontiers.  $\square$

The exponent in the canonical product has a finite, base-generic double-counting origin that is useful independently of sinc products. For  $b, N, J \in \mathbb{N}$ , put

$$c_J(b, m) := \#\{j \in \{0, \dots, J\} : b^j \mid m\}.$$

If  $q > 0$ , multiplication by  $q$  identifies the positive multiples of  $q$  not exceeding  $N$  with  $1, \dots, \lfloor N/q \rfloor$ . Consequently, for every function  $f$  into a commutative monoid,

$$\prod_{1 \leq n \leq \lfloor N/q \rfloor} f(qn) = \prod_{\substack{1 \leq m \leq N \\ q \mid m}} f(m), \quad (140)$$

$$\prod_{j=0}^J \prod_{1 \leq n \leq \lfloor N/b^j \rfloor} f(b^j n) = \prod_{m=1}^N f(m)^{c_J(b, m)} \quad (b > 0). \quad (141)$$

For  $b > 1$  and  $m \neq 0$ , if the displayed layer range reaches the multiplicity, that is, if  $\text{multiplicity}(b, m) \leq J$ , then

$$c_J(b, m) = \text{multiplicity}(b, m) + 1. \quad (142)$$

Thus, under the explicit hypothesis  $\text{multiplicity}(b, m) \leq J$  for every  $1 \leq m \leq N$ ,

$$\prod_{j=0}^J \prod_{1 \leq n \leq \lfloor N/b^j \rfloor} f(b^j n) = \prod_{m=1}^N f(m)^{\text{multiplicity}(b, m) + 1}. \quad (143)$$

The same three identities hold in every additive commutative monoid, with products and powers replaced by sums and natural scalar multiples. In Lean, the underlying finite-set equality is `filter_dvd_eq_image`; (140) and its additive companion are `prod_multiples_eq_prod_filter` and `sum_multiples_eq_sum_filter`; (141) and its companion are `prod_layers_eq_prod_pow_card` and `sum_layers_eq_sum_nsmul_card`; and (142)–(143) and the final additive companion are `card_filter_pow_dvd`, `prod_layers_eq_prod_pow_multiplicity`, and `sum_layers_eq_sum_nsmul_multiplicity`, all in `BaseLayerRegrouping.lean`. These are finite algebraic identities. In particular they assert no infinite-product convergence, and the sufficient layer-coverage condition must be supplied explicitly rather than inferred by the API from a bound such as  $N < b^{J+1}$ .

### 5.3 Total local factorization and exact complex zero jets

The complex sinc identities needed at an integer center are most useful after their removable denominators have been cleared. With the convention  $\text{sinc } z = \sin z/z$ , continuously extended at zero, one has for every  $z \in \mathbb{C}$ , every  $n \in \mathbb{N}$ , and every  $a \in \mathbb{C} \setminus \{0\}$ ,

$$z \text{sinc}(\pi z) = \frac{\sin(\pi z)}{\pi}, \quad (144)$$

$$(n + z) \text{sinc}(\pi(n + z)) = (-1)^n z \text{sinc}(\pi z), \quad (145)$$

$$(an + z) \text{sinc}\left(\frac{\pi(an + z)}{a}\right) = (-1)^n z \text{sinc}\left(\frac{\pi z}{a}\right). \quad (146)$$

Unlike identities obtained by dividing through by  $z$  or  $n + z$ , these equalities retain the points where either factor vanishes. They are respectively `mul_complexSinc_pi`, `mul_complexSinc_pi_add_nat`, and `mul_complexSinc_pi_div_add_nat` in `IntegerZeroLocalFactorization.lean`; the last theorem allows an arbitrary nonzero *complex* scale, with no order or reality hypothesis.

Here is the reusable all-order differential principle. Let  $\mathbb{K}$  be any nontrivially normed field, let  $E$  be a normed  $\mathbb{K}$ -space, and let  $U : \mathbb{K} \rightarrow E$  be  $C^k$  at zero. For every  $n, k \in \mathbb{N}$ ,

$$D^k(z \mapsto z^n \cdot U(z))(0) = \begin{cases} \left(\binom{k}{n} : \mathbb{K}\right)(n! : \mathbb{K}) \cdot D^{k-n}U(0), & n \leq k, \\ 0, & k < n, \end{cases} \quad (147)$$

where  $\cdot$  denotes scalar action on  $E$ . This is the vector-valued theorem `iteratedDeriv_pow_smul_zero_eq_ite` in `LeadingJet.lean`. At the index  $k = n + r$ , its conditional disappears:

$$D^{n+r}(z \mapsto z^n \cdot U(z))(0) = \left(\binom{n+r}{n} : \mathbb{K}\right)(n! : \mathbb{K}) \cdot D^rU(0). \quad (148)$$

The vector and scalar forms of (148) are `iteratedDeriv_pow_smul_zero_add` and `iteratedDeriv_pow_mul_zero_add`. If a scalar-valued function  $f$  agrees with  $z^n U(z)$  merely on a neighborhood of zero, the same formula is `iteratedDeriv_eq_choose_factorial_mul_of_eventuallyEq_pow_mul`; no separate regularity hypothesis on  $f$  is needed. At an arbitrary center  $z_0$ , a vector-valued germ  $f(z) = (z - z_0)^n \cdot U(z)$  has the all-index form `iteratedDeriv_eq_ite_of_eventuallyEq_sub_pow_smul` and the  $n + r$  form `iteratedDeriv_eq_choose_factorial_smul_of_eventuallyEq_sub_pow_smul`; the scalar specialization is `iteratedDeriv_eq_choose_factorial_mul_of_eventuallyEq_sub_pow_mul`.

The coefficient is deliberately the product  $\binom{n+r}{n}n!$ , with both natural numbers cast into  $\mathbb{K}$ , not the quotient  $(n+r)!/r!$ . Thus none of these theorems requires factorial division, characteristic zero, completeness, or analyticity. In positive characteristic the displayed scalar can vanish, so the generic statement identifies the first *potentially* nonzero jet rather than asserting nonvanishing. Setting  $r = 0$  recovers the compatibility formula

$$\left. \frac{d^n}{dz^n} (z^n U(z)) \right|_{z=0} = n! U(0), \quad (149)$$

exported as `iteratedDeriv_pow_mul_zero`; its zero-germ companion is `iteratedDeriv_eq_factorial_mul_of_eventuallyEq_pow_mul`.

We now apply the total sinc identities to the standalone product  $\Phi = \text{rvachevFourierProduct}$ . Put

$$P_d(w) := \prod_{h=0}^{d-1} \text{sinc}\left(\frac{\pi w}{2^h}\right), \quad P_0(w) = 1. \quad (150)$$

For arbitrary  $k, q \in \mathbb{N}$  and  $w \in \mathbb{C}$ , the global, denominator-cleared factorization at the center  $2^k q$  is

$$(2^k q + w)^{k+1} \Phi(2^k q + w) = (-1)^q w^{k+1} P_{k+1}(w) \Phi\left(\frac{q}{2} + \frac{w}{2^{k+1}}\right). \quad (151)$$

This is `rvachevFourierProduct_two_pow_mul_add_factorization` in `IntegerZeroLocalFactorization.lean`. It includes even  $q$ , when the tail can retain further zero multiplicity, and also the degenerate center  $q = 0$ . If  $q$  is odd, its sign is exactly  $-1$ ; the corresponding specialization is `rvachevFourierProduct_two_pow_mul_odd_add_factorization`.

For a positive integer  $m$ , let  $d = v_2(m) + 1$  and let  $q = m/2^{v_2(m)}$  be its canonical odd part. Then

$$(m + w)^d \Phi(m + w) = -w^d P_d(w) \Phi\left(\frac{q}{2} + \frac{w}{2^d}\right), \quad (152)$$

which is `rvachevFourierProduct_nat_add_factorization`. The fully signed form is just as clean. For  $m \in \mathbb{Z} \setminus \{0\}$ , set

$$d_m := v_2(|m|) + 1, \quad q_m := \frac{|m|}{2^{v_2(|m|)}}, \quad \sigma_m := \text{sgn}(m). \quad (153)$$

Then  $q_m$  is odd and, for every complex  $w$ ,

$$(m + w)^{d_m} \Phi(m + w) = -w^{d_m} P_{d_m}(w) \Phi\left(\frac{q_m}{2} + \sigma_m \frac{w}{2^{d_m}}\right). \quad (154)$$

This is `rvachevFourierProduct_int_add_factorization`; the sign of the integer center occurs only in the tail coordinate. Equations (151)–(154) are polynomially cleared identities on all of  $\mathbb{C}$ , not punctured-neighborhood statements.

The remaining analytic argument can therefore be packaged into a single local unit. The product  $\Phi$  is entire by `rvachevFourierProduct_differentiable`, and, for every odd natural  $q$ ,

$$\Phi(q/2) \neq 0, \quad (155)$$

by `rvachevFourierProduct_nat_div_two_ne_zero_of_odd`. Define the total cofactor

$$U_m(w) := -\frac{P_{d_m}(w) \Phi(q_m/2 + \sigma_m w/2^{d_m})}{(m + w)^{d_m}}. \quad (156)$$

The Lean definition `integerZeroLocalCofactor` is total for every integer  $m$ . For  $m \neq 0$  and  $m + w \neq 0$ ,

$$\Phi(m + w) = w^{d_m} U_m(w), \quad (157)$$

by `rvachevFourierProduct_int_add_eq_pow_mul_cofactor`; the same equality holds as a germ at  $w = 0$  by `rvachevFourierProduct_int_add_eventuallyEq_pow_mul_cofactor`. In the original coordinate this is the centered germ

$$\Phi(z) = (z - m)^{d_m} U_m(z - m) \quad \text{near } z = m,$$

exported as `rvachevFourierProduct_int_eventuallyEq_sub_pow_mul_cofactor`.

For every integer parameter  $m$ , with no parity or nonzero hypothesis, the total quotient  $U_m$  is analytic at every point  $w_0$  satisfying  $m + w_0 \neq 0$ , by `integerZeroLocalCofactor_analyticAt_of_add_ne`. For nonzero  $m$ , its local-origin specialization is `integerZeroLocalCofactor_analyticAt`. Its exact central value is

$$U_m(0) = -\frac{\Phi(q_m/2)}{m^{d_m}} \neq 0. \quad (158)$$

These three facts are `integerZeroLocalCofactor_analyticAt`, `integerZeroLocalCofactor_or_zero`, and `integerZeroLocalCofactor_zero_ne`, respectively. More precisely, the middle theorem is the all-integer totalized identity

$$U_m(0) = -\frac{\Phi(\text{divMaxPow}(|m|, 2)/2)}{m^{\text{padicValNat}(2, |m|)+1}};$$

for  $m \neq 0$  it reduces to (158), and the last theorem makes it an analytic unit.

Combining the germ factorization, the nonzero unit, and (149) gives the exact complex multiplicity and its entire leading jet:

$$\text{ord}_{z=m} \Phi(z) = d_m, \quad (159)$$

$$\Phi^{(j)}(m) = 0 \quad (0 \leq j < d_m), \quad (160)$$

$$\Phi^{(d_m)}(m) = -\frac{d_m! \Phi(q_m/2)}{m^{d_m}} \neq 0. \quad (161)$$

In fact the division-free all-order identity is, for every  $r \in \mathbb{N}$ ,

$$\Phi^{(d_m+r)}(m) = \binom{d_m+r}{d_m} d_m! U_m^{(r)}(0). \quad (162)$$

The order statement is `analyticOrderAt_rvachevFourierProduct_int`; the lower derivatives are `iteratedDeriv_rvachevFourierProduct_int_eq_zero_of_1t`; the displayed all-order formula is `iteratedDeriv_rvachevFourierProduct_int_add_order`; the displayed leading coefficient combines its  $r = 0$  case with `integerZeroLocalCofactor_zero` and is exported as `iteratedDeriv_rvachevFourierProduct_int`; and its nonvanishing is also exported directly as `iteratedDeriv_rvachevFourierProduct_int_ne_zero`, all in `IntegerZeroAnalyticOrder.lean`. This strengthens the real positive-integer limit in [Theorem 5.7](#): it is a signed-integer, complex-local theorem with the first surviving derivative evaluated exactly.

#### 5.4 Finite weighted scale arithmetic behind the zero orders

The valuation in (127) belongs to a more general finite additive calculus; neither primality of the scale base nor an analytic infinite product is needed for this layer. Let  $(A, +, 0)$  be an additive commutative monoid, let  $w : \mathbb{N} \rightarrow A$ , and define the inclusive weighted prefix

$$P_w(n) := \sum_{h=0}^n w(h). \quad (163)$$

It satisfies

$$P_w(0) = w(0), \quad P_w(n+1) = P_w(n) + w(n+1), \quad (164)$$

$$P_w(m+n) = \sum_{h=0}^{m-1} w(h) + P_{w^{(m)}}(n), \quad w^{(m)}(h) := w(m+h). \quad (165)$$

Empty sums have their usual value. More generally, if  $S$  is finite,  $a : S \rightarrow \mathbb{N}$ , and  $a(x) < H$  for every  $x \in S$ , then

$$\sum_{x \in S} P_w(a(x)) = \sum_{h=0}^{H-1} \#\{x \in S : h \leq a(x)\} \odot w(h), \quad (166)$$

where  $q \odot y$  means the sum of  $q$  copies of  $y$ . This finite layer-cake identity requires no subtraction, order, or multiplication on  $A$ .

For a natural base  $b$ , put

$$\nu_b(n) := \text{padicValNat}(b, n), \quad W_{b,w}(n) := P_w(\nu_b(n)) = \sum_{h=0}^{\nu_b(n)} w(h). \quad (167)$$

Thus, for  $b > 1$ ,  $n > 0$ , and  $k \in \mathbb{N}$ ,

$$W_{b,w}(b^k n) = \sum_{h=0}^{k-1} w(h) + W_{b,w^{(k)}}(n), \quad (168)$$

$$W_{b,w}(bn) = w(0) + W_{b,w^{(1)}}(n). \quad (169)$$

For the divisibility form, the optimal algebraic hypothesis is only  $b \neq 1$ . If  $n > 0$  and  $\nu_b(n) < H$ , then

$$W_{b,w}(n) = \sum_{\substack{0 \leq h < H \\ b^h | n}} w(h). \quad (170)$$

Since  $\nu_b(n) < n$ , the canonical choice  $H = n$  gives

$$W_{b,w}(n) = \sum_{\substack{0 \leq h < n \\ b^h | n}} w(h). \quad (171)$$

Summing over the positive integers through  $N$  and exchanging the two finite sums yields

$$\sum_{m=0}^{N-1} W_{b,w}(m+1) = \sum_{h=0}^{N-1} (N \operatorname{div} b^h) \odot w(h) \quad (b \neq 1), \quad (172)$$

where  $\operatorname{div}$  denotes natural-number quotient. For  $b > 1$  this is the familiar floor  $\lfloor N/b^h \rfloor$ .

Pascal weights now give the hockey-stick identities

$$P_{h \mapsto \binom{h+r}{r}}(n) = \binom{n+r+1}{r+1}, \quad (173)$$

$$P_{h \mapsto \binom{h}{r}}(n) = \binom{n+1}{r+1}. \quad (174)$$

Consequently, with no restriction on  $b, n, r$ ,

$$W_{b, h \mapsto \binom{h}{r}}(n) = \binom{\nu_b(n)+1}{r+1}, \quad (175)$$

and, for  $b \neq 1$ ,

$$\sum_{m=0}^{N-1} \binom{\nu_b(m+1)+1}{r+1} = \sum_{h=0}^{N-1} (N \operatorname{div} b^h) \binom{h}{r}. \quad (176)$$

*Formal proof.* The definition (163), its zero and successor laws, and its shifted splitting law are `inclusivePrefixSum`, `inclusivePrefixSum_zero`, `inclusivePrefixSum_succ`, and `inclusivePrefixSum_add`. The monoid-valued Fubini identity (166) is `sum_inclusivePrefixSum_eq_sum_filter_card_nsmul`.

The scale definition (167) is `weightedScaleMultiplicity`. Equations (168) and (169) are `weightedScaleMultiplicity_base_pow_mul` and `weightedScaleMultiplicity_base_mul`. The bounded and canonical divisibility sums (170) and (171) are `weightedScaleMultiplicity_eq_sum_filter_pow_dvd_of_lt` and `weightedScaleMultiplicity_eq_sum_filter_pow_dvd`; the cumulative identity (172) is `sum_range_weightedScaleMultiplicity`.

Finally, (173) and (174) are `inclusivePrefixSum_add_choose` and `inclusivePrefixSum_choose`. Their valuation specialization and cumulative floor sum are `weightedScaleMultiplicity_choose` and `sum_range_padicValNat_choose`. All declarations are in `WeightedScaleMultiplicity.lean`.  $\square$

Lean totalizes the boundary cases:  $\nu_b(0) = \nu_0(n) = \nu_1(n) = 0$ , and natural division by zero is zero. Accordingly, the resulting  $W$ -values at those degenerate inputs are algebraic conventions. More importantly, this subsection proves finite arithmetic identities only. Formula (175) supplies the binomial expression proposed for higher-rank Rvachev products, but it does not assert convergence of such a product or identify its analytic zero order. Those analytic bridges remain separate obligations.

On the real axis the origin is the unique global point of unit modulus:

$$|\widehat{\operatorname{up}}(x)| = 1 \iff x = 0, \quad x \neq 0 \implies |\widehat{\operatorname{up}}(x)| < 1. \quad (177)$$

These are `norm_rvachevFourier_eq_one_iff` and `norm_rvachevFourier_lt_one_of_ne_zero`. On each positive real lobe the sign and peak are also exact. If  $m \in \mathbb{N}$  and  $m < x < m+1$ , then

$$\widehat{\operatorname{up}}(x) = \varepsilon_m |\widehat{\operatorname{up}}(x)|, \quad (178)$$

by `rvachevFourier_eq_thueMorse_sign_mul_norm`. If moreover  $m \geq 1$ , there is exactly one point of  $(m, m+1)$  maximizing the modulus over that open lobe and exactly one point of the reflected lobe  $-(m+1), -m)$  maximizing it there. These are `existsUnique_isMaxOn_rvachevFourier_lobe` and `existsUnique_isMaxOn_rvachevFourier_neg_lobe`. All five facade declarations are in `RvachevFourierDecay.lean`.

Globally, the origin is the unique real frequency of unit modulus:

$$|\widehat{\text{up}}(x)| \leq 1, \quad |\widehat{\text{up}}(x)| = 1 \iff x = 0. \quad (179)$$

Equivalently, the first inequality is strict whenever  $x \neq 0$ . These statements are `norm_rvachevFourier_le_one`, `norm_rvachevFourier_lt_one_of_ne_zero`, and `norm_rvachevFourier_eq_one_iff` in `RvachevFourierDecay.lean`.

## 5.5 Rapid decay and inversion

For every  $N \in \mathbb{N}$  there is a constant  $C_N > 0$  such that

$$|\xi|^N |\widehat{\text{up}}(\xi)| \leq C_N \quad (\xi \in \mathbb{R}), \quad (180)$$

and the companion formal theorem gives the corresponding polynomially weighted bound for every iterated real-frequency derivative. Fourier inversion holds in the exact form

$$\text{up}(x) = \int_{\mathbb{R}} \widehat{\text{up}}(\xi) e^{2\pi i x \xi} d\xi. \quad (181)$$

These are `rvachevFourier_real_rapidDecay`, `rvachevFourier_real_iteratedDeriv_rapidDecay`, and `rvachev_fourier_inversion_analytic` in `PoissonSummation.lean` and `FourierAnalytic.lean`.

## 5.6 Poisson summation and partitions of unity

For  $u > 0$  and  $t \in \mathbb{R}$ , the formal Poisson identity is

$$\sum_{k \in \mathbb{Z}} \text{up}(t + uk) = \frac{1}{u} \sum_{k \in \mathbb{Z}} \widehat{\text{up}}(k/u) e^{2\pi i kt/u}. \quad (182)$$

It is `rvachev_poisson_summation`. Combining it with (126) gives, for every positive integer  $m$ ,

$$\sum_{k \in \mathbb{Z}} \text{up}\left(t + \frac{k}{m}\right) = m, \quad (183)$$

and in particular

$$\sum_{k \in \mathbb{Z}} \text{up}(t + k) = 1. \quad (184)$$

These are `rvachev_partition_one_over_nat` and `rvachev_partition_unity` in `PoissonSummation.lean`.

## 5.7 Banach-free cyclic aliasing

**Theorem 5.9** (Finite-range twists and cyclic aliasing). *The convergence mechanism behind a root-of-unity filter does not require a norm. Let  $R$  be any scalar type, let  $B$  be any index type, let  $I$  be finite, and let  $E$  be a complete uniform additive commutative group with a distributive  $R$ -action such that multiplication by each fixed scalar is continuous. If  $a : B \rightarrow E$  is summable,  $\ell : B \rightarrow I$ , and  $c : I \rightarrow R$ , then*

$$b \mapsto c(\ell(b)) a(b) \quad \text{is summable.} \quad (185)$$



Thus a scalar multiplier taking only finitely many values preserves unconditional summability.

Separately, let  $E$  be a complete Hausdorff uniform additive commutative group, write  $\bar{n}$  for the class of  $n \in \mathbb{Z}$  in  $\text{ZMod}(q)$ , and define

$$\mathcal{R}_r(a) := \sum_{\substack{n \in \mathbb{Z} \\ \bar{n} = r}} a(n). \quad (186)$$

For every  $q \in \mathbb{N}$ , including Lean’s convention  $\text{ZMod}(0) = \mathbb{Z}$ , the family  $r \mapsto \mathcal{R}_r(a)$  has sum  $\sum_{n \in \mathbb{Z}} a(n)$ . In particular, for  $q > 0$ ,

$$\sum_{r \in \mathbb{Z}/q\mathbb{Z}} \mathcal{R}_r(a) = \sum_{n \in \mathbb{Z}} a(n). \quad (187)$$

Suppose in addition that  $E$  is a complex module and multiplication by each fixed complex scalar is continuous. Let  $\chi_q$  be the standard additive character of  $\mathbb{Z}/q\mathbb{Z}$ . For  $q > 0$ , put

$$\mathcal{A}_k(a) := \sum_{n \in \mathbb{Z}} \chi_q(-\bar{n}k) a(n). \quad (188)$$

Every twisted series in (188) is summable. With the unnormalized discrete Fourier transform

$$(\mathcal{D}_q f)(k) := \sum_{r \in \mathbb{Z}/q\mathbb{Z}} \chi_q(-rk) f(r),$$

one has the exact forward and inverse identities

$$\mathcal{D}_q(\mathcal{R}_\bullet(a))(k) = \mathcal{A}_k(a), \quad (189)$$

$$\mathcal{R}_r(a) = \frac{1}{q} \sum_{k \in \mathbb{Z}/q\mathbb{Z}} \chi_q(kr) \mathcal{A}_k(a). \quad (190)$$

*Formal proof.* The finite-range convergence principle (185) is `summable_finite_range_smul`. The two sums are `intResidueTsum` and `intFourierTsum`; character twisting is `intFourierSummand_summable`. The fiberwise summation statement and its finite-modulus consequence are `hasSum_intResidueTsum` and `sum_intResidueTsum`. The forward transform and normalized inverse are `dft_intResidueTsum` and `intResidueTsum_eq_dft_average`. All eight declarations are in `SummableCyclicAlias.lean`.

The proof partitions the bilateral family into its residue fibers, observes that each cyclic character is constant on a fiber, and applies the finite DFT and its inverse. Summability of  $a$  is the only convergence hypothesis; norm-summability is not assumed.  $\square$

## 5.8 Moment expansion of the Fourier transform

With

$$c_n = \int_{-1}^1 x^{2n} \text{up}(x) \, dx. \quad (191)$$

the exact entire series is

$$\widehat{\text{up}}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n c_n}{(2n)!} (2\pi z)^{2n}. \quad (192)$$

This is `rvachevFourier_eq_momentSeries` in `AnalyticMoments.lean`. For later finite filters the individual mode is named

$$A_n(z) := \frac{(-1)^n c_n}{(2n)!} (2\pi z)^{2n} = \text{rvachevFourierMomentTerm } z \, n. \quad (193)$$

The declarations `rvachevFourierMomentTerm_zero` and `rvachevFourierMomentTerm_scale` prove

$$A_0(z) = 1, \quad A_n(cz) = (c^2)^n A_n(z) \quad (c, z \in \mathbb{C}, n \in \mathbb{N}).$$

More strongly than the equality of total sums in (192), `hasSum_rvachevFourierMomentTerm` proves `HasSum(A_•(z), up(z))` for every bounded Fabius solution. These four declarations are in `AnalyticMoments.lean`; the canonical product specialization used below is in `RvachevQBinomialFilter.lean`.

## 6 The global extension, derivatives, flatness, and inverse

### 6.1 The signed extension and all iterated derivatives

The translate series (11) is locally finite. It agrees with the bounded  $F$  on  $[0, 1]$ , vanishes on  $(-\infty, 0]$ , and satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}). \quad (194)$$

These facts are `extendedFabius_eq_fabiusReal`, `extendedFabius_eq_zero_of_nonpos`, and `extendedFabius_hasDerivAt` in `GlobalExtension.lean`.

Iterating the dilation law gives the exact identities

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x), \quad (195)$$

$$\text{up}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m(x+1)) \quad (-1 \leq x \leq 1). \quad (196)$$

They are `iteratedDeriv_extendedFabius` and `iteratedDeriv_rvachev`. The global extension obeys

$$|\mathcal{F}(x)| \leq 1, \quad (197)$$

by `abs_extendedFabius_le_one`. Hence, for all  $m \in \mathbb{N}$  and  $x \in \mathbb{R}$ ,

$$|F^{(m)}(x)| \leq 2^{\binom{m+1}{2}}, \quad (198)$$

and this bound is attained at  $x = 2^{-m}$ :

$$F^{(m)}(2^{-m}) = 2^{\binom{m+1}{2}}. \quad (199)$$

The exact bound, attainment, and supremum statements are `abs_iteratedDeriv_fabiusReal_le`, `iteratedDeriv_fabiusReal_inv_two_pow_all`, and `isGreatest_abs_iteratedDeriv_fabiusReal_all` in `BoundedDerivatives.lean`.

### 6.2 Normalized Volterra calculus and exact Fabius primitives

Repeated integration is most useful here in a form that remembers the base point, works for either orientation of the endpoints, and does not introduce a new normalization at order zero. Let  $E$  be a real normed vector space,  $f : \mathbb{R} \rightarrow E$ , and  $a \in \mathbb{R}$ . Write

$$(V_a f)(x) := \int_a^x f(t) dt, \quad (200)$$

$$L_a^0 f := f, \quad L_a^{n+1} f := V_a(L_a^n f), \quad (201)$$

$$J_a^0 f(x) := f(x), \quad J_a^{n+1} f(x) := \int_a^x \frac{(x-t)^n}{n!} f(t) dt. \quad (202)$$

Scalar multiplication in  $E$  is understood in the last display. These are oriented interval integrals: no hypothesis such as  $a \leq x$  occurs. In Lean they are the three public definitions `volterraPrimitive`, `iteratedPrimitive`, and `normalizedVolterra` in `NormalizedVolterra.lean`.

**Theorem 6.1** (Literal primitive ladder). *For every  $m, n \in \mathbb{N}$ , literal iteration satisfies*

$$L_a^0 f = f, \quad L_a^{n+1} f = V_a(L_a^n f), \quad (203)$$

$$L_a^{m+n} f = L_a^m(L_a^n f), \quad L_a^{n+1} f(a) = 0. \quad (204)$$

*If  $E$  is complete and  $f$  is continuous, then every  $L_a^n f$  is continuous and*

$$D(L_a^{n+1} f)(x) = L_a^n f(x), \quad (205)$$

$$D^k(L_a^{k+m} f) = L_a^m f, \quad (206)$$

$$D^k(L_a^n f) = L_a^{n-k} f \quad (k \leq n). \quad (207)$$

*The derivative in (205) exists in the strict sense, and  $L_a^n f$  is  $C^n$ .*

The four algebraic identities are respectively `iteratedPrimitive_zero`, `iteratedPrimitive_succ`, `iteratedPrimitive_add`, and `iteratedPrimitive_succ_self`. Continuity is `continuous_volterraPrimitive` and `continuous_iteratedPrimitive`. The strict and ordinary pointwise FTC statements, and its equality-of-functions form, are `iteratedPrimitive_succ_hasStrictDerivAt`, `iteratedPrimitive_succ_hasDerivAt`, and `deriv_iteratedPrimitive_succ`. The two cancellation formulas and the smoothness statement are `iteratedDeriv_iteratedPrimitive_add`, `iteratedDeriv_iteratedPrimitive`, and `contDiff_iteratedPrimitive`.

**Theorem 6.2** (Canonical Cauchy kernel). *The normalized kernel has the total order-zero and positive-order rules*

$$J_a^0 f(x) = f(x), \quad J_a^{n+1} f(x) = \int_a^x \frac{(x-t)^n}{n!} f(t) dt, \quad (208)$$

$$J_a^1 f(x) = \int_a^x f(t) dt, \quad J_a^{n+1} f(a) = 0. \quad (209)$$

*If  $f = g$  on the closed unoriented interval between  $a$  and  $x$ , then  $J_a^n f(x) = J_a^n g(x)$ . Real scalar multiplication commutes with  $J_a^n$ , and any finite sum of interval-integrable inputs may be passed through it.*

*If  $E$  is complete and  $f$  is continuous, then literal iteration and the kernel agree,*

$$L_a^n f = J_a^n f, \quad (210)$$

*and hence*

$$D(J_a^{n+1} f)(x) = J_a^n f(x) \quad \text{strictly}, \quad (211)$$

$$D^k(J_a^{k+m} f) = J_a^m f, \quad (212)$$

$$J_a^{m+n} f = J_a^m(J_a^n f). \quad (213)$$

*Moreover  $J_a^n f$  is  $C^n$ .*

The four unfolding and base-point declarations are `normalizedVolterra_zero`, `normalizedVolterra_succ`, `normalizedVolterra_one`, and `normalizedVolterra_succ_self`; locality, scalar compatibility, and finite additivity are `normalizedVolterra_congr`, `normalizedVolterra_smul`, and `normalizedVolterra_finsetSum`. Equation (210) is `iteratedPrimitive_eq_normalizedVolterra`. The remaining conclusions are, in the order displayed, `normalizedVolterra_succ_hasStrictDerivAt`, `deriv_normalizedVolterra_succ`, `iteratedDeriv_normalizedVolterra_add`, `normalizedVolterra_add`, and `contDiff_normalizedVolterra`.

The normalization is exactly covariant under every real affine map  $A(t) = ct + d$ :

$$J_{ca+d}^n f(cx + d) = c^n J_a^n (f \circ A)(x), \quad (214)$$

$$J_a^n (f \circ A)(x) = (c^n)^{-1} J_{ca+d}^n f(cx + d) \quad (c \neq 0). \quad (215)$$

The first, division-free form includes negative scales, whose sign is handled by the oriented integral, and even  $c = 0$ ; it also includes order zero. Neither form assumes integrability or completeness. They are `normalizedVolterra_affine` and `normalizedVolterra_comp_affine`.

Changing the base point also has an exact finite form. If  $f$  is interval integrable on the two oriented pieces from  $a$  to  $b$  and from  $b$  to  $x$ , then

$$J_a^n f(x) = J_b^n f(x) + \sum_{k=0}^{n-1} \frac{(x-b)^k}{k!} J_a^{n-k} f(b). \quad (216)$$

No ordering of  $a, b, x$  is required. At  $n = 0$  the sum is empty and both Volterra terms reduce to  $f(x)$ . This is `normalizedVolterra_basepoint_shift`. If, in addition,  $f$  vanishes on the open unoriented interval between  $b$  and  $x$ , then the local positive-order tail is zero and

$$J_a^{n+1} f(x) = \sum_{k=0}^n \frac{(x-b)^k}{k!} J_a^{n+1-k} f(b). \quad (217)$$

Only interval integrability from  $a$  to  $b$  is assumed in this corollary; no value condition is imposed at either endpoint, and the vanishing hypothesis is vacuous when  $b = x$ . The declaration is `normalizedVolterra_succ_eq_taylor_of_eq_zero`.

The hypothesis boundary is therefore sharp: affine covariance, the base-point identities, and the direct kernel algebra take values in an arbitrary real normed vector space. Completeness is used only for the strict-FTC and smooth literal-iteration layer, its identification with the direct kernel, and the Taylor reconstruction below. The four affine and base-point declarations just named are post-mainline additions checked at code checkpoint `e109088ed`.

The finite algebra of polynomial weights is especially clean when written with rising factorials. Put

$$n^{\bar{r}} := n(n+1) \cdots (n+r-1), \quad n^{\bar{0}} := 1,$$

and write  $P^{[r]}$  for the  $r$ -th Hasse derivative of a real polynomial  $P$ .

**Theorem 6.3** (Kernel raising, polynomial commutator, and Taylor reconstruction). *For every  $n, r \in \mathbb{N}$  and fixed endpoint  $x$ ,*

$$J_a^n ((x - \cdot)^r f)(x) = n^{\bar{r}} J_a^{n+r} f(x). \quad (218)$$

*This includes  $n = 0$ : the rising factorial retains the identity when  $r = 0$  and annihilates the positive shifts when  $r > 0$ . If  $f$  is interval integrable between  $a$  and  $x$ , then*

$$J_a^n (P f)(x) = \sum_{r=0}^{\text{natdeg } P} (-1)^r n^{\bar{r}} P^{[r]}(x) J_a^{n+r} f(x), \quad (219)$$

$$J_a^n (t^p f(t))(x) = \sum_{r=0}^p (-1)^r \binom{p}{r} n^{\bar{r}} x^{p-r} J_a^{n+r} f(x). \quad (220)$$

*Here the second left side means  $J_a^n (t \mapsto t^p f(t))(x)$ . The formulas have separate positive-order companions in which  $n$  is written  $n+1$ ; those companions have the same hypotheses and retain the literal Cauchy-kernel order.*

*Finally, if  $E$  is complete and  $g : \mathbb{R} \rightarrow E$  is  $C^{n+1}$ , then*

$$J_a^{n+1} (D^{n+1} g)(x) = g(x) - \sum_{k=0}^n \frac{(x-a)^k}{k!} D^k g(a). \quad (221)$$

*In particular, if  $D^k g(a) = 0$  for  $0 \leq k \leq n$ , then the left side is exactly  $g(x)$ .*

The positive-order and all-order kernel identities are `normalizedVolterra_succ_kernel_pow` and `normalizedVolterra_kernel_pow`. The Hasse-derivative formulas are `normalizedVolterra_succ_polynomial` and `normalizedVolterra_polynomial`; their monomial specializations are `normalizedVolterra_succ_monomial` and `normalizedVolterra_monomial`. Taylor’s formula and its zero-jet reconstruction are `normalizedVolterra_succ_iteratedDeriv_eq_sub_taylor` and `normalizedVolterra_succ_iteratedDeriv_eq_of_zero_jet`. Thus every declaration in the generic module has a literal mathematical counterpart above, including the order-zero and endpoint cases.

For a real order  $\alpha$ , define the total oriented operator

$$(I_{a+}^{\alpha}f)(x) := \int_a^x \frac{(x-t)^{\alpha-1}}{\Gamma(\alpha)} f(t) dt. \quad (222)$$

Here the real power is Lean’s total real-power convention. Its classical Riemann–Liouville interpretation below is restricted to  $0 < \alpha$  and  $a \leq x$ .

**Theorem 6.4** (Positive-real fractional Volterra foundation, covariance, and order raising). *Let  $E$  be a real normed vector space and  $f, g : \mathbb{R} \rightarrow E$ . For arbitrary real  $\alpha, a, x, c, d$ , the operator vanishes at its base point, depends only on the input on the closed unoriented interval between  $a$  and  $x$ , and satisfies*

$$I_{a+}^{\alpha}(cf)(x) = cI_{a+}^{\alpha}f(x), \quad I_{a+}^1f(x) = \int_a^x f(t) dt. \quad (223)$$

For every  $n \in \mathbb{N}$ , without endpoint-order or regularity assumptions,

$$I_{a+}^{n+1}f(x) = J_{a+}^{n+1}f(x). \quad (224)$$

If  $0 < \alpha$ ,  $a \leq x$ , and  $f$  and  $g$  are continuous on  $[a, x]$ , then the two fractional kernels are interval-integrable and

$$I_{a+}^{\alpha}(f+g)(x) = I_{a+}^{\alpha}f(x) + I_{a+}^{\alpha}g(x). \quad (225)$$

More generally, let  $a \leq b$ ,  $a \leq x$ , and let  $h$  be interval-integrable on  $[a, x]$ . If  $h = 0$  on the open tail  $(b, x)$ , then

$$\int_a^x h(t) dt = \int_a^{\min(x,b)} h(t) dt. \quad (226)$$

If  $0 < \alpha$ ,  $f$  is continuous on  $[a, x]$ , and  $f = 0$  on  $(b, x)$ , this specializes to

$$I_{a+}^{\alpha}f(x) = \int_a^{\min(x,b)} \frac{(x-t)^{\alpha-1}}{\Gamma(\alpha)} f(t) dt. \quad (227)$$

For  $0 < \alpha$ ,  $0 < \beta$ , and  $s \leq x$ , the shifted real beta kernel is interval-integrable. If  $s < x$ , its raw value is

$$\int_s^x (x-u)^{\alpha-1}(u-s)^{\beta-1} du = (x-s)^{\alpha+\beta-1} \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}. \quad (228)$$

Its Gamma-normalized form is

$$\int_s^x \frac{(x-u)^{\alpha-1}}{\Gamma(\alpha)} \frac{(u-s)^{\beta-1}}{\Gamma(\beta)} du = \frac{(x-s)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)}. \quad (229)$$

If, moreover,  $E$  is complete,  $a \leq x$ , and  $f$  is continuous on  $[a, x]$ , then

$$I_{a+}^{\alpha+\beta}f(x) = I_{a+}^{\alpha} \left[ u \mapsto I_{a+}^{\beta}f(u) \right] (x). \quad (230)$$

For  $v \in E$ , completeness of  $E$ , and a strict endpoint  $a < x$ , the same beta calculation gives the normalized shifted-power rule

$$I_{a+}^{\alpha} \left[ t \mapsto \frac{(t-a)^{\beta-1}}{\Gamma(\beta)} v \right] (x) = \frac{(x-a)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)} v \quad (\alpha, \beta > 0), \quad (231)$$

and, for  $\rho > -1$ ,

$$I_{a+}^{\alpha} [t \mapsto (t-a)^{\rho} v] (x) = \frac{\Gamma(\rho+1)(x-a)^{\alpha+\rho}}{\Gamma(\alpha+\rho+1)} v \quad (\alpha > 0). \quad (232)$$

The constant specialization includes the degenerate endpoint:

$$I_{a+}^{\alpha} [t \mapsto v] (x) = \frac{(x-a)^{\alpha}}{\Gamma(\alpha+1)} v \quad (\alpha > 0, a \leq x). \quad (233)$$

For arbitrary real  $\alpha$ ,  $0 < c$ , and  $a \leq x$ , increasing-affine covariance requires no regularity, integrability, or completeness hypothesis:

$$I_{(ca+d)+}^{\alpha} f(cx+d) = c^{\alpha} I_{a+}^{\alpha} [t \mapsto f(ct+d)](x), \quad (234)$$

$$I_{a+}^{\alpha} [t \mapsto f(ct+d)](x) = (c^{\alpha})^{-1} I_{(ca+d)+}^{\alpha} f(cx+d). \quad (235)$$

Finally, let  $E$  be complete, let  $0 < \alpha$ ,  $a \leq x$ , and suppose that  $g$  is continuous on  $[a, x]$ , has the right-sided interior derivative  $g'$ , and that  $g'$  is interval-integrable. Gamma-normalized integration by parts gives

$$I_{a+}^{\alpha+1} g'(x) = I_{a+}^{\alpha} g(x) - \frac{(x-a)^{\alpha}}{\Gamma(\alpha+1)} g(a). \quad (236)$$

This includes  $a = x$ ; when  $g(a) = 0$ , the boundary term vanishes. Only the right derivative on the open interval is required; no two-sided derivative hypothesis is used.

The definition and ten exact theorems in `FractionalVolterra.lean` are `fractionalVolterra`, `fractionalVolterra_self`, `fractionalVolterra_congr`, `fractionalVolterra_smul`, `intervalIntegrable_fractionalVolterra_kernel`, `intervalIntegral_eq_integral_min_of_eq_zero`, `fractionalVolterra_eq_intervalIntegral_min_of_eq_zero`, `fractionalVolterra_add_input`, `fractionalVolterra_one`, and `fractionalVolterra_nat_succ`, together with `intervalIntegral_fractionalVolterra_betaKernel`. The six theorems in `FractionalVolterraSemigroup.lean` are `intervalIntegrable_fractionalVolterra_betaKernel`, `intervalIntegral_fractionalVolterra_normalizedBetaKernel`, `fractionalVolterra_normalized_rpow_smul`, `fractionalVolterra_rpow_smul`, `fractionalVolterra_const`, and `fractionalVolterra_add`. The four theorems in `FractionalVolterraCalculus.lean` are `fractionalVolterra_affine`, `fractionalVolterra_comp_affine`, `fractionalVolterra_add_one_deriv`, and `fractionalVolterra_add_one_deriv_of_eq_zero`. Thus these three reusable operator modules have one definition and twenty public theorems. They prove no classical order-zero law, reversed-endpoint fractional interpretation, semigroup theorem for merely interval-integrable inputs or noncomplete targets, Caputo or Riemann–Liouville derivative operator, complex-order theorem, nonpositive-scale affine covariance, or hierarchy-specific formula by themselves. The positive-real Fabius–CDF specialization is recorded in [Section 7.3](#); the following total definition and six specialization theorems bring the four-module total to two definitions and twenty-six public theorems.

The signed Fabius dilation law now enters only at the specialization step.

**Theorem 6.5** (Causal Rvachev fractional primitives and exact first shifts). *For every real  $\beta$ ,  $x$ , define the total causal primitive*

$$U_{\beta}(x) := I_{(-1)+}^{\beta} \text{up}(x). \quad (237)$$

If  $0 < \beta$  and  $-1 \leq x$ , compact support gives the exact cutoff

$$U_\beta(x) = \frac{1}{\Gamma(\beta)} \int_{-1}^{\min(x,1)} (x-t)^{\beta-1} \text{up}(t) dt. \quad (238)$$

For every  $n \in \mathbb{N}$  and every real  $x$ ,

$$U_{n+1}(x) = J_{-1}^{n+1} \text{up}(x). \quad (239)$$

If  $0 < \alpha, 0 < \beta$ , and  $-1 \leq x$ , then

$$U_{\alpha+\beta}(x) = I_{(-1)+}^\alpha U_\beta(x). \quad (240)$$

For  $0 < \alpha$ , the signed extension satisfies, for every  $x \geq 0$ ,

$$I_{0+}^{\alpha+1} \mathcal{F}(x) = 2^\alpha I_{0+}^\alpha \mathcal{F}\left(\frac{x}{2}\right). \quad (241)$$

The bounded function satisfies, for every  $x \in [0, 1]$ ,

$$I_{0+}^{\alpha+1} F(x) = 2^\alpha I_{0+}^\alpha F\left(\frac{x}{2}\right). \quad (242)$$

Finally the compactly supported fold, integrated from its left support endpoint, obeys for every  $x \geq -1$

$$I_{(-1)+}^{\alpha+1} \text{up}(x) = 2^\alpha I_{0+}^\alpha F\left(\frac{x+1}{2}\right). \quad (243)$$

The total definition and six theorems in `FabiusFractionalVolterra.lean` are `rvachevFractionalPrimitive`, `rvachevFractionalPrimitive_eq_intervalIntegral_min`, `rvachevFractionalPrimitive_nat_succ`, `rvachevFractionalPrimitive_add`, `fractionalVolterra_add_one_extendedFabius_of_nonneg`, `fractionalVolterra_add_one_fabiusReal`, and `fractionalVolterra_add_one_rvachevUp`. Their domains are part of the statements: the signed formula does not cover  $x < 0$ , the bounded formula is confined to  $[0, 1]$ , the cutoff and semigroup formulas start at  $-1$ , and the support-truncated formula is asserted only at positive order.

**Theorem 6.6** (Complete integer primitive ladder of the Fabius functions). *For every  $n, p \in \mathbb{N}$  and  $x \in \mathbb{R}$ ,*

$$J_0^n \mathcal{F}(x) = 2^{\binom{n}{2}} \mathcal{F}\left(\frac{x}{2^n}\right), \quad (244)$$

$$J_0^n (t^p \mathcal{F}(t))(x) = \sum_{r=0}^p (-1)^r \binom{p}{r} n^{\bar{r}} x^{p-r} 2^{\binom{n+r}{2}} \mathcal{F}\left(\frac{x}{2^{n+r}}\right). \quad (245)$$

For the bounded function the same formulas hold whenever  $x \leq 1$ :

$$J_0^n F(x) = 2^{\binom{n}{2}} F\left(\frac{x}{2^n}\right), \quad (246)$$

$$J_0^n (t^p F(t))(x) = \sum_{r=0}^p (-1)^r \binom{p}{r} n^{\bar{r}} x^{p-r} 2^{\binom{n+r}{2}} F\left(\frac{x}{2^{n+r}}\right). \quad (247)$$

All four identities include  $n = 0$ , arbitrary endpoint orientation, and the natural power convention  $0^0 = 1$ .

The four displayed statements are exactly `normalizedVolterra_extendedFabius`, `normalizedVolterra_pow_mul_extendedFabius`, `normalizedVolterra_fabiusReal_of_1e_one`, and `normalizedVolterra_pow_mul_fabiusReal_of_1e_one` in `FabiusAntiderivatives.lean`. Their proof architecture is deliberately triangular: the global dilation equation identifies the  $n$ -th derivative of the scaled candidate; its initial Taylor jet vanishes; (221) reconstructs the candidate; and (220) supplies every natural polynomial weight. The bounded formulas use `fabiusReal_eq_extendedFabius_of_1e_one`: the oriented segment from 0 to any  $x \leq 1$  remains inside the half-line on which the two functions agree.

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The generic positive-real operator is formalized in [Theorem 6.4](#), including its positive-real, ordered-endpoint, continuous-Banach-input semigroup, shifted-power and constant evaluations, positive-scale affine covariance, and right-derivative order-raising identity. The positive-real Fabius-CDF stopped-power and unit-endpoint moment formulas are formalized in [Theorem 7.3](#), and the first signed, bounded, and up-to- $F$  fractional shifts in [Theorem 6.5](#). What remains includes the signed branch for  $x < 0$ , iterated fractional shifts, the shifted dyadic lattice and its endpoint-up wrapper, reflected or piecewise support-tail formulas, weighted hierarchy-specific monomials, semigroups for shifted or reflected hierarchy variants, a full two-branch refinement law, fractional inverse/quantile formulas, classical order-zero or reversed-endpoint interpretations, Caputo and Riemann–Liouville derivative operators, complex powers and complex order, and normal or Mellin continuation. At integer order, separately named packages for the complete up ladder based at  $-1$ , the reflected  $1 - F$  ladder, arbitrary affine copies of  $F$ , and the full piecewise polynomial tail after the support of up also remain frontier targets.

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### 6.3 Dyadic derivative cells and exact value distributions

Write

$$T_n := \binom{n+1}{2}, \quad \varepsilon_k := (-1)^{\text{wt}_2(k)}.$$

For  $n, k \in \mathbb{N}$ , the formal cell coordinate is the total function

$$c_{n,k}(y) := -1 + \frac{2k+1+y}{2^n}. \quad (248)$$

Its geometric use is  $0 \leq k < 2^n$  and  $-1 \leq y \leq 1$ . In that range it maps  $[-1, 1]$  into  $[-1, 1]$ , and

$$c_{n,0}(-1) = -1, \quad c_{n,k}(1) = c_{n,k+1}(-1), \quad c_{n,2^n}(-1) = 1. \quad (249)$$

Thus the images are exactly the  $2^n$  adjacent closed cells from  $-1$  to  $1$ .

**Theorem 6.7** (Exact cellwise derivative tiling). *For every  $n \in \mathbb{N}$ ,  $0 \leq k < 2^n$ , and  $-1 \leq y \leq 1$ ,*

$$\text{up}^{(n)}(c_{n,k}(y)) = 2^{T_n} \varepsilon_k \text{up}(y), \quad (250)$$

$$\left| \text{up}^{(n)}(c_{n,k}(y)) \right| = 2^{T_n} \text{up}(y). \quad (251)$$

*In particular, the midpoint of the  $k$ -th cell has the exact signed value*

$$\text{up}^{(n)}\left(-1 + \frac{2k+1}{2^n}\right) = 2^{T_n} \varepsilon_k. \quad (252)$$

*All three formulas include  $n = 0$ , and the cell identities include both endpoints.*

*Formal proof.* The coordinate (248) is `rvachevDerivativeCell`. Its affine form, adjacent-endpoint identity, two outer endpoint identities, and closed-interval range theorem are `rvachevDerivativeCell_eq_div_add`, `rvachevDerivativeCell_one_eq_succ_neg_one`, `rvachevDerivativeCell_zero_neg_one`, `rvachevDerivativeCell_two_pow_neg_one`, and `rvachevDerivativeCell_mem_Icc`. Equations (250)–(252) are respectively `iteratedDeriv_rvachev_cell`, `abs_iteratedDeriv_rvachev_cell`, and `iteratedDeriv_rvachev_cell_zero`, all in `RvachevDerivativeDistribution.lean`.  $\square$



**Theorem 6.8** (Continuous-test derivative distributions). *Let  $E$  be a real Banach space and let  $H : \mathbb{R} \rightarrow E$  be continuous. For every  $n \in \mathbb{N}$ , the Bochner interval integrals satisfy*

$$\int_{-1}^1 H(\text{up}^{(n)}(x)) \, dx = 2^{-n} \sum_{k=0}^{2^n-1} \int_{-1}^1 H(2^{T_n} \varepsilon_k \text{up}(y)) \, dy, \quad (253)$$

$$\int_{-1}^1 H(|\text{up}^{(n)}(x)|) \, dx = \int_{-1}^1 H(2^{T_n} \text{up}(y)) \, dy, \quad (254)$$

$$\int_{-1}^1 H(2^{-T_n} |\text{up}^{(n)}(x)|) \, dx = \int_{-1}^1 H(\text{up}(y)) \, dy. \quad (255)$$

The signed formula is the exact  $2^n$ -term Thue–Morse mixture and includes order zero.

Let  $\lambda_{[-1,1]}$  denote Lebesgue measure restricted to the closed interval  $[-1, 1]$ , and let  $f_{\#}\mu$  denote the pushforward of  $\mu$  by  $f$ . The normalized absolute identity is also proved as the exact equality of Borel measures

$$\left( x \mapsto 2^{-T_n} |\text{up}^{(n)}(x)| \right)_{\#} \lambda_{[-1,1]} = \text{up}_{\#} \lambda_{[-1,1]}. \quad (256)$$

*Formal proof.* Equations (253)–(255) are `intervalIntegral_comp_iteratedDeriv_rvachev`, `intervalIntegral_comp_abs_iteratedDeriv_rvachev`, and `intervalIntegral_comp_normalized_abs_iteratedDeriv_rvachev`. The pushforward equality (256) is `map_normalized_abs_iteratedDeriv_rvachev_restrict_Icc`. These declarations are in `RvachevDerivativeDistribution.lean`.  $\square$

**Corollary 6.9** (All nonnegative real absolute moments). *For every  $n \in \mathbb{N}$  and every real  $p \geq 0$ ,*

$$\int_{-1}^1 |\text{up}^{(n)}(x)|^p \, dx = (2^{T_n})^p \int_{-1}^1 \text{up}(y)^p \, dy. \quad (257)$$

Here the powers are Lean’s real-power operation. Thus  $p = 0$  is included, with the convention  $0^0 = 1$ .

*Formal proof.* This is `intervalIntegral_abs_iteratedDeriv_rvachev_rpow` in `RvachevDerivativeDistribution.lean`; its hypothesis is exactly  $0 \leq p$ .  $\square$

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**Exact current boundary.** The signed API above is the finite continuous-test-function mixture (253); a separately packaged Borel pushforward law simplified to an independent Rademacher sign is not yet formalized. Nor does the current module state the `eLpNorm` or general rearrangement-invariant norm ladder, the exact inverse-Fabius level sets, the beta-transform identities, or the later spectral layers. Those claims remain in the research-frontier report. Equation (256) is the formal measure-theoretic base from which such absolute-value corollaries can subsequently be proved.

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## 6.4 Effective flatness

For every  $n \in \mathbb{N}$ ,  $x \geq 0$ , and  $2^n x \leq 1$ , the formal development proves both

$$F(x) \leq 2^{\binom{n+1}{2}} x^n, \quad (258)$$

$$F(x) \leq \frac{2^{\binom{n+1}{2}}}{n!} x^n. \quad (259)$$

These are `fabiusReal_le_two_pow_mul_pow` in `EffectiveFlatness.lean` and `fabiusReal_le_two_pow_div_factorial_mul_pow` in `SharpFlatness.lean`. Their reflected forms at 1 and their folded forms for `up` are also formalized in those modules. In particular, for each natural power,

$$F(x) = o(x^n) \quad (x \rightarrow 0),$$

as stated by `fabiusReal_isLittleO_pow_at_zero` in `FabiusFlatness.lean`.

## 6.5 The precise analytic statements

The exact real-analytic locus of the bounded function is

$$F \text{ is analytic at } x \iff x \notin [0, 1]. \quad (260)$$

This is `fabius_analyticAt_iff`, with canonical specialization `canonical_fabius_analyticAt_iff`. For the other global forms the current proved statements are deliberately one-sided:

$$\text{up is not analytic at any } x \in [-1, 1], \quad (261)$$

$$\mathcal{F} \text{ is not analytic at any } x \in [0, 2]. \quad (262)$$

They are `canonical_rvachev_not_analyticAt` and `extendedFabius_not_analyticAt` in `NowhereAnalytic.lean`. No stronger global locus is inferred here.

## 6.6 The totaled inverse

Let  $I : \mathbb{R} \rightarrow [0, 1]$  be the inverse order isomorphism on  $[0, 1]$ , extended constantly by 0 on the left and 1 on the right. This is the formal `fabiusInv` constructed from `fabiusOrderIso`. It is continuous and nondecreasing; on  $[0, 1]$  it is a strictly increasing bijection and satisfies

$$F(I(y)) = y, \quad I(F(x)) = x \quad (x, y \in [0, 1]). \quad (263)$$

For  $0 < y < 1$ , it is smooth and

$$I'(y) = \frac{1}{F'(I(y))} = \frac{1}{2 \text{ up}(2I(y) - 1)} > 0. \quad (264)$$

*Formal proof.* The range and inverse identities are `fabiusInv_mem_Icc`, `fabiusReal_fabiusInv`, and `fabiusInv_fabiusReal`. The order, continuity, and bijectivity statements are `strictMonoOn_fabiusInv`, `monotone_fabiusInv`, `continuous_fabiusInv`, and `bijOn_fabiusInv`. Interior smoothness and the derivative chain are `fabiusInv_contDiffOn_Ioo`, `deriv_fabiusInv`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, and `deriv_fabiusInv_pos`, all in `FabiusInverse.lean`.  $\square$

The same module controls the curvature of the inverse without asserting differentiability at the clamped endpoints. For  $0 < y < 1$ ,

$$I''(y) = -\frac{F''(I(y))}{F'(I(y))^3}. \quad (265)$$

This is `deriv_deriv_fabiusInv`, with the corresponding derivative witness `deriv_fabiusInv_hasDerivAt`. On the closed half intervals, continuity at the endpoints and the interior derivative criterion give

$$I \text{ is strictly concave on } [0, \tfrac{1}{2}] \text{ and strictly convex on } [\tfrac{1}{2}, 1]. \quad (266)$$

The exact declarations are `strictConcaveOn_fabiusInv_firstHalf` and `strictConvexOn_fabiusInv_secondHalf`.

The pointwise statement is sharper. The midpoint jet is

$$I(1/2) = 1/2, \quad I'(1/2) = 1/2, \quad I''(1/2) = 0, \quad (267)$$

and, for  $0 < y < 1$ ,

$$\begin{aligned} I''(y) < 0 &\iff 0 < y < \tfrac{1}{2}, \\ I''(y) > 0 &\iff \tfrac{1}{2} < y < 1, \\ I''(y) = 0 &\iff y = \tfrac{1}{2}. \end{aligned} \quad (268)$$

The exact declarations are `fabiusInv_half`, `deriv_fabiusInv_half`, `deriv_deriv_fabiusInv_half`, and the `deriv_deriv_fabiusInv_neg_iff`, `deriv_deriv_fabiusInv_pos_iff`, and `deriv_deriv_fabiusInv_eq_zero_iff` family. The hypothesis  $0 < y < 1$  is essential in the last three statements: the totalized inverse is not differentiable at the clamping endpoints.

The inverse inherits the exact symmetry and midpoint

$$I(1 - y) = 1 - I(y), \quad I(1/2) = 1/2, \quad (269)$$

and for  $0 \leq a \leq 2^n$ ,

$$I\left(F\left(\frac{a}{2^n}\right)\right) = \frac{a}{2^n}. \quad (270)$$

These are `fabiusInv_one_sub`, `fabiusInv_half`, and `fabiusInv_fabiusDyadicUnit_of_1e`. Its exact analytic locus is likewise

$$I \text{ is analytic at } y \iff y \notin [0, 1], \quad (271)$$

by `fabiusInv_analyticAt_iff`; in the open interval its derivative never vanishes by `deriv_fabiusInv_ne_zero`.

Flatness of  $F$  becomes infinite endpoint steepness of  $I$ . For every  $n \in \mathbb{N}$ ,

$$y = o(I(y)^n) \quad (y \downarrow 0), \quad (272)$$

$$1 - y = o((1 - I(y))^n) \quad (y \uparrow 1). \quad (273)$$

These are `id_isLittleO_fabiusInv_pow_at_zero_right` and `one_sub_isLittleO_one_sub_fabiusInv_pow_at_one_left`. In particular, the two one-sided difference quotients diverge:

$$\frac{I(y)}{y} \longrightarrow +\infty \quad (y \downarrow 0), \quad \frac{1 - I(y)}{1 - y} \longrightarrow +\infty \quad (y \uparrow 1). \quad (274)$$

The formally proved counterparts are `tendsto_fabiusInv_div_atTop` and `tendsto_one_sub_fabiusInv_div_one_sub_atTop`. Thus the totalized inverse is not differentiable at either clamping point, exactly as stated by `fabiusInv_not_differentiableAt_zero` and `fabiusInv_not_differentiableAt_one`; no finite endpoint derivative is being smuggled into the curvature statement above. The stronger limits of the positive interior derivative itself are also formalized:

$$I'(y) \longrightarrow +\infty \quad (y \downarrow 0), \quad I'(y) \longrightarrow +\infty \quad (y \uparrow 1). \quad (275)$$

They are `tendsto_deriv_fabiusInv_atTop_at_zero_right` and `tendsto_deriv_fabiusInv_atTop_at_one_left`; these are limits through the open interval, not finite derivatives at 0 or 1.

## 6.7 Precise non-elementarity

The inductive predicate `IsElementary` contains constants and the identity and is closed under arithmetic operations, composition, exponential, logarithm, trigonometric operations, and variable real powers whose elementary base is globally nonvanishing. The last closure statement is `IsElementary.rpow_of_ne_zero`; the positive-base and positive-region interfaces are `IsElementary.rpow_of_pos` and `eqOn_rpow_of_pos`. Every expression in this class has a dense real-analytic locus:

$$\text{IsElementary } g \implies \overline{\{x : g \text{ is analytic at } x\}} = \mathbb{R}.$$

This is `IsElementary.dense_analyticLocus`. It yields the proved global obstructions

$$F, \quad \text{up}, \quad \mathcal{F}, \quad I \quad \text{are not elementary.} \quad (276)$$

The exact declarations are `canonical_fabius_not_isElementary`, `rvachev_not_isElementary`, `extendedFabius_not_isElementary`, and `fabiusInv_not_isElementary`; local versions for  $F$  and  $I$  on  $(0, 1)$  are `canonical_fabius_not_isElementary_on_Ioo` and `canonical_fabiusInv_not_isElementary_on_Ioo`.

The larger formal class `IsElementaryOrInverse` admits an inverse-branch constructor only with an open branch domain, continuity and a right-inverse identity there, and analyticity of the chosen total extension on the interior of the complement. The canonical global functions themselves are not members of that larger class:

$$\neg \text{IsElementaryOrInverse } F, \quad \neg \text{IsElementaryOrInverse } I. \quad (277)$$

These are `fabius_not_isElementaryOrInverse` and `fabiusInv_not_isElementaryOrInverse`. More strongly in a different direction, no member of the class can represent  $F$  or  $I$  even merely on  $(0, 1)$ , by `canonical_fabius_not_isElementaryOrInverse_on_Ioo` and `canonical_fabiusInv_not_isElementaryOrInverse_on_Ioo` in `InverseNotElementary.lean`.

## 7 Moments, generating functions, and reciprocal powers

### 7.1 Even and half moments

Retain the even moments  $c_n$  from (191); in particular,  $c_0 = 1$ . Define

$$d_0 = 1, \quad d_n = n \int_0^1 t^{n-1} \text{up}(t) dt \quad (n \geq 1). \quad (278)$$

Both are positive rational sequences. Their exact triangular recurrences, for  $n \geq 1$ , are

$$(2n+1)(2^{2n}-1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k, \quad (279)$$

$$(n+1)(2^n-1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k. \quad (280)$$

These are the rational declarations `moment_succ` and `halfMoment_succ` in `Arithmetic.lean`; positivity is `moment_pos` in `Paper06487Supplement.lean` and `halfMoment_pos` in `Arithmetic.lean`. The displayed integral definitions are `integral_even_pow_mul_rvachev_eq_moment`, `moment_eq_integral_formula`, and `halfMoment_eq_integral_formula_all` in `AnalyticMoments.lean`.

The entire generating function

$$G(z) = 1 + z \int_0^1 \text{up}(t) e^{zt} dt = \sum_{n=0}^{\infty} d_n \frac{z^n}{n!} \quad (281)$$

satisfies

$$G(2z) = \frac{e^z - 1}{z} G(z), \quad (282)$$

with the removable value at  $z = 0$ . The exact formal statements are `proposition_two_formula` and `complexGeneratingFunction_eq_series_formula` in `AnalyticMoments.lean`; entire differentiability is `differentiable_complexGeneratingFunction` in `NegativeLaplaceVerticalLog.lean`.

The two sequences are linked by

$$d_n = 2^{-n} \sum_{0 \leq k \leq n/2} \binom{n}{2k} c_k, \quad (283)$$

and, at odd indices,

$$d_{2k+1} = \frac{2k+1}{2} c_k. \quad (284)$$

These are `halfMoment_eq_evenMomentSum` and `halfMoment_odd_eq_moment` in `MomentPowerSeries.lean`. Division-free natural numerators for  $c_n$  and  $d_n$ , together with their exact recurrences and odd-index bridge, are formalized by `momentNumerator`, `halfMomentNumerator`, `halfMomentNumerator_succ`, `halfMomentNumerator_odd_index_eq`, and `halfMomentNumerator_odd_index_formula`. No unproved numerical table is needed.

More explicitly, if  $C_n = \text{momentNumerator}(n)$ , then  $C_n$  is a positive integer and

$$c_n = \frac{C_n}{(2n+1)!! \prod_{j=1}^n (2^{2j} - 1)}. \quad (285)$$

The exact rational identity is `moment_eq_momentNumerator_div`; the product reindexing is `evenMersenneProduct_eq_prod_Ico`, and positivity is `momentNumerator_pos`. Thus (285) records the proved natural normalization, not a claim that the displayed denominator is already reduced.

## 7.2 Universal Bell and cumulant transforms

The algebra converting moments to cumulants is independent of the particular Fabius law. Let  $R$  be any commutative  $\mathbb{Q}$ -algebra and let  $a : \mathbb{N} \rightarrow R$ . Write

$$(\mathcal{N}a)_n = \frac{a_n}{n!}, \quad (\mathcal{D}a)_n = n!a_n. \quad (286)$$

These changes of coefficient convention are mutually inverse. For cumulants  $\kappa : \mathbb{N} \rightarrow R$  and moments  $\mu : \mathbb{N} \rightarrow R$ , define the complete Bell and moment–cumulant transforms by the formal identities

$$\mathcal{B}(\kappa)_n = n![X^n] \exp \left( \sum_{j \geq 1} \kappa_j \frac{X^j}{j!} \right), \quad (287)$$

$$\mathcal{K}(\mu)_n = n![X^n] \log \left( 1 + \sum_{j \geq 1} \mu_j \frac{X^j}{j!} \right). \quad (288)$$

Here the exponential and logarithm are formal power series implemented by finite coefficient recurrences; no topology or field structure on  $R$  is being assumed.

**Proposition 7.1** (Bell recurrence and exact inversion). *For every  $n \in \mathbb{N}$ ,*

$$\mathcal{B}(\kappa)_0 = 1, \quad \mathcal{K}(\mu)_0 = 0, \quad (289)$$

$$\mathcal{B}(\kappa)_{n+1} = \sum_{j=0}^n \binom{n}{j} \kappa_{j+1} \mathcal{B}(\kappa)_{n-j}. \quad (290)$$

*At every index, including zero, the two composites are exactly*

$$\mathcal{B}(\mathcal{K}(\mu))_n = \begin{cases} 1, & n = 0, \\ \mu_n, & n > 0, \end{cases} \quad \mathcal{K}(\mathcal{B}(\kappa))_n = \begin{cases} 0, & n = 0, \\ \kappa_n, & n > 0. \end{cases} \quad (291)$$

*Consequently the transforms are inverse as functions when  $\mu_0 = 1$  and  $\kappa_0 = 0$ , respectively. More precisely, if  $f : R \rightarrow S$  is a morphism of commutative  $\mathbb{Q}$ -algebras, then*

$$f(\mathcal{B}_R(\kappa)_n) = \mathcal{B}_S(f \circ \kappa)_n, \quad f(\mathcal{K}_R(\mu)_n) = \mathcal{K}_S(f \circ \mu)_n. \quad (292)$$

More generally, if  $E$  and  $A$  are ordinary coefficient sequences for the formal exponent and mass series, then the coefficient-convention bridge is

$$\mathcal{B}(\mathcal{D}E) = \mathcal{D}(\text{expCoeff}(E)), \quad \mathcal{K}(\mathcal{D}A) = \mathcal{D}(\text{logCoeff}(A)). \quad (293)$$

Thus the same theorem applies without modification to polynomial rings, function rings, and scalar extensions, not only to real- or rational-valued moment sequences.

*Formal proof.* The operations  $\mathcal{N}, \mathcal{D}, \mathcal{B}, \mathcal{K}$  are respectively `factorialNormalize`, `factorialDenormalize`, `completeBellPolynomial`, and `momentCumulant`. The two identities following (286) are `factorialNormalize_factorialDenormalize` and `factorialDenormalize_factorialNormalize`. The zero values and (290) are `completeBellPolynomial_zero`, `momentCumulant_zero`, and `completeBellPolynomial_succ`.

Equation (291) is `completeBellPolynomial_momentCumulant` together with `momentCumulant_completeBellPolynomial`; its normalized function-level forms are `completeBellPolynomial_momentCumulant_of_zero_eq_one` and `momentCumulant_completeBellPolynomial_of_zero_eq_zero`. The bridge (293) is `completeBellPolynomial_factorialDenormalize` and `momentCumulant_factorialDenormalize`. Finally, functoriality is (292), formalized by `map_completeBellPolynomial` and `map_momentCumulant`. All declarations are in `MomentCumulantAlgebra.lean` and assume only a commutative  $\mathbb{Q}$ -algebra.  $\square$

**Proposition 7.2** (Finite polynomial integrals from moments and formal cumulants). *Let  $\Omega$  be a measurable space, let  $\mu$  be any measure on it, let  $S$  be a semiring with a semiring homomorphism  $\varphi : S \rightarrow \mathbb{R}$ , and let  $Z : \Omega \rightarrow \mathbb{R}$ . For  $p \in S[X]$ , write*

$$p_n = [X^n]p, \quad p_\varphi(x) = \sum_{n \in \text{supp}(p)} \varphi(p_n)x^n, \quad m_n = \text{moment}_\mu(Z; n).$$

Here  $\text{moment}_\mu$  uses Mathlib’s totalized Bochner-integral convention; whenever  $Z^n$  is integrable it is the ordinary integral  $\int_\Omega Z(\omega)^n d\mu(\omega)$ . Assume only that  $Z^n$  is integrable for  $n \in \text{supp}(p)$ , and put  $\kappa = \mathcal{K}(m)$ . Then

$$\int_\Omega p_\varphi(Z(\omega)) d\mu(\omega) = \sum_{n \in \text{supp}(p)} \varphi(p_n)m_n, \quad (294)$$

$$= \sum_{n \in \text{supp}(p)} \varphi(p_n)\mathcal{B}(\kappa)_n + \varphi(p_0)(m_0 - 1). \quad (295)$$

If  $m_0 = 1$ , the correction in (295) vanishes. This holds automatically when  $\mu$  is a probability measure.

*Formal proof.* All four public identifiers share the prefix `integral_eval2_eq_sum_`. In the order of the two displayed equations, the  $m_0 = 1$  corollary, and the probability-measure corollary, their suffixes are

- (i) `moment`;
- (ii) `completeBell_momentCumulant_with_mass_correction`;
- (iii) `completeBell_momentCumulant_of_moment_zero_eq_one`;
- (iv) `completeBell_momentCumulant`.

They are declared in `PolynomialExpectationCumulant.lean`. They require neither boundedness nor independence, and no integrability assumptions for powers outside the finite support of  $p$ . They formalize the finite moment-to-Bell rewrite used by endpoint transfer, not the analytic transfer expansion or its remainder estimates.  $\square$

### 7.3 The probability–Laplace bridge

Let  $\mu_X$  be the distribution of the weighted sum  $X$  from (33). Its survival function on the nonnegative half-line is exactly

$$\mu_X((t, \infty)) = \text{up}(t) \quad (t \geq 0). \quad (296)$$

If  $g, g' : \mathbb{R} \rightarrow \mathbb{R}$  are continuous and  $g$  has derivative  $g'(t)$  at every  $t \in [0, 1]$ , then

$$\int_{[0,1]} g(x) \, d\mu_X(x) = g(0) + \int_0^1 g'(t) \text{up}(t) \, dt. \quad (297)$$

More generally, let  $E$  be a real Banach space and let  $0 \leq c \leq 1$ . For a map  $k : \mathbb{R} \rightarrow E$ , with interval integrability on  $[0, c]$  in the first line and on  $[0, 1]$  in the second, the partial and full survival layer-cake identities are

$$\int_0^c \text{up}(t) k(t) \, dt = \int_{\mathbb{R}} \left( \int_0^{\min\{z, c\}} k(t) \, dt \right) d\mu_X(z), \quad (298)$$

$$\int_0^1 \text{up}(t) k(t) \, dt = \int_{\mathbb{R}} \left( \int_0^z k(t) \, dt \right) d\mu_X(z). \quad (299)$$

No positivity assumption is imposed on  $k$ ; taking  $E = \mathbb{C}$  gives the complex-kernel form used by transform arguments.

The atom-preserving closed-supergraph dual is the lower-CDF layer cake. For any finite measure  $\nu$  on  $\mathbb{R}$ ,  $a \leq b$ , and Banach-valued  $k$  interval-integrable on  $[a, b]$ ,

$$\int_a^b \nu((-\infty, t]) k(t) \, dt = \int_{\mathbb{R}} \left( \int_{\min\{\max\{z, a\}, b\}}^b k(t) \, dt \right) d\nu(z). \quad (300)$$

For the partial terminal form on  $[c, b]$ , only  $c \leq b$  and the one-sided almost-everywhere bound  $z \leq b$  are needed. Compact support on  $[a, b]$  gives the convenient partial corollary and the full form; the closed rays retain any atoms of  $\nu$ .

**Theorem 7.3** (Fractional stopped-power formula for the Fabius CDF). *For every positive real  $\alpha$  and every  $x \in \mathbb{R}$ ,*

$$I_{0+}^\alpha F(x) = \frac{1}{\Gamma(\alpha + 1)} \int_{\mathbb{R}} (x - z)_+^\alpha d\mu_X(z) = \frac{1}{\Gamma(\alpha + 1)} \int_{\Omega} (x - X(\omega))_+^\alpha d\mathbf{P}(\omega). \quad (301)$$

*This includes  $x \leq 0$ , where both sides vanish. At the unit endpoint, reflection of the weighted-sum law gives*

$$I_{0+}^\alpha F(1) = \frac{1}{\Gamma(\alpha + 1)} \int_{\mathbb{R}} z^\alpha d\mu_X(z) = \frac{1}{\Gamma(\alpha + 1)} \int_{\Omega} X(\omega)^\alpha d\mathbf{P}(\omega). \quad (302)$$

The generic closed-supergraph engine is `integral_smul_setIntegral_supergraph` in `SubgraphFubini.lean`. Its finite-measure clamped, one-sided-upper-supported partial, compactly supported partial, and full forms are `intervalIntegral_cdf_smul_eq_integral_clamp`, `intervalIntegral_cdf_smul_eq_integral_max_of_ae_le`, `intervalIntegral_cdf_smul_eq_integral_max_of_ae_mem_Icc`, and `intervalIntegral_cdf_smul_eq_integral_of_ae_mem_Icc` in `CDFLayerCake.lean`. The positive-real fractional forms—respectively without a support hypothesis, with only an almost-everywhere lower support bound, directly for a probability CDF under that same bound, and at the endpoint of an almost-everywhere compact support interval—are `fractionalVolterra_measureReal_Iic_eq_integral_clamp`, `fractionalVolterra_measureReal_Iic_eq_integral_posPart`, `fractionalVolterra_cdf_eq_integral_posPart`, and `fractionalVolterra_measureReal_Iic_eq_integral`

`_rpow_of_ae_mem_Icc` in `FractionalCDFLayerCake.lean`. Finally, `ProbabilityRepresentation.fractionalVolterra_fabiusReal_eq_integral_posPart`, `ProbabilityRepresentation.fractionalVolterra_fabiusReal_eq_uniformProduct_integral_posPart`, `ProbabilityRepresentation.fractionalVolterra_fabiusReal_one_eq_integral_rpow`, and `ProbabilityRepresentation.fractionalVolterra_fabiusReal_one_eq_uniformProduct_integral_rpow` in `FabiusFractionalIntegral.lean` are exactly (301)–(302). These declarations require neither a density nor atomlessness. They prove positive-real integral identities, not a fractional derivative, a Caputo formula, a complex-order continuation, or a folded-up, quantile, or inverse-function specialization.

For  $s \in \mathbb{R}$  and  $k \in \mathbb{N}$ , put

$$L_k(s) = \int_{[0,1]} e^{-sx} x^k d\mu_X(x). \quad (303)$$

These raw tilted moments agree with the formal analytic Fabius Laplace moments, and at zero tilt

$$L_n(0) = \int_{[0,1]} (1-x)^n d\mu_X(x) = d_n. \quad (304)$$

On the real axis, every derivative of the generating function is a tilted moment:

$$G^{(n)}(s) = L_n(-s), \quad G^{(n)}(0) = d_n. \quad (305)$$

Reflection of the law about  $1/2$  gives

$$L_0(s) = e^{-s} L_0(-s), \quad e^{s/2} L_0(s) = e^{-s/2} L_0(-s), \quad (306)$$

so  $s \mapsto e^{s/2} L_0(s)$  is even. Two useful quantitative forms, with all variables real, are

$$L_0\left(\frac{a+b}{2}\right)^2 \leq L_0(a)L_0(b) \quad (a, b \in \mathbb{R}), \quad (307)$$

$$L_k(s) \leq (1/a)^k k! L_0(s-a) \quad (a > 0). \quad (308)$$

*Formal proof.* The survival identity and integration-by-parts formula are `weightedSumDistribution_real_Ioi_eq_rvachevUp_of_nonneg` and `integral_unit_eq_zero_add_integral_deriv_mul_rvachevUp`. The Banach-valued partial and full formulas (298)–(299) are `ProbabilityRepresentation.intervalIntegral_rvachevUp_smul_eq_integral_min` and `ProbabilityRepresentation.intervalIntegral_rvachevUp_smul_eq_integral`. Their finite-measure, support-free and interval-supported engines are the three `intervalIntegral_survival_smul_eq_integral_clamp`, `intervalIntegral_survival_smul_eq_integral_min_of_ae_mem_Icc`, and `intervalIntegral_survival_smul_eq_integral_of_ae_mem_Icc` declarations in `SurvivalLayerCake.lean`. The tilted and zero-tilt identities are `unitLaplaceMoment_weightedSumDistribution_eq_fabiusLaplaceMoment`, `unitLaplaceMoment_weightedSumDistribution_zero_eq_halfMoment`, and `unitEndpointMoment_weightedSumDistribution_eq_halfMoment` in `ProbabilityLaplaceMoments.lean`. The derivative identities are `iteratedDeriv_generatingFunction` and `iteratedDeriv_generatingFunction_zero` in `LaplaceMoments.lean`; (306) is `fabiusLaplaceMoment_zero_reflection` and `fabiusLaplaceMoment_zero_centered_even`. Finally, (307)–(308) are `fabiusLaplaceMoment_midpoint_sq_le_all` and `fabiusLaplaceMoment_le_shift` in `UnitLaplaceMomentBounds.lean`.  $\square$



## 7.4 Exact values at reciprocal powers of two

For every  $n \in \mathbb{N}$ ,

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}), \quad F(2^{-n}) = \frac{d_n}{n! 2^{\binom{n}{2}}}. \quad (309)$$

This is `halfMoment_eq_fabius_formula` in `AnalyticMoments.lean`. The numerator-cleared version is `fabiusAtInverseTwoPow_eq_halfMomentNumerator_formula` in `ExactInversePower.lean`.

Writing  $V_n = F(2^{-n})$ , one has  $V_0 = 1$ , and for  $n \geq 1$ ,

$$V_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!} V_k. \quad (310)$$

This is `fabiusAtInverseTwoPow_recurrence`. The same sequence has exact finite path and ordered-composition descriptions, proved by `fabiusAtInverseTwoPow_eq_pathSum`, `fabiusAtInverseTwoPow_eq_composition_formula`, `pathWeight_eq_boundary_product`, and `fabiusRecurrenceSequence_eq_sum_compositions` in `FabiusInverseDyadicClosedForm.lean`. Their positivity and recurrence uniqueness are `fabiusCompositionSum_pos` and `triangularRecurrence_unique`.

The totalized inverse sends both exact rational presentations straight back to their arguments. In particular, for every  $n \in \mathbb{N}$ , including  $n = 0$ ,

$$I(\text{fabiusAtInverseTwoPow } n) = 2^{-n}, \quad (311)$$

$$I\left(2^{-\binom{n}{2}} \text{fabiusCompositionSum } n\right) = 2^{-n}. \quad (312)$$

These are `fabiusInv_fabiusAtInverseTwoPow` in `FabiusInverse.lean` and `fabiusInv_two_pow_choose_mul_fabiusCompositionSum` in `FabiusInverseDyadicClosedForm.lean`.

## 7.5 Parity and the exact two-adic valuation

Lucas parity is packaged as `odd_binomial_coefficient_counts` and `card_odd_inner_binomial_coefficients`. It implies the exact numerator theorem `momentNumerator_odd`. For every  $n \geq 1$ , the reduced numerator and denominator of  $2d_n$  are both odd, equivalently

$$\nu_2(d_n) = -1. \quad (313)$$

This is `two_mul_halfMoment_num_den_odd` together with `halfMoment_padicVal_two`.

Consequently, for  $n \geq 1$ ,

$$\nu_2(F(2^{-n})) = -\binom{n}{2} - 1 - \nu_2(n!), \quad (314)$$

by `fabiusAtInverseTwoPow_padicVal_two`. The binary form uses the proved Legendre identity

$$\nu_2(n!) = n - \text{wt}_2(n), \quad (315)$$

namely `factorial_padicVal_two`; the combined declaration is `fabiusAtInverseTwoPow_padicVal_two_closed` in `TwoAdic.lean`.

The central-binomial specialization gives, for every  $n \geq 0$ ,

$$\nu_2\binom{2n}{n} = \text{wt}_2(n), \quad (316)$$

$$\varepsilon_n = (-1)^{\nu_2\binom{2n}{n}}. \quad (317)$$

These are `padicValNat_two_centralBinom` and `thueMorseSign_eq_neg_one_pow_centralBinom`. The companion theorem `padicValNat_two_centralBinom_eq_zero_iff` says that the valuation vanishes exactly when the binary weight is zero, hence only at  $n = 0$ ; a positive power of two has binary weight, and therefore valuation, one. These three declarations exhaust the public surface of `CentralBinomialValuation.lean` and give the direct formal crosswalk for the central-binomial theorem in the Thue–Morse atlas.

## 7.6 Bernoulli recurrences

The formal development also proves Bernoulli convolution and recurrence forms for both sequences. For  $n \geq 1$ ,

$$c_n = \frac{1}{2^{2n} - 1} \sum_{k=1}^n 2^{2n-2k} (2^{2k} - 2) \binom{2n}{2k} B_{2k} c_{n-k}, \quad (318)$$

$$d_n = \frac{n 2^n}{4(2^n - 1)} d_{n-1} - \frac{1}{2^n - 1} \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{2k} 2^{n-2k} B_{2k} d_{n-2k}. \quad (319)$$

Here  $B_1 = -1/2$ . These are `moment_bernoulli_recurrence` and `halfMoment_bernoulli_recurrence`; their convolution forms and combined source-paper statement are `moment_bernoulli_convolution`, `halfMoment_bernoulli_convolution`, and `proposition_twenty_two_formula` in `BernoulliRecurrences.lean`.

## 8 Exact evaluation at arbitrary dyadic rationals

The rationality theorem for dyadic arguments comes with two complementary exact presentations:

1. a closed finite formula involving Thue–Morse signs and the rational moments  $c_k$ ;
2. an executable bit-recursive Taylor evaluator.

### 8.1 The closed Thue–Morse formula

**Theorem 8.1** (Exact dyadic formula). *Let  $n, a \in \mathbb{N}$  with  $0 \leq a \leq 2^n$ . Then*

$$F\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} (-1)^{\text{wt}_2(h)} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2a - 2h - 1)^{n-2k}. \quad (320)$$

*An empty outer sum has value 0.*

*Formal proof.* The rational expression on the right is the definition `fabijsDyadic`. Its equality, after coercion to  $\mathbb{R}$ , with  $F(a/2^n)$  under the displayed bound on  $a$  is `fabijsDyadic_cast` in `DyadicAnalytic.lean`.  $\square$

Every quantity on the right of (320) is rational, so dyadic rationality is immediate. More importantly, the formula lives entirely in  $\mathbb{Q}$  and can be evaluated without roundoff.

**Theorem 8.2** (Global dyadic formula). *For all  $n, a \in \mathbb{N}$  the signed global extension satisfies*

$$\mathcal{F}\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2a - 2h - 1)^{n-2k}. \quad (321)$$

*No upper bound on  $a$  is imposed. An empty outer sum has value 0.*

*Formal proof.* The displayed rational identity is `fabiusDyadic_cast_extended_nat` in `GlobalDyadic.lean`, after unfolding the closed form `fabiusDyadic`.  $\square$

This closed form depends only on the represented dyadic rational, not on the chosen numerator–exponent pair. Writing  $\text{Dyad}(n, m) = \text{fabiusDyadic } n \, m$ , Lean proves

$$\text{Dyad}(n + k, 2^k m) = \text{Dyad}(n, m), \quad (322)$$

$$\frac{m_1}{2^{n_1}} = \frac{m_2}{2^{n_2}} \text{ in } \mathbb{Q} \implies \text{Dyad}(n_1, m_1) = \text{Dyad}(n_2, m_2). \quad (323)$$

The exact declarations are `fabiusDyadic_refine_pow` and `fabiusDyadic_eq_of_rat_eq` in `GlobalDyadic.lean`.

Two special cases are the ones used in practice.

**Corollary 8.3** (Bounded case). *If  $0 \leq a \leq 2^n$  then  $\mathcal{F}(a/2^n) = F(a/2^n)$ , so (321) reduces to (320).*

*Proof.* The two functions agree on  $[0, 1]$  and  $a/2^n \in [0, 1]$ .  $\square$

**Corollary 8.4** (Bump case). *Let  $n \in \mathbb{N}$  and  $q \in \mathbb{Z}$ . If  $|q| < 2^n$ , then*

$$\text{up}\left(\frac{q}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{q+2^n-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2q + 2^{n+1} - 2h - 1)^{n-2k}, \quad (324)$$

*that is, (321) evaluated at  $a = q + 2^n$ . If  $|q| \geq 2^n$ , then  $\text{up}(q/2^n) = 0$ .*

*Proof.* Taking  $b = 0$  in (12) gives  $\mathcal{F}(x) = \text{up}(x - 1)$  for  $0 \leq x \leq 2$ . With  $x = 1 + q/2^n = (q + 2^n)/2^n$  and  $|q| < 2^n$  we have  $0 < x < 2$ , so  $\text{up}(q/2^n) = \mathcal{F}((q + 2^n)/2^n)$ , and (321) with  $a = q + 2^n$  applies. The last statement is the support condition  $\text{supp up} = [-1, 1]$ .  $\square$

Corollary 8.4 is the global form invoked in the proof of the uniform denominator multiplier theorem below. The point is that it genuinely requires Theorem 8.2 and not (320): for  $0 < q < 2^n$  the numerator  $a = q + 2^n$  exceeds  $2^n$ , so the bounded statement does not apply. At the endpoints  $|q| = 2^n$  the two clauses agree, both giving 0; for  $q = -2^n$  the outer sum is empty and for  $q = 2^n$  the value is  $\mathcal{F}(2) = 0$ .

*Remark 8.5* (Why the sum is finite). Both index sets in (321) are finite: the outer range is  $0 \leq h < a$ , and the inner range is  $0 \leq k \leq \lfloor n/2 \rfloor$ . The moments  $c_k$  are rational by the proved recurrence (279). Thus the formula is an exact computation in  $\mathbb{Q}$ , with no analytic truncation or floating-point roundoff.

*Remark 8.6* (Exact evaluation without uncertified examples). The finite sum is an exact rational algorithm, and its agreement with the analytic Fabius function is `fabiusDyadic_cast`. Concrete numerical specializations that do not yet have their own named Lean theorems are recorded only in the consolidated research-frontier volume.

## 8.2 Thue–Morse cancellation

The mechanism that makes the dyadic formulas simplify is the exact Prouhet cancellation carried by the Thue–Morse signs.

**Lemma 8.7** (Affine power cancellation). *If  $d < b$ , then for all  $X, Y \in \mathbb{Q}$ ,*

$$\sum_{h=0}^{2^b-1} (-1)^{\text{wt}_2(h)} (X + Yh)^d = 0. \quad (325)$$

*Formal proof.* This is the strict-degree case of `thueMorse_affine_power_sum_of_1e` in `ThueMorsePrefix.lean`; the monomial statement used internally is `thueMorsePowerSum_eq_zero_of_1t`.  $\square$

The finite block polynomial packages the same signs without introducing an analytic exponential identity:

$$\sum_{h < 2^b} \varepsilon_h z^h = \prod_{j=0}^{b-1} (1 - z^{2^j}). \quad (326)$$

This is `thueMorseBlockPolynomial_eq_product` in `ThueMorseGenerating.lean`. At the level of the centered formal exponential series, the corresponding zero has exact order  $b$ , by `order_thueMorseCenteredPowerSeries` and `thueMorseCenteredPowerSeries_eq_product` in `ThueMorseExponential.lean`.

Writing  $P_b(z)$  for either side of (326), its zero at  $z = 1$  has exact multiplicity  $b$  over  $\mathbb{Z}[z]$ :

$$(1 - z)^b \mid P_b(z), \quad (1 - z)^{b+1} \nmid P_b(z). \quad (327)$$

The divisibility and matching nondivisibility statements are respectively `one_sub_X_pow_dvd_thueMorseBlockPolynomial` and `one_sub_X_pow_succ_not_dvd_thueMorseBlockPolynomial` in `ThueMorseBlockAlgebra.lean`.

The ordinary derivatives of the finite block polynomial make that exact order visible without passing to formal power series.

**Proposition 8.8** (Block-polynomial derivatives at one). *For every  $b, r \geq 0$ ,*

$$\left. \frac{d^r}{dz^r} \sum_{h < 2^b} \varepsilon_h z^h \right|_{z=1} = r! \sum_{h < 2^b} \varepsilon_h \binom{h}{r}. \quad (328)$$

Consequently this derivative is zero when  $r < b$ , while the first surviving derivative is

$$\left. \frac{d^b}{dz^b} \prod_{j=0}^{b-1} (1 - z^{2^j}) \right|_{z=1} = (-1)^b b! 2^{\binom{b}{2}}. \quad (329)$$

*Formal proof.* The derivative–binomial dictionary is `iterate_derivative_thueMorseBlockPolynomial_eval_one`; its strict vanishing and first surviving cases are respectively `iterate_derivative_thueMorseBlockPolynomial_eval_one_of_lt` and `iterate_derivative_thueMorseBlockPolynomial_eval_one_self`, all in `ThueMorseBlockAlgebra.lean`.  $\square$

**Proposition 8.9** (The first surviving Thue–Morse moment). *For every  $k \geq 0$ ,*

$$\sum_{r=0}^{2^k-1} \varepsilon_r r^k = (-1)^k 2^{\binom{k}{2}} k!. \quad (330)$$

*Formal proof.* This is `thueMorsePowerSum_self` in `ThueMorsePrefix.lean`. Together with `thueMorsePowerSum_eq_zero_of_lt`, it is also the coefficient calculation behind the exact-order theorem above.  $\square$

**Corollary 8.10** (Translation invariance). *For every  $k \geq 0$  and  $Q \in \mathbb{Q}$ ,*

$$\sum_{r=0}^{2^k-1} \varepsilon_r (r + Q)^k = (-1)^k 2^{\binom{k}{2}} k!. \quad (331)$$

*Formal proof.* Apply `thueMorse_affine_power_sum_self` with  $X = Q$  and  $Y = 1$ .  $\square$

**Proposition 8.11** (Centered reflection and the characteristic-two boundary). *Let  $R$  be a commutative ring, let  $c \in R$  satisfy  $c + c = 2^m - 1$ , and put*

$$S_{m,r}(c) = \sum_{n=0}^{2^m-1} \varepsilon_n (n - c)^r \in R. \quad (332)$$

Then

$$S_{m,r}(c) = (-1)^{m+r} S_{m,r}(c). \quad (333)$$

Consequently, if  $m + r$  is odd, then  $2S_{m,r}(c) = 0$ . This is the characteristic-free conclusion: in a ring with 2-torsion one cannot in general cancel the factor 2. Over  $\mathbb{Q}$ , taking  $c = (2^m - 1)/2$  gives the familiar parity rule

$$\sum_{n=0}^{2^m-1} \varepsilon_n \left( n - \frac{2^m - 1}{2} \right)^r = 0 \quad (m + r \text{ odd}). \quad (334)$$

*Formal proof.* The fixed-point identity is `sum_thueMorseSign_mul_centered_pow_reflect`, and its universal 2-torsion consequence is `two_mul_sum_thueMorseSign_mul_centered_pow_eq_zero`. The rational specialization, including the legitimate cancellation of  $2 \neq 0$ , is `sum_thueMorseSign_mul_midpoint_pow_eq_zero`. All three declarations are in `ThueMorseMents.lean`.  $\square$

The centered specialization  $Q = -2^k$  is therefore

$$\sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k)^k = (-1)^k 2^{\binom{k}{2}} k!, \quad (335)$$

the exact declaration `thueMorseCenteredPowerSum_self`.

### 8.3 The exact leading-bit recurrence

Let  $e, a \in \mathbb{N}$  with  $0 < a \leq 2^e$ , and write

$$b = \lfloor \log_2 a \rfloor, \quad r = e - b, \quad q = a - 2^b, \quad y = \frac{q}{2^e}. \quad (336)$$

Thus  $0 \leq q < a$ : the leading binary power has been removed. Put  $V_m = F(2^{-m})$ , whose exact half-moment form is (309), and define

$$T_r(y) = \sum_{j=0}^r \frac{2^{\binom{j+1}{2}} V_{r-j}}{j!} y^j. \quad (337)$$

Then the exact rational recurrence is

$$\boxed{F\left(\frac{a}{2^e}\right) = T_r\left(\frac{q}{2^e}\right) - F\left(\frac{q}{2^e}\right)}. \quad (338)$$

The formal route stays entirely inside the rational dyadic model: `fabiusTaylorHorner_eq_sum` identifies the Horner routine with (337); `fabiusInversePowTwoTable_hasBlockTaylor` proves the block identity; `blockTaylor_implies_bitRecurrence` turns it into the leading-bit recurrence; and `fabiusInversePowTwoTable_hasBitRecurrence` instantiates that recurrence for the precomputed inverse-power table. Finally, `fabiusDyadicUnit_eq_fabiusDyadic` and `fabiusDyadic_cast` identify the executable rational result with the analytic function. These declarations are in `DyadicClosedForm.lean` and `DyadicAnalytic.lean`.

### 8.4 The bit-recursive evaluator

Use  $b, r, q, y$ , and  $T_r$  from (336)–(337). Equation (338) reduces the numerator from  $a$  to the strictly smaller  $q$ .

**Algorithm 8.12** (Exact dyadic evaluator). Precompute  $V_j = F(2^{-j})$  for  $0 \leq j \leq e$  from (309). Define  $\text{Eval}(e, a)$  recursively by

$$\text{Eval}(e, a) = \begin{cases} 0, & a \leq 0, \\ 1, & a \geq 2^e, \\ T_r(q/2^e) - \text{Eval}(e, q), & 0 < a < 2^e, \end{cases}$$

where  $b, r, q$  are given by (336); evaluate  $T_r$  by Horner's rule.

A literal pseudocode version is:

```

Eval(e, a, V):
  if a <= 0:      return 0
  if a >= 2^e:    return 1

  b = floor_log2(a)
  r = e - b
  q = a - 2^b
  y = q / 2^e

  p = V[0]                // highest Taylor coefficient, j = r
  for m = 1 .. r:
    j = r - m
    p = V[m] + (2^(j+1) * y / (j+1)) * p

  return p - Eval(e, q, V)

```

The displayed loop is one convenient Horner ordering; any exact evaluation of (337) is equivalent. In the formal source this evaluator is `fabiusDyadicValue`.

**Theorem 8.13** (Correctness and termination). *For every  $e \in \mathbb{N}$  and  $a \in \mathbb{Z}$ ,*

$$\text{Eval}(e, a) = F(a/2^e) \tag{339}$$

*under the bounded convention  $F = 0$  to the left and  $F = 1$  to the right.*

*Formal proof.* The pseudocode is the rational routine `fabiusDyadicValue`, whose analytic correctness is `fabiusDyadicValue_cast` in `PaperStatements.lean`. The interior recursion is `fabiusInversePowTwoTable_hasBitRecurrence`; the corresponding total recursive correctness theorem is `fabiusDyadicUnitAux_eq_of_bitRecurrence`.  $\square$

### 8.5 Reduction of an arbitrary dyadic argument

Algorithm 8.12 evaluates the bounded function  $F$  on  $[0, 1]$ . A total evaluator for the signed global extension first folds the argument back into the unit interval and records the associated sign. The fold is a reduction modulo 2 followed by a reflection.

**Theorem 8.14** (Triangular reduction). *Let  $n \in \mathbb{N}$  and put  $s = 2^n$ . Let  $a \in \mathbb{N}$  and write*

$$a = 2sB + R, \quad B = \left\lfloor \frac{a}{2s} \right\rfloor, \quad 0 \leq R < 2s, \tag{340}$$

and set

$$C = \begin{cases} R, & R \leq s, \\ 2s - R, & R > s. \end{cases} \quad (341)$$

Then  $0 \leq C \leq s$  and

$$\boxed{\mathcal{F}\left(\frac{a}{2^n}\right) = \varepsilon_B F\left(\frac{C}{2^n}\right)}. \quad (342)$$

For  $a \in \mathbb{Z}$  with  $a < 0$  one has  $\mathcal{F}(a/2^n) = 0$ .

*Proof.* Write  $x = a/2^n = 2B + R/s$ . Since  $0 \leq R < 2s$  we have  $2B \leq x < 2B + 2$ , so  $x$  lies in the  $B$ th block and (12) gives

$$\mathcal{F}(x) = \varepsilon_B \text{up}(x - 2B - 1) = \varepsilon_B \text{up}\left(\frac{R}{s} - 1\right).$$

Put  $t = R/s - 1$ , so that  $-1 \leq t < 1$ . The fold relation (7) says  $\text{up}(t) = F(t + 1)$  for  $t \leq 0$  and  $\text{up}(t) = F(1 - t)$  for  $t > 0$ ; on  $[-1, 1]$  both clauses read  $\text{up}(t) = F(1 - |t|)$ . If  $R \leq s$  then  $|t| = 1 - R/s$  and  $F(1 - |t|) = F(R/s)$ . If  $R > s$  then  $|t| = R/s - 1$  and  $F(1 - |t|) = F(2 - R/s) = F((2s - R)/2^n)$ . In both cases the argument is  $C/2^n$  with  $C$  as in (341), and  $0 \leq C \leq s$  is immediate from  $0 \leq R < 2s$ . The final statement is the vanishing of  $\mathcal{F}$  on the negative half-line.  $\square$

Combining (342) with Algorithm 8.12 gives a total exact evaluator for  $\mathcal{F}$  at every dyadic rational: reduce with (340)–(341), compute the block sign  $\varepsilon_B = (-1)^{\text{wt}_2(B)}$  from the binary digits of  $B = \lfloor a/2^{n+1} \rfloor$ , and call  $\text{Eval}(n, C)$ .

The reduction is compatible with unreduced presentations. Replacing  $(n, a)$  by  $(n + 1, 2a)$  doubles  $s$  and  $R$ , leaves  $B$  unchanged, and doubles  $C$ ; the right side of (342) is therefore unchanged, as it must be.

*Remark 8.15* (Recognizing dyadic arguments exactly). A reduced rational  $p/q$  with  $q > 0$  is dyadic exactly when  $q$  is a power of two, and in that case  $n = \log_2 q$  and  $a = p$  are read off directly. An exact rational interface to  $F$ ,  $\text{up}$ , and  $\mathcal{F}$  therefore tests a single integer condition on the reduced denominator, returns “not dyadic” when it fails, and otherwise invokes the signed integer evaluator above. No floating-point recognition of powers of two is involved anywhere, so the interface is exact on every input and cannot be defeated by rounding in the recognition step.

## 8.6 Denominator bounds

Let

$$M_m = \prod_{j=1}^m (2^{2^j} - 1), \quad (2m + 1)!! = 1 \cdot 3 \cdots (2m + 1). \quad (343)$$

**Theorem 8.16** (Uniform denominator multiplier). *Let  $n \geq 1$  and  $a \in \mathbb{Z}$  with  $|a| < 2^n$ . Then*

$$\text{up}(a/2^n) \, n! \, 2^{\binom{n+1}{2}} (2 \lfloor n/2 \rfloor + 1)!! \, M_{\lfloor n/2 \rfloor} \quad (344)$$

*is an integer.*

*Formal proof.* This is `rvachevDyadic_mul_denominatorBound_isNatural_all` in `DenominatorBound.lean`. Its proof uses `moment_mul_common_denominator` for the moments and `fabiusDyadic_mul_denominatorBound_isInteger` for the bounded dyadic values.  $\square$

Let  $\Delta_n$  be the least common multiple of the reduced denominators of the values  $\text{up}(a/2^n)$  with odd  $a$  in the support. The theorem gives the explicit divisibility bound

$$\Delta_n \mid n! \, 2^{\binom{n+1}{2}} (2 \lfloor n/2 \rfloor + 1)!! \, M_{\lfloor n/2 \rfloor}. \quad (345)$$

A complementary lower statement is also formalized.

**Proposition 8.17** (The reciprocal power is one of the samples). *For every  $n \geq 1$  the reduced denominator of  $F(2^{-n})$  divides  $\Delta_n$ . Consequently*

$$v_2(\Delta_n) \geq \binom{n}{2} + 1 + v_2(n!). \quad (346)$$

*Formal proof.* The denominator divisibility is `proposition_fifteen` in `PaperStatements.lean`, after the bridge `dyadicDenominator_eq_fabiusDyadicDenominator`. The exact two-adic exponent is `fabiusAtInverseTwoPow_padicVal_two_closed`.  $\square$

Together, (345) and (346) give the formally proved denominator information used in this article. Proposed exact factorizations of the least common denominator, together with the associated numerical tables, are kept in the consolidated research-frontier volume until their individual statements are formalized.

### 8.7 The exact half-base $q$ -binomial layer

The dyadic dilation forces the specialization  $q = 1/2$ . For  $a, q \in \mathbb{Q}$ , define

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (347)$$

and write  $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$  for the Gaussian coefficient at that fixed base. For  $k \leq n$ , its exact Mersenne normalization is

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = \frac{\mathfrak{m}_n}{\mathfrak{m}_k \mathfrak{m}_{n-k} 2^{k(n-k)}}, \quad \mathfrak{m}_j = \prod_{\ell=1}^j (2^\ell - 1). \quad (348)$$

In particular the row is symmetric under  $k \mapsto n - k$ .

**Theorem 8.18** (Finite half-base  $q$ -binomial theorem). *For every  $n \in \mathbb{N}$  and  $z \in \mathbb{Q}$ ,*

$$(z; 1/2)_n = \sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k. \quad (349)$$

*At every dyadic node this gives the single closed evaluation*

$$\sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} (2^m)^k = \begin{cases} 0, & m < n, \\ (-1)^n \mathfrak{m}_m / \mathfrak{m}_{m-n}, & n \leq m. \end{cases} \quad (350)$$

*In particular, the sum vanishes if and only if  $m < n$ . Moreover,*

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n. \quad (351)$$

*Formal proof.* These are `halfQBinomial_theorem`, `qBinomial_half_two_pow_sum_eq_ite`, `qBinomial_half_two_pow_sum_eq_zero_iff`, `halfQPochhammer_eq_mersenne_div`, `halfQBinomial_eq_mersenne`, and `halfQBinomial_symm` in `HalfQBinomial.lean`.  $\square$

**Remark 8.19** (The denominator-free finite  $q$ -binomial theorem). The half-base identity is a specialization of a ring-valued statement that does not divide by a  $q$ -Pochhammer symbol. For a commutative ring  $R$  and  $q \in R$ , define the polynomial Gaussian coefficients recursively by

$$(z; q)_0 = 1, \quad (z; q)_{n+1} = (z; q)_n (1 - zq^n),$$



and

$$\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}_q = \begin{bmatrix} n \\ k+1 \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_q, \quad \begin{bmatrix} n \\ 0 \end{bmatrix}_q = \begin{bmatrix} n \\ n \end{bmatrix}_q = 1, \quad \begin{bmatrix} n \\ k \end{bmatrix}_q = 0 \quad (n < k).$$

The recursive triangle is already defined over every semiring and is functorial: for semirings  $R, S$  and every homomorphism  $\varphi : R \rightarrow S$ ,

$$\varphi \left( \begin{bmatrix} n \\ k \end{bmatrix}_q \right) = \begin{bmatrix} n \\ k \end{bmatrix}_{\varphi(q)}. \quad (352)$$

Over a commutative semiring it also satisfies, for all natural  $n, k$ , the second Pascal orientation

$$\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}_q = q^{k+1} \begin{bmatrix} n \\ k+1 \end{bmatrix}_q + \begin{bmatrix} n \\ k \end{bmatrix}_q, \quad (353)$$

and, whenever  $k \leq n$ , the reflection law

$$\begin{bmatrix} n \\ n-k \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q. \quad (354)$$

None of these structural laws uses division or cancellation; the zero extension above the diagonal is what makes (353) valid without an index side condition.

Then for every  $z \in R$ ,

$$(z; q)_n = \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \quad (355)$$

The recursive coefficient already makes sense over an arbitrary semiring, while the displayed subtraction requires a commutative ring. No denominator, nonzero assumption, or exclusion of  $q = 0$  and roots of unity occurs. The recurrence, triangular boundaries, and (355) are the definition `finiteQPochhammerIn`, its boundary and recurrence `finiteQPochhammerIn_zero` and `finiteQPochhammerIn_succ`, the coefficient definition `gaussianBinomial`, and the theorems `gaussianBinomial_zero_zero`, `gaussianBinomial_zero_succ`, `gaussianBinomial_succ_succ`, `gaussianBinomial_zero_right`, `gaussianBinomial_eq_zero_of_lt`, `gaussianBinomial_self`, and `finite_qBinomial_theorem` in `FiniteQBinomialCore.lean`. The functoriality, alternate Pascal, and symmetry statements (352)–(354) are `map_gaussianBinomial`, `gaussianBinomial_succ_succ_alt`, and `gaussianBinomial_symm`, respectively, in the same module.

Four denominator-free product identities make this recursion especially useful downstream. For natural  $m, n$ , and for  $k \leq n$ ,

$$(a; q)_{n+1} = (1-a)(aq; q)_n, \quad (356)$$

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n, \quad (357)$$

$$(q; q)_k \begin{bmatrix} n \\ k \end{bmatrix}_q = (q^{n-k+1}; q)_k, \quad (358)$$

$$(q; q)_n = (q; q)_k (q; q)_{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_q. \quad (359)$$

They are `finiteQPochhammerIn_succ_shift`, `finiteQPochhammerIn_add`, `finiteQPochhammerIn_self_mul_gaussianBinomial`, and `finiteQPochhammerIn_self_eq_mul_mul_gaussianBinomial`, respectively, again in `FiniteQBinomialCore.lean`. All four hold over every commutative ring, including empty products, zero divisors, positive characteristic,  $q = 0$ , and

roots of unity. The final identity is the full denominator-free replacement for the usual quotient definition of a Gaussian coefficient.

Over the rationals, `finiteQPochhammerIn_rat_eq` identifies the generic product with the established finite  $q$ -Pochhammer notation. At  $q = 1/2$ , `gaussianBinomial_half_eq_halfQBinomial` identifies the recursive coefficient with the established rational quotient, and `halfQBinomial_theorem` is now the direct specialization of the ring theorem.

**Theorem 8.20** (Denominator-free  $q$ -Vandermonde and shifted central forms). *Let  $R$  be a commutative semiring, let  $q \in R$ , and let all displayed indices be natural numbers. The two zero-extended Vandermonde orientations are*

$$\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_{j=0}^r q^{(m-j)(r-j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q, \quad (360)$$

$$= \sum_{j=0}^r q^{j(n-(r-j))} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q. \quad (361)$$

Here subtraction is natural subtraction; outside the admissible support one of the zero-extended Gaussian coefficients vanishes. Consequently no condition  $r \leq m+n$  is needed. The central specialization is

$$\begin{bmatrix} m+n \\ m \end{bmatrix}_q = \sum_{j=0}^m q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q. \quad (362)$$

Equivalently, removing exactly the zero-extended terms gives the literal Durfee support

$$\begin{bmatrix} m+n \\ m \end{bmatrix}_q = \sum_{j=0}^{\min(m,n)} q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q. \quad (363)$$

For every  $N, k \in \mathbb{N}$ , including  $N < k$ , the positive shifted form is

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \sum_{\ell=0}^N q^{\ell(\ell+k)} \begin{bmatrix} N \\ \ell \end{bmatrix}_q \begin{bmatrix} N \\ \ell+k \end{bmatrix}_q. \quad (364)$$

If  $k \leq N$ , row reflection gives the same right side for  $\begin{bmatrix} 2N \\ N-k \end{bmatrix}_q$ , and its literal-support form is

$$\begin{bmatrix} 2N \\ N-k \end{bmatrix}_q = \sum_{\ell=k}^N q^{\ell(\ell-k)} \begin{bmatrix} N \\ \ell \end{bmatrix}_q \begin{bmatrix} N \\ \ell-k \end{bmatrix}_q. \quad (365)$$

No identity in this theorem uses division, cancellation, a nonvanishing hypothesis, or an exclusion of  $q = 0$ , roots of unity, positive characteristic, or zero divisors.

*Formal proof.* Equations (360) and (361) are `gaussianBinomial_add_vandermonde` and `gaussianBinomial_add_vandermonde'`. The full-range and literal- $\min(m, n)$  central identities are `gaussianBinomial_add_central` and `gaussianBinomial_add_central_min`; the positive shift is `gaussianBinomial_two_mul_add_shifted_central`; and the two negative-shift presentations are `gaussianBinomial_two_mul_sub_shifted_central` and `gaussianBinomial_two_mul_sub_shifted_central ICC`. These seven declarations exhaust the public surface of `QBinomialVandermonde.lean`. The positive shift is total in  $k$ ; the last two declarations assume exactly  $k \leq N$ .  $\square$

**Theorem 8.21** (Reversed Gaussian row, Lagrange weights, and all residual moments). *Let  $R$  be a commutative ring and  $q \in R$ . The zero-extended numerator row is*

$$N_{n,k}(q) = \begin{cases} (-1)^{n-k} q^{\binom{n-k+1}{2}} [n]_q, & k \leq n, \\ 0, & n < k. \end{cases} \quad (366)$$

*The Lean definition is initially written with the reversed index  $n - k$ , but the division-free symmetry (354) proves the forward form displayed here. In fact the zero-extended identity remains valid for every pair  $n, k$  when natural subtraction is read with Lean’s truncation convention. For every  $z \in R$ ,*

$$\sum_{k=0}^n N_{n,k}(q) z^k = \prod_{j=0}^{n-1} (z - q^{j+1}), \quad (367)$$

*and, for every  $0 \leq d \leq n$ ,*

$$\sum_{k=0}^n N_{n,k}(q) (q^k)^d = (q; q)_n 0^d. \quad (368)$$

*Now let  $R$  be a field and suppose that the nodes  $1, q, \dots, q^n$  are distinct. If  $\lambda_{n,k}^{(q)}$  is the evaluation-at-zero Lagrange weight belonging to  $q^k$ , then  $(q; q)_n \neq 0$  and, for  $k \leq n$ ,*

$$\lambda_{n,k}^{(q)} = \frac{N_{n,k}(q)}{(q; q)_n}, \quad (q; q)_n \lambda_{n,k}^{(q)} = N_{n,k}(q). \quad (369)$$

*Finally every positive residual moment has the single closed form*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} (q^k)^d = (-1)^n q^{n(n+1)/2} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q, \quad d > 0. \quad (370)$$

*All statements include  $n = 0$ . The field-level formulas impose only distinctness of the displayed finite node set; they do not add a redundant global ban on  $q = 0$  or on roots of unity.*

*Formal proof.* The numerator and its two branches are `geometricQBinomialWeightNumerator`, `geometricQBinomialWeightNumerator_eq_of_le`, and `geometricQBinomialWeightNumerator_eq_zero_of_lt`. Its forward-index forms on the natural range and globally are `geometricQBinomialWeightNumerator_eq_forward_of_le` and `geometricQBinomialWeightNumerator_eq_forward`. Equations (367) and (368) are `reversed_finite_qBinomial_theorem` and `sum_geometricQBinomialWeightNumerator_mul_pow`. Node distinctness gives the nonvanishing denominator by `finiteQPochhammerIn_self_ne_zero_of_injOn`; the quotient and cleared-denominator identities in (369) are `geometricLagrangeWeight_eq_gaussianBinomial_div` and `finiteQPochhammerIn_mul_geometricLagrangeWeight_eq`. The complete positive-degree formula (370) is `sum_geometricLagrangeWeight_mul_pow_eq_gaussianBinomial`. These declarations are all in `GeometricQBinomialLagrange.lean`; the finite-product identities in the preceding remark are in `FiniteQBinomialCore.lean`.  $\square$

Here  $q$  is the Gaussian base and the ratio of the geometric nodes; it is unrelated to the additive translation  $Q$  in the translated Fabius formulas below. Because Theorem 8.21 is base-generic, the explicit Gaussian–Lagrange identification and its residual moments are no longer research-frontier obligations at Gaussian base  $q = 1/4$ . Over rational  $0 < q < 1$ , `GeometricLagrangeQBinomial.lean` and `GeometricLagrangeQMoments.lean` additionally express the weights and all residual moments in the established quotient-defined  $q$ -Pochhammer and  $q$ -binomial notation, prove the residual sign, and compute the exact finite  $\ell^1$  norm of the row. Their dedicated quarter-base corollaries include `quarterGeometricLagrangeQMoment_eq_residual_qBinomial`

1 and `sum_abs_quarterGeometricLagrangeWeight_eq_qPochhammer_ratio`. The generic unconditionally summable geometric filter and its exact Gaussian tsum tail are proved in [Proposition 10.9](#). The finite row variation, with  $w_q(n, j)$  the  $q$ -Toeplitz weight `qToeplitzWeight` from `GeneralQConditionNumber.lean`, also has its exact infinite limit: for every  $0 \leq q < 1$ ,

$$\sum_{j \leq n} |w_q(n, j)| \longrightarrow \frac{(-q; q)_\infty}{(q; q)_\infty}.$$

This is `tendsto_sum_abs_qToeplitzWeight` in `LimitConditionNumber.lean`; the same module proves positivity of the denominator on the full range and the effective lower bound  $1/(1 - q) \leq (-q; q)_\infty / (q; q)_\infty$ , and promotes that bound to the literal divergence theorem `tendsto_qConditionNumberLimit_atTop_at_one_left`. The actual infinite Rvachev sinc-product instantiation, at the quarter base and at a generic complex squared base under finite-node injectivity, is now the exact theorem [theorem 10.10](#). What remains in the research-frontier volume is the different finite-prefix problem for  $P_{b,n}$ : its quotient and Bell-coefficient package, the analytic sign and remainder/bracketing bounds, uniform accelerated and derivative convergence, and asymptotics.

The polynomial in (349) has the complete rational root locus

$$\sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} z^k = 0 \iff \exists j < n \text{ such that } z = 2^j, \quad z \in \mathbb{Q}. \quad (371)$$

This is `qBinomial_half_sum_eq_zero_iff`. The same statement is stable under scalar extension at embedded rational arguments. Let  $K$  be a field with a  $\mathbb{Q}$ -algebra structure, and write  $\iota : \mathbb{Q} \rightarrow K$  for its canonical map. Then, for every  $z \in \mathbb{Q}$ ,

$$\sum_{k=0}^n \iota \left( (-1)^k 2^{-\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} \right) (\iota z)^k = 0 \iff \exists j < n \text{ such that } \iota z = 2^j \text{ in } K. \quad (372)$$

This is `qBinomial_half_sum_algebraMap_eq_zero_iff` in `HalfQBinomial.lean`. The scalar statement classifies the embedded rational arguments  $\iota z$ ; it does not assert a root classification for arbitrary elements of  $K$ .

The same Gaussian coefficients form a geometric Prouhet functional. Set

$$w_{n,k} = (-1)^k (1/2)^{\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2}, \quad \xi_k = -2^k.$$

For every rational polynomial  $p$  with  $\deg p \leq n$ ,

$$\sum_{k=0}^n w_{n,k} p(\xi_k) = [X^n] p \, 2^{\binom{n+1}{2}} (1/2; 1/2)_n = [X^n] p \, \mathfrak{m}_n. \quad (373)$$

Thus the functional annihilates every polynomial of degree below  $n$ . At  $n = 0$ , the degree-valued extractor covers all constant polynomials; the strict-degree cancellation statement admits exactly the zero polynomial. The two forms of (373) are `halfQBinomial_negativeDyadic_polynomial_1_sum_eq_coeff` and `halfQBinomial_negativeDyadic_polynomial_sum_eq_mersenne`; strict-degree cancellation is `halfQBinomial_negativeDyadic_polynomial_sum_eq_zero_of_degree_lt`, all in `HalfQBinomial.lean`.

The inner cancellation in the dyadic formula is the exact Thue–Morse block identity

$$\sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} z^r = \prod_{j=0}^{k-1} (1 - z^{2^j}), \quad (374)$$

formalized as `thueMorseBlockPolynomial_eq_product`. Its affine power-sum form annihilates every degree below  $k$ , while its first surviving moment is  $(-1)^k 2^{\binom{k}{2}} k!$ ; see `thueMorse_affine_power_sum_of_1e` and `thueMorse_affine_power_sum_self`.

For comparison with the moment recurrence, put

$$a_n = \frac{d_n}{n!}, \quad \mathcal{A}(X) = \sum_{n \geq 0} a_n X^n, \quad E(X) = \frac{e^X - 1}{X}, \quad (375)$$

with the removable value  $E(0) = 1$ . The refinement is

$$\mathcal{A}(2X) = E(X) \mathcal{A}(X), \quad (376)$$

and its coefficient recurrence is

$$(2^n - 1)a_n = \sum_{j=0}^{n-1} \frac{a_j}{(n-j+1)!} \quad (n \geq 1), \quad a_0 = 1. \quad (377)$$

The functional identity follows from `proposition_two_formula` and `complexGeneratingFunction_eq_series_formula` in `AnalyticMoments.lean`; the coefficient recurrence is `halfMoment_original_recurrence` in `BernoulliRecurrences.lean`. These are the formal-series bridge used by the specialized dyadic theorem.

For  $m, k, d \in \mathbb{N}$  and  $Q \in \mathbb{C}$ , write

$$\sigma_{m;k,d}(Q) = \sum_{r=0}^{m2^k-1} (-1)^{\text{wt}_2(r)} (r - m2^k + Q)^d. \quad (378)$$

The exact result is the following finite identity.

**Theorem 8.22** (Translated global dyadic formula). *For all  $m, n \in \mathbb{N}$  and  $Q \in \mathbb{C}$ ,*

$$\begin{aligned} \mathcal{F}(m/2^n) &= \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \\ &\quad \times \sum_{r=0}^{m2^k-1} (-1)^{\text{wt}_2(r)} (r - m2^k + Q)^{n+k}. \end{aligned} \quad (379)$$

*The real number on the left is embedded in  $\mathbb{C}$ . The entire right-hand side is independent of the common translation  $Q$ . When  $m \leq 2^n$ , the left side is the bounded value  $F(m/2^n)$ .*

*Formal proof.* The complete normalized finite expression is packaged as a polynomial in the translation and proved equal to the constant centered formula by `qBinomialThueMorseDyadicTranslatedFormulaPolynomial_eq_const`. Its identification with the signed dyadic value over  $\mathbb{C}$  is `globalFabius_dyadic_eq_qBinomialThueMorseDyadic_translated_sum_complex`; the rational core and real, complex, and Gaussian-rational specializations are in `FabiusQBinomialFormula.lean` and `FabiusDyadicQBinomialScalar.lean`.  $\square$

The rational specialization is also independent of how the dyadic argument is written. If

$$\frac{m_1}{2^{n_1}} = \frac{m_2}{2^{n_2}},$$

then the two translated rational formulas agree even when their common translations  $Q_1, Q_2 \in \mathbb{Q}$  differ. This is `qBinomialThueMorseDyadicTranslatedFormula_eq_of_rat_eq`; the one-step refinement  $(m, n) \mapsto (2m, n+1)$  is `qBinomialThueMorseDyadicTranslatedFormula_refine`.

Over every field over  $\mathbb{Q}$ , both conclusions persist even when the scalar translations and the dyadic presentations are changed simultaneously; these stronger wrappers are `qBinomialThueMorseDyadicTranslatedFormulaIn_eq_of_rat_eq` and `qBinomialThueMorseDyadicTranslatedFormulaIn_refine` in `FabiusDyadicQBinomialScalar.lean`.

Binary complementation gives the raw inverse-dyadic form

$$\mathcal{F}(2^{-n}) = \frac{1}{(-2)^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r+Q)^{n+k}. \quad (380)$$

Its translated numerator is constant by `qBinomialThueMorseRawTranslatedNumeratorPolynomial_eq_const`. The algebraic statement is in fact valid over every field  $K$  equipped with an  $\mathbb{Q}$ -algebra structure and every scalar translation  $Q \in K$ : the fully expanded right-hand side equals the canonical inverse-dyadic rational value after its scalar embedding by `fabiusAtInverseTwoPow_cast_eq_qBinomialThueMorse_rawTranslated_sum`. Translation independence and the intermediate value identification are `qBinomialThueMorseRawTranslatedFormulaIn_eq_centered` and `qBinomialThueMorseRawTranslatedFormulaIn_eq_fabiusAtInverseTwoPow`. For every scalar type  $K$  carrying Mathlib’s `RCLike` structure, the corresponding theorem for an arbitrary bounded Fabius solution is `fabiusFunction_inverse_two_pow_eq_qBinomialThueMorseRawTranslatedFormulaIn`. All five declarations are in `FabiusRawQBinomialScalar.lean`; the final one uses the `RCLike` hypothesis only to embed the real Fabius value. Variable-base  $q$ -interpolation formulas, barycentric identities, numerical cancellation studies, and complexity comparisons are maintained in the consolidated research-frontier volume rather than used as theorems here.

## 9 The global binary-reduction telescope

The Taylor-block recursion also applies to the signed global extension  $\mathcal{F}$ . It can therefore be iterated along the binary expansion of an arbitrary nonnegative real number, rather than only along the finite expansion of a dyadic rational. The reductions need not terminate, but their residuals tend geometrically to zero.

Define the finite reduction polynomial

$$\mathcal{T}_m(y) = \sum_{j=0}^m 2^{\binom{j+1}{2} - \binom{m-j}{2}} \frac{d_{m-j}}{(m-j)!} \frac{y^j}{j!}. \quad (381)$$

All powers of 2 here are integer powers. In particular, negative exponents are allowed, and  $\mathcal{T}_0(y) = 1$ . The signed Taylor-block identity is

$$\mathcal{F}(2^{-m} + y) = -\mathcal{F}(y) + \mathcal{T}_m(y), \quad 0 \leq y < 2^{-m}. \quad (382)$$

Fix  $x \geq 0$ . At scale  $m$ , put

$$p_m = \lfloor 2^m x \rfloor, \quad y_m = x - \frac{p_m}{2^m}, \quad 0 \leq y_m < 2^{-m}. \quad (383)$$

The previous prefix requires a genuine half-scale value at  $m = 0$ :

$$p_{m-1}^* = \begin{cases} \lfloor x/2 \rfloor, & m = 0, \\ p_{m-1}, & m \geq 1. \end{cases} \quad (384)$$

Uniformly,  $p_{m-1}^* = \lfloor p_m/2 \rfloor$ . Thus  $p_m = 2p_{m-1}^* + \delta_m$  with  $\delta_m \in \{0, 1\}$ . Define

$$\alpha_m(x) = (-1)^{\text{wt}_2(p_m)} (2p_{m-1}^* - p_m), \quad R_m(x) = (-1)^{\text{wt}_2(p_m)} \mathcal{F}(y_m). \quad (385)$$

The coefficient  $\alpha_m$  is zero or a sign, and it is nonzero precisely when the newly exposed binary digit is 1.

**Lemma 9.1** (One-step residual identity). *For  $m \geq 1$ ,*

$$\alpha_m(x)\mathcal{T}_m(y_m) = R_{m-1}(x) - R_m(x). \quad (386)$$

*Proof.* If  $p_m = 2p_{m-1}$ , then  $y_m = y_{m-1}$ , the Thue–Morse signs agree, and  $\alpha_m = 0$ , so both sides vanish. If  $p_m = 2p_{m-1} + 1$ , then

$$y_{m-1} = 2^{-m} + y_m, \quad (-1)^{\text{wt}_2(p_m)} = -(-1)^{\text{wt}_2(p_{m-1})}, \quad \alpha_m = (-1)^{\text{wt}_2(p_{m-1})}.$$

Apply (382) to  $y_{m-1}$  and multiply by the old prefix sign; the two copies of  $\mathcal{F}(y_m)$  combine exactly into  $R_{m-1} - R_m$ .  $\square$

The scale-zero identity is separate and retains the parity of the integer block:

$$\mathcal{F}(x) = \alpha_0(x)\mathcal{T}_0(y_0) + R_0(x). \quad (387)$$

Indeed, write  $p_0 = 2p_{-1}^* + \delta_0$  and use the two-unit translation law (12). When  $\delta_0 = 0$ , the polynomial term vanishes. When  $\delta_0 = 1$ , the identity  $\mathcal{F}(1 + y_0) = -\mathcal{F}(y_0) + 1$  supplies it.

**Theorem 9.2** (Exact finite binary telescope). *For  $x \geq 0$  and  $N \in \mathbb{N}$ ,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^N \alpha_m(x)\mathcal{T}_m(y_m(x)) + R_N(x).} \quad (388)$$

*Proof.* Start with (387). Summing (386) for  $m = 1, \dots, N$  telescopes all intermediate residuals. The case  $N = 0$  is exactly the scale-zero identity.  $\square$

For  $N \geq 1$ ,  $0 \leq y_N < 2^{-N} \leq 1/2$ . On this interval  $0 \leq \mathcal{F}(y) = F(y) \leq 2y$ , so

$$|R_N(x)| \leq 2^{1-N}. \quad (389)$$

The estimate is independent of  $x$ . Therefore, if  $B_N(x)$  denotes the finite sum in (388), then for every  $x \in \mathbb{R}$  and  $N \geq 1$ ,

$$|B_N(x) - \mathcal{F}(x)| \leq 2^{1-N}. \quad (390)$$

This is `norm_globalBinaryReductionSum_sub_extendedFabius_le`. Lean’s natural-floor convention extends every  $B_N$  by zero on  $x \leq 0$ , where  $\mathcal{F}$  also vanishes, and `globalBinaryReductionSum_tendstoUniformly_extendedFabius` proves that the finite telescopes converge uniformly on the whole line. The one-step identity then gives, for  $m \geq 2$ ,

$$|\alpha_m(x)\mathcal{T}_m(y_m)| \leq |R_{m-1}| + |R_m| \leq 4 \cdot 2^{-(m-1)}. \quad (391)$$

Thus the limiting series is absolutely, not merely conditionally, convergent.

**Theorem 9.3** (Absolutely convergent global binary series). *For every  $x \geq 0$ ,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x)\mathcal{T}_m(y_m(x)).} \quad (392)$$

*On  $0 \leq x \leq 1$ , the sum is equivalently  $F(x)$ .*

Lean’s natural-floor definitions make the telescope genuinely all-real: when  $x \leq 0$ , every binary prefix, summand, residual, and  $\mathcal{F}(x)$  vanishes termwise. Thus the finite identity and the  $m = 0$  infinite series hold for every  $x \in \mathbb{R}$ , as `extendedFabius_eq_globalBinaryReductionSum_add_remainder_all` and `extendedFabius_eq_tsum_globalBinaryReductionSummand_all` in `FabiusBinaryReductionSeries.lean`. The fully expanded parity-power version has the same all-real scope by `globalFabius_eq_tsum_fabiusParityPower_literal_all` in `FabiusParityPowerSeries.lean`. The displayed floor formulas above remain ordinary nonnegative-floor formulas on their stated domain.

The index  $m = 0$  is indispensable globally. For  $0 \leq x < 1$  it vanishes, so a series starting at  $m = 1$  happens to be valid there. At  $x = 1$ , however, the zeroth term is 1 and all later coefficients vanish. More generally it is the zeroth term that distinguishes even and odd integer blocks of the signed extension.

### 9.1 The corrected parity-power presentation

The same summands have a more literal presentation in terms of prefix parity and inverse dyadic values. Define, using integer exponents,

$$\kappa_j = \frac{((j : \mathbb{Z}) + 2)((j : \mathbb{Z}) - 1)}{2} = \binom{j+1}{2} - 1. \quad (393)$$

The integer interpretation matters:  $\kappa_0 = -1$ , hence  $2^{\kappa_0} = 1/2$ ; natural subtraction would silently give the wrong term. Put

$$\mathcal{I}_m(y) = \sum_{j=0}^m 2^{\kappa_j} \mathcal{F}(2^{(j:\mathbb{Z})-(m:\mathbb{Z})}) \frac{y^j}{j!}. \quad (394)$$

**Lemma 9.4.** *For every  $m$  and  $y$ ,*

$$\mathcal{I}_m(y) = \frac{1}{2} \mathcal{T}_m(y). \quad (395)$$

*Proof.* For  $j \leq m$ , set  $r = m - j$ . The inverse-dyadic identity (309) gives

$$\mathcal{F}(2^{j-m}) = \frac{d_{m-j}}{(m-j)! 2^{\binom{m-j}{2}}}.$$

Together with  $\kappa_j = \binom{j+1}{2} - 1$ , this identifies each summand with one half of the corresponding summand in (381).  $\square$

Let  $\tau(a) = \text{wt}_2(a) \bmod 2$ . Since

$$\frac{(-1)^a - 1}{2} = 2 \left\lfloor \frac{a}{2} \right\rfloor - a,$$

the literal summand

$$\begin{aligned} \Lambda_m(x) &:= ((-1)^{\lfloor 2^m x \rfloor} - 1) (-1)^{\tau(\lfloor 2^m x \rfloor)} \\ &\quad \times \sum_{j=0}^m 2^{((j+2)(j-1))/2} \mathcal{F}(2^{j-m}) \frac{(x - \lfloor 2^m x \rfloor / 2^m)^j}{j!} \end{aligned} \quad (396)$$

is exactly  $\alpha_m(x) \mathcal{T}_m(y_m)$ . Both exponents  $((j+2)(j-1))/2$  and  $j - m$  in this display are integer exponents, not truncated natural subtractions.

**Warning 9.5.** In (378) and (379), the summation indices are cast into the ambient characteristic-zero field before the subtraction in  $r - m2^k + Q$  is performed. This is the typing used by the formal theorem; the displayed subtraction is therefore field subtraction, never truncated subtraction on natural numbers.



**Theorem 9.6** (Corrected parity-power series). *For every  $x \geq 0$ ,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \Lambda_m(x)}, \quad (397)$$

*and the series converges absolutely. On the half-open unit interval  $0 \leq x < 1$ , the zeroth summand vanishes and one may start at  $m = 1$ .*

*Proof.* The elementary parity identity and (395) prove  $\Lambda_m = \alpha_m \mathcal{T}_m(y_m)$  term by term. The claim follows from (392). The half-open corollary follows from  $\lfloor x \rfloor = 0$ ; the closed endpoint  $x = 1$  requires the restored zeroth term.  $\square$

## 9.2 The global complex $q$ -binomial series

The finite constant-polynomial identity from section 8.7 can be inserted at every scale of the binary telescope. For  $Q \in \mathbb{C}$ , define

$$\mathcal{N}_n^{(Q)} = \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r - 2^k + Q)^{n+k}. \quad (398)$$

Translation invariance and the inverse-dyadic normalization give

$$\frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}} (1/2; 1/2)_n} = \frac{2^n d_n}{n!}. \quad (399)$$

The normalization differs from the  $2^{n^2}$  in (379): here it produces a Taylor coefficient rather than a function value.

For  $m \in \mathbb{N}$ , put

$$\mathcal{Q}_m^{(Q)}(y) = 2^{-\binom{m+1}{2}} \sum_{n=0}^m \frac{(2^{m+1}y)^{m-n}}{(m-n)!} \frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}} (1/2; 1/2)_n}. \quad (400)$$

**Theorem 9.7** ( $q$ -binomial Taylor identity). *For every  $m$ , every  $y \in \mathbb{R}$ , and every  $Q \in \mathbb{C}$ ,*

$$\mathcal{Q}_m^{(Q)}(y) = \mathcal{T}_m(y), \quad (401)$$

*where the real polynomial on the right is embedded in  $\mathbb{C}$ . In particular, each finite summand is pointwise independent of  $Q$  before any infinite sum is taken.*

*Proof.* Insert (399) into (400) and set  $j = m - n$ . Collecting the powers of 2 gives

$$2^{-\binom{m+1}{2}} 2^{(m+1)j} 2^{m-j} = 2^{\binom{j+1}{2} - \binom{m-j}{2}}.$$

The remaining factorials and moments are exactly those in (381).  $\square$

**Theorem 9.8** (Global complex  $q$ -binomial series). *For fixed  $Q \in \mathbb{C}$  and every  $x \geq 0$ ,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{Q}_m^{(Q)}(y_m(x))}, \quad (402)$$

*with the left side embedded in  $\mathbb{C}$ . The series converges absolutely.*

The displayed floor presentation is stated on the nonnegative half-line. The literal Lean summand has a stronger all-real scope: for every  $x \in \mathbb{R}$ , its series is absolutely summable and sums to the signed global extension, while for  $x \leq 0$  every term and the target vanish. These are `summable_norm_qBinomialFabiusGlobalSummand_all`, `hasSum_qBinomialFabiusGlobalSummand_all`, `extendedFabius_eq_tsum_qBinomialFabiusGlobalSummand_all`, `globalFabius_eq_tsum_qBinomialFabiusGlobalSummand_all`, and `qBinomialFabiusGlobalSummand_eq_zero_of_nonpos` in `FabiusGlobalQBinomialSeries.lean`. The older nonnegative signatures are retained as compatibility wrappers.

This is not a new infinite basic-hypergeometric identity. The outer sum is the analytic binary telescope, while at each scale a finite  $q$ -binomial identity presents the same Taylor polynomial. Expanding one summand makes the architecture explicit:

$$\begin{aligned} \alpha_m(x) 2^{-\binom{m+1}{2}} \sum_{n=0}^m \frac{(2^{m+1}x - 2p_m(x))^{m-n}}{(m-n)!} \frac{1}{2^{\binom{n}{2}}(1/2; 1/2)_n} \\ \times \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r - 2^k + Q)^{n+k}. \end{aligned} \quad (403)$$

Convergence comes entirely from the binary-tail majorant (391); the special-function manipulations are finite and algebraic.

## 10 A complex $q$ -binomial discrete limit

There is a second way in which the same Gaussian row produces  $\mathcal{F}$ . It is a genuine limit of finite shifted splines rather than a sequence of partial sums of (402). Reuse the safe midpoint cutoff  $N_p(x)$  from (94). For  $x \geq 0$  it is equivalent to the inclusive source range

$$0 \leq r \leq \left\lfloor 2^p x - \frac{1}{2} \right\rfloor, \quad (404)$$

including the empty case where the upper endpoint is  $-1$ .

For fixed  $Q \in \mathbb{C}$  and  $x \in \mathbb{R}$ , define

$$\begin{aligned} \mathcal{D}_n^{(Q)}(x) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ &\times \sum_{r=0}^{N_{n+k}(x)-1} (-1)^{\text{wt}_2(r)} (r - 2^{n+k}x + Q)^{n+k}. \end{aligned} \quad (405)$$

**Theorem 10.1** (Complex discrete-limit formula). *For every fixed  $Q \in \mathbb{C}$  and every  $x \in \mathbb{R}$ ,*

$$\boxed{\mathcal{D}_n^{(Q)}(x) \longrightarrow \mathcal{F}(x)} \quad (406)$$

*in  $\mathbb{C}$ . If  $0 \leq x \leq 1$ , the limit is equivalently  $F(x)$ .*

*Proof.* If  $x \leq 0$ , every safe cutoff  $N_p(x)$  is zero. Hence every shifted spline and every approximant (405) is zero, as is  $\mathcal{F}(x)$ . It remains to treat  $x \geq 0$ .

For a degree  $p$  and complex shift  $Q$ , introduce the finite shifted spline

$$S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}}p!} \sum_{r=0}^{N_p(x)-1} (-1)^{\text{wt}_2(r)} (r - 2^p x + Q)^p. \quad (407)$$

At  $Q = 1/2$  this is exactly the centered spline (95), so by (102),

$$S_p^{(1/2)}(x) \longrightarrow \mathcal{F}(x).$$

**Lemma 10.2** (Fixed-cutoff Taylor expansion of a complex shift). *Read (95) at  $p = 0$  as*

$$S_0(x) = \sum_{r=0}^{N_0(x)-1} \varepsilon_r, \quad (408)$$

so that a centered branch  $S_d$  is defined for every integer  $d \geq 0$ . Fix  $p \geq 1$  and  $x \geq 0$ , and abbreviate  $M = N_p(x)$ . For every  $d$  with  $0 \leq d \leq p$ , the degree- $d$  branch carried by that one fixed cutoff satisfies

$$\frac{1}{2^{\binom{d}{2}} d!} \sum_{r=0}^{M-1} \varepsilon_r \left( 2^p x - \frac{1}{2} - r \right)^d = S_d(2^{p-d} x), \quad (409)$$

and in particular  $|S_d(2^{p-d} x)| \leq 1$ . Moreover, for every  $Q \in \mathbb{C}$ ,

$$S_p^{(Q)}(x) - S_p^{(1/2)}(x) = \sum_{d=0}^{p-1} \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}} (p-d)!} \left( \frac{1}{2} - Q \right)^{p-d} S_d(2^{p-d} x). \quad (410)$$

*Proof.* Write the shifted spline in its descending orientation. Since  $(-1)^p (r - 2^p x + Q)^p = (2^p x - Q - r)^p$ , definition (407) reads

$$S_p^{(Q)}(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r=0}^{M-1} \varepsilon_r (2^p x - Q - r)^p, \quad M = N_p(x), \quad (411)$$

and at  $Q = 1/2$  this is (95). The cutoff  $M$  depends on  $p$  and  $x$  only. It is unaffected by  $Q$ , and it is held fixed in what follows while the degree of the summand varies; that is the whole point of (409).

We prove (409). The dilation  $x \mapsto 2^{p-d} x$  compensates exactly for the drop in degree, in the sense that

$$N_d(2^{p-d} x) = \max \left( 0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor \right) = N_p(x) = M \quad (412)$$

by (94). If  $M = 0$ , both sides of (409) are empty sums and there is nothing to prove. Assume  $M \geq 1$ . Every index in the range satisfies

$$r \leq M - 1 = \left\lfloor 2^p x + \frac{1}{2} \right\rfloor - 1 \leq 2^p x - \frac{1}{2},$$

so every base  $2^p x - \frac{1}{2} - r$  occurring in (409) is nonnegative and the truncation in (96) is inactive. For  $d \geq 1$ , formula (96) evaluated at the point  $2^{p-d} x$ , whose own cutoff is  $M$  by (412), is therefore literally the left side of (409). For  $d = 0$  both sides equal  $\sum_{r=0}^{M-1} \varepsilon_r$ , by (408) and (412).

For the uniform bound, the rescaled argument  $2^{p-d} x$  is nonnegative. The all-degree declaration `abs_fabiusUniformSpline_le_one` therefore gives the claim directly, including  $d = 0$ .

Now expand. Put  $w = 2^p x - \frac{1}{2}$  for the descending argument at  $Q = 1/2$  and  $t = \frac{1}{2} - Q$  for its increment, so that  $2^p x - Q = w + t$ . The binomial theorem gives

$$(w + t - r)^p = \sum_{d=0}^p \binom{p}{d} t^{p-d} (w - r)^d.$$

Multiply by  $\varepsilon_r / (2^{\binom{p}{2}} p!)$ , sum over  $0 \leq r \leq M - 1$ , and interchange the two finite sums. Using (411) on the left and (409) on each inner sum,

$$\begin{aligned} S_p^{(Q)}(x) &= \sum_{d=0}^p \frac{t^{p-d}}{2^{\binom{p}{2}} p!} \binom{p}{d} \sum_{r=0}^{M-1} \varepsilon_r (w - r)^d \\ &= \sum_{d=0}^p \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}} (p-d)!} t^{p-d} S_d(2^{p-d} x), \end{aligned} \quad (413)$$

because  $\binom{p}{d}/p! = 1/(d!(p-d)!)$  and the inner sum is  $2^{\binom{d}{2}}d!S_d(2^{p-d}x)$ . The term  $d = p$  is  $S_p(x) = S_p^{(1/2)}(x)$ ; subtracting it leaves (410).  $\square$

*Remark 10.3* (Which convention makes the sign disappear). The expansion (410) is free of alternating signs only because the increment is measured in the *descending* variable  $2^p x - Q$  of (411), that is, because the coefficient carries  $(\frac{1}{2} - Q)^{p-d}$  and not  $(Q - \frac{1}{2})^{p-d}$ . If one instead measures the shift by the ascending increment  $q = Q - \frac{1}{2}$ , which is the natural reading when the branch is written as a function of the ascending argument  $r - 2^p x + Q$  of (407), then the identity acquires exactly the factor  $(-1)^{p+d}$ :

$$S_p^{(Q)}(x) - S_p^{(1/2)}(x) = \sum_{d=0}^{p-1} (-1)^{p+d} \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}}(p-d)!} q^{p-d} S_d(2^{p-d}x), \quad q = Q - \frac{1}{2}. \quad (414)$$

The two displays are the same statement, since  $(-1)^{p+d} = (-1)^{p-d}$  and therefore  $(-1)^{p+d}q^{p-d} = (\frac{1}{2} - Q)^{p-d}$ . It is worth being explicit about where the sign does *not* live: it is not a matter of renormalizing the lower branches. The alternating prefactor  $(-1)^d$  of (95) is cancelled exactly by the  $(-1)^d$  produced when the ascending argument is reversed, so the branch values  $S_d(2^{p-d}x)$  appearing in (410) and in (414) are the same numbers. The sign belongs to the increment alone, and the convention that removes it is the descending one.

*Remark 10.4* (Formal bound). Only the exponential bound (415) is asserted here; it is `norm_fabiusComplexShiftSpline_sub_center_le_half_pow_mul_exp` in `FabiusComplexShiftSpline.lean`. A proposed sharper constant is recorded in the research-frontier gap register pending a corrected named theorem.

Expanding a fixed cutoff branch about  $Q = 1/2$  gives an exact finite Taylor combination of lower-degree centered branches. The normalized Thue–Morse bounds then give, for  $p \geq 1$  and  $x \geq 0$ ,

$$\left| S_p^{(Q)}(x) - S_p^{(1/2)}(x) \right| \leq 2^{-(p-1)} \exp(|Q - \frac{1}{2}|). \quad (415)$$

The nonpositive half-line is handled separately by exact vanishing, and `fabiusComplexShiftSpline_tendsto_globalFabius_all` proves that every fixed complex shift has the same pointwise spline limit for every real  $x$ .

Now reindex the outer sum in (405) by  $j = n - k$ . Direct collection of signs, powers, and factorials gives the exact finite identity

$$\mathcal{D}_n^{(Q)}(x) = \sum_{j=0}^n \omega_{n,j} S_{2n-j}^{(Q)}(x), \quad (416)$$

where

$$\omega_{n,j} = \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_n}. \quad (417)$$

The finite  $q$ -binomial theorem gives the row normalization

$$\sum_{j=0}^n \omega_{n,j} = 1. \quad (418)$$

The same theorem identifies the entire row rather than only its value at one.

**Proposition 10.5** (Toeplitz row polynomial and module-valued  $q$ -Richardson cancellation). *For  $z \in \mathbb{Q}$ , put  $W_n(z) = \sum_{j=0}^n \omega_{n,j} z^j$ . Then*

$$W_n(z) = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}. \quad (419)$$

Consequently its rational zeros are exactly

$$W_n(z) = 0 \iff z \in \{2, 2^2, \dots, 2^n\}. \quad (420)$$

Moreover, let  $M$  be any additive commutative monoid carrying a  $\mathbb{Q}$ -module structure. If an  $M$ -valued sequence has the finite geometric-error model  $u_p = L + \sum_{r=1}^n 2^{-rp} \cdot a_r$ , where  $\cdot$  denotes scalar multiplication, then

$$\sum_{j=0}^n \omega_{n,j} \cdot u_{2n-j} = L. \quad (421)$$

Thus the cancellation needs no multiplication, topology, order, or norm on the target. Taking  $M = \mathbb{Q}$  recovers the rational formula  $u_p = L + \sum_{r=1}^n a_r 2^{-rp}$ . This module-valued extension concerns cancellation only; the root classification above remains a statement for rational  $z$ . All statements include  $n = 0$ : the row polynomial is then 1, the root set is empty, and the error sum is empty.

*Formal proof.* The full generating identity and its root classification are `sum_range_discreteLimitWeight_mul_pow` and `sum_range_discreteLimitWeight_mul_pow_eq_zero_iff`. The reusable single-mode identity is `sum_range_discreteLimitWeight_mul_geometricMode_smul_eq_zero`, and the module-valued cancellation theorem is `discreteLimitWeight_qRichardson_exact_module`; its  $M = \mathbb{Q}$  specialization is `discreteLimitWeight_qRichardson_exact`. All are in `FabijsDiscreteLimitToeplitz.lean`. The former proof of (418) is now its specialization at  $z = 1$ .  $\square$

*Remark 10.6* (Finite moments, Lagrange weights, and local exactness). The row calculation is an instance of two reusable algebraic layers. First, let  $I$  be finite, let  $\varphi : R \rightarrow S$  be a homomorphism from a semiring to a commutative semiring, let  $w_i, x_i \in S$  for  $i \in I$ , and let  $p \in R[X]$  have degree at most  $N$ . If  $p^\varphi$  denotes coefficientwise mapping, then the finite weighted functional expands as

$$\sum_{i \in I} w_i p^\varphi(x_i) = \sum_{d=0}^N \varphi([X^d]p) \sum_{i \in I} w_i x_i^d.$$

Thus prescribed moments select a prescribed coefficient; repeated nodes and a zero surviving moment are allowed, and no field or subtraction is needed. The same-ring coefficient and strict-cancellation forms are `sum_weight_mul_eval_eq_coeff_mul_of_moments` and `sum_weight_mul_eval_eq_zero_of_degree_lt` in `FinitePolynomialFunctional.lean`; that module also contains the scalar-extension forms used by the later Prouhet arguments.

Second, over a field, distinct nodes  $v_i$  have evaluation weights  $\lambda_i(x)$  given by the Lagrange basis at  $x$ . They reproduce every polynomial of degree less than the number of nodes, are characterized uniquely by their monomial moments, and at the first omitted degree satisfy

$$\sum_i \lambda_i(x) v_i^{|I|} = x^{|I|} - \prod_i (x - v_i).$$

These are respectively `sum_lagrangeEvalWeight_mul_eval`, `eq_lagrangeEvalWeight_of_moments`, and `sum_lagrangeEvalWeight_mul_pow_card`. On the distinct geometric nodes  $1, q, \dots, q^n$  at the target 0, the moments through degree  $n$  are  $0^d$ ; in particular degree zero has mass one, while positive degrees vanish. The declarations `sum_geometricLagrangeWeight_mul_pow` and `geometricLagrange_richardson_exact_of_eq` in `GeometricLagrange.lean` include both  $n = 0$  and an arbitrary block start, including start zero.

The first omitted degree is only the first member of an exact all-order family. Let  $h_r(x, (v_i)_{i \in I})$  be the complete homogeneous symmetric polynomial of degree  $r$  in the target  $x$  and the node family. Then

$$\sum_i \lambda_i(x) v_i^{|I|+r} = x^{|I|+r} - \prod_i (x - v_i) h_r(x, (v_i)_{i \in I}) \quad (r \geq 0). \quad (422)$$

More generally, the same formula holds over any commutative ring for arbitrary weights and possibly repeated nodes whenever the moments below  $|I|$  reproduce evaluation at  $x$ . No division or distinctness is used in this representation-independent form. The universal symmetric-polynomial evaluation and its adjoining-variable recurrence are `completeHomogeneousEval_eq_eval_hsymm` and `completeHomogeneousEval_option_succ`; the quotient evaluation is `eval_completeHomogeneousQuotient` in `CompleteHomogeneous.lean`. The companion `CompleteHomogeneousGenerating.lean` packages these evaluations as the formal series `completeHomogeneousGeneratingSeriesOn`. Its coefficient and adjoining-variable laws are `coeff_completeHomogeneousGeneratingSeriesOn` and `completeHomogeneousGeneratingSeriesOn_insert`; over every commutative ring, `prod_one_sub_mul_completeHomogeneousGeneratingSeriesOn` clears the finite linear-denominator product and `completeHomogeneousGeneratingSeriesOn_eq_invOfUnit_prod` identifies the series with its canonical `PowerSeries.invOfUnit` reciprocal. This is a formal coefficient identity and asserts no analytic evaluation, convergence, or Bell/generalized-harmonic conversion. The generic residual theorem `sum_weight_mul_pow_card_add` and its distinct-node Lagrange specialization `sum_lagrangeEvalWeight_mul_pow_card_add` are in `LagrangeResidualMoments.lean`. Thus (422) extends the preceding first-error formula rather than replacing its low-degree exactness foundation.

*Proposition 10.7 (Principal specialization and denominator-free geometric residuals). Let  $R$  first be a commutative semiring. For arbitrary  $c, q \in R$  and  $p, r \in \mathbb{N}$ , complete homogeneous evaluation on a geometric alphabet is*

$$h_r(1, q, \dots, q^p) = \begin{bmatrix} p+r \\ p \end{bmatrix}_q, \quad (423)$$

$$h_r(c, cq, \dots, cq^p) = c^r \begin{bmatrix} p+r \\ p \end{bmatrix}_q. \quad (424)$$

*These identities are recursive and denominator-free: they remain valid at  $q = 0$ , at roots of unity, in positive characteristic, and in the zero semiring.*

*Now let  $R$  be a commutative ring and let  $w_0, \dots, w_p \in R$ . If*

$$\sum_{k=0}^p w_k (q^k)^d = 0^d \quad (0 \leq d \leq p), \quad (425)$$

*then every positive moment is*

$$\sum_{k=0}^p w_k (q^k)^m = (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} m-1 \\ p \end{bmatrix}_q, \quad m > 0. \quad (426)$$

*There is a strictly more general scaled-node form. If exactness is assumed directly on the displayed nodes,*

$$\sum_{k=0}^p w_k (cq^k)^d = 0^d \quad (0 \leq d \leq p), \quad (427)$$

*then*

$$\sum_{k=0}^p w_k (cq^k)^m = c^m (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} m-1 \\ p \end{bmatrix}_q, \quad m > 0. \quad (428)$$

*No nonzeroness or unit hypothesis is imposed on  $c$ ; this direct scaled hypothesis is therefore meaningful even for zero divisors.*

*Over a field, distinctness of  $1, q, \dots, q^p$  supplies (425) for the Lagrange weights  $\lambda_{p,k}^{(q)}$ . Consequently, for every block start  $s \in \mathbb{N}$ ,*

$$\sum_{k=0}^p \lambda_{p,k}^{(q)} (q^{s+k})^m = (-1)^p q^{sm + \binom{p+1}{2}} \begin{bmatrix} m-1 \\ p \end{bmatrix}_q, \quad m > 0. \quad (429)$$

*Formal proof.* The semiring-level infrastructure is `completeHomogeneousEval_smul`, `completeHomogeneousEval_option_zero`, `completeHomogeneousEval_fin_succ`, `completeHomogeneousEvalOn_range`, and `completeHomogeneousEvalAt_zero` in `CompleteHomogeneous.lean`. Splitting off the leading variable 1 and scaling the remaining alphabet by  $q$  turns the head–tail recurrence into  $q$ -Pascal’s recurrence. This proves `completeHomogeneousEval_geometric`; its common-scale and range-indexed forms are `completeHomogeneousEval_scaled_geometric` and `completeHomogeneousEvalOn_range_pow_eq_gaussianBinomial`, all in `GeometricCompleteHomogeneous.lean`.

The offset-degree engines for  $m = p+1+r$  are `sum_weight_mul_geometric_pow_succ_add` and `sum_weight_mul_scaled_geometric_pow_succ_add`. Above-diagonal vanishing of the recursive Gaussian coefficient folds the cancelled degrees and every higher residual into `sum_weight_mul_geometric_pow_of_pos` and `sum_weight_mul_scaled_geometric_pow_of_pos`. Their field-valued Lagrange specializations are `sum_geometricLagrangeWeight_mul_pow_succ_add`, `sum_geometricLagrangeWeight_mul_pow_of_pos`, `sum_geometricLagrangeWeight_mul_scaled_geometric_pow_of_pos`, and `sum_geometricLagrangeWeight_mul_shifted_pow_of_pos` in `GeometricResidualMoments.lean`. The pre-existing pointwise closed weight and cleared-denominator formulas remain in `GeometricQBinomialLagrange.lean`; its positive-moment theorem is now a thin corollary of this denominator-free engine.  $\square$

The same moment calculation acts diagonally on every formal power series, not only on a polynomial of bounded degree. This produces a reusable operator in which low-degree cancellation and every surviving geometric residual are visible coefficient by coefficient.

*Proposition 10.8* (Finite formal-series filters and the Gaussian residual). *Let  $R$  be a commutative semiring, let  $\iota$  be an arbitrary index type, let  $S \subseteq \iota$  be finite, and choose nodes  $x_i \in R$ , weights  $w_i \in R$ , and a formal power series  $f \in R[[X]]$ . Repeated nodes are allowed. Define*

$$\mathcal{T}_{S,x,w}f(X) = \sum_{i \in S} w_i f(x_i X). \quad (430)$$

Then, for every  $m \in \mathbb{N}$ ,

$$[X^m] \mathcal{T}_{S,x,w}f = \left( \sum_{i \in S} w_i x_i^m \right) [X^m]f. \quad (431)$$

The operator commutes with a further rescaling of  $X$ , and with every commutative-semiring homomorphism when the nodes, weights, and coefficients are transported together.

More generally, suppose a target  $t \in R$  and an order  $p \in \mathbb{N}$  satisfy

$$\sum_{i \in S} w_i x_i^d = t^d \quad (0 \leq d \leq p). \quad (432)$$

Degree zero in this hypothesis gives  $\sum_i w_i = 1$ . Hence the filter preserves the constant coefficient, and for every  $m \leq p$ ,

$$[X^m] \mathcal{T}_{S,x,w}f = t^m [X^m]f. \quad (433)$$

At  $t = 0$ , this preserves degree zero and cancels every degree  $1 \leq m \leq p$ , including the boundary case  $p = 0$  without pretending that  $0^0$  vanishes.

Now let  $R$  be a commutative ring, take the nodes  $1, q, \dots, q^p$ , and assume the exactness condition (425). For

$$\mathcal{G}_{q,p,w}f(X) = \sum_{j=0}^p w_j f(q^j X),$$



all coefficients, including the mass-one boundary, are

$$[X^m]\mathcal{G}_{q,p,w}f = \begin{cases} [X^0]f, & m = 0, \\ (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} m-1 \\ p \end{bmatrix}_q [X^m]f, & m > 0. \end{cases} \quad (434)$$

Thus the above-diagonal zero of the recursive Gaussian coefficient gives all cancelled degrees, and the same formula gives every later residual. No division, convergence, distinctness, nonzeroness, or field structure is used. Over a field, distinctness of the geometric nodes makes the Lagrange row a canonical specialization of the same identity.

The generic operator and its diagonal action are `finitePowerSeriesFilter` and `coeff_finitePowerSeriesFilter` in `FinitePowerSeriesFilter.lean`. Its rescaling and scalar-transport laws are `finitePowerSeriesFilter_rescale` and `map_finitePowerSeriesFilter`; exactness, mass one, constant preservation, and low-degree cancellation are `sum_weight_eq_one_of_finite_exact`, `constantCoeff_finitePowerSeriesFilter_of_exact`, `coeff_finitePowerSeriesFilter_of_exact_of_le`, and `coeff_finitePowerSeriesFilter_eq_zero_of_pos_of_le`. The geometric operator is `geometricSeriesFilter`; its positive-degree coefficient theorem and the all-index identity (434) are `coeff_geometricSeriesFilter_of_exact` and `geometricSeriesFilter_eq_residual_mk` in `GeometricPowerSeriesFilter.lean`. The distinct-node Lagrange specializations are `coeff_geometricLagrangeSeriesFilter_of_pos` and `geometricLagrangeSeriesFilter_eq_residual_mk`. Every series in this paragraph is formal: none of these declarations asserts analytic convergence, evaluation at a nonzero argument, or an infinite acceleration formula.

There is now a separate convergent-series companion. It deliberately uses Mathlib’s unconditional summability rather than silently identifying a natural-order boundary sum with a tsum.

*Proposition 10.9* (Unconditionally summable analytic filters and their Gaussian tail). *Let  $K$  be a commutative semiring, let  $E$  be a normed additive commutative group with a  $K$ -module structure, let  $S$  be finite, and choose nodes  $x_i \in K$ , weights  $w_i \in K$ , coefficients  $a_m \in E$ , and a scale  $z \in K$ . Define*

$$\mathcal{A}_{S,x,w}(a; z) := \sum_{i \in S} \sum_{m \geq 0} w_i (x_i z)^m a_m. \quad (435)$$

*Assume for every  $i \in S$  that the displayed weighted inner series is unconditionally summable. Then the finite and infinite sums interchange and*

$$\mathcal{A}_{S,x,w}(a; z) = \sum_{m \geq 0} \left( \sum_{i \in S} w_i x_i^m \right) z^m a_m. \quad (436)$$

*If, moreover,  $\sum_{i \in S} w_i x_i^d = t^d$  for  $0 \leq d \leq p$ , then*

$$\begin{aligned} \mathcal{A}_{S,x,w}(a; z) &= \sum_{d=0}^p (tz)^d a_d \\ &\quad + \sum_{r \geq 0} \left( \sum_{i \in S} w_i x_i^{p+1+r} \right) z^{p+1+r} a_{p+1+r}. \end{aligned} \quad (437)$$

*At target  $t = 0$ , the finite head is exactly  $a_0$ , including  $p = 0$ ; the positive degrees never pass through the  $0^0$  boundary.*

*Now let  $K$  be a commutative ring with continuous constant scalar multiplication on  $E$  (each fixed scalar acts continuously), take  $x_j = q^j$  for  $0 \leq j \leq p$ , and suppose the row is exact at zero through*



degree  $p$ ; write  $\mathcal{A}_{q,p,w}$  for this geometric specialization. It is enough to assume unconditional summability of  $\sum_m (q^j z)^m a_m$  only when  $w_j \neq 0$ . Then

$$\mathcal{A}_{q,p,w}(a; z) = a_0 + \sum_{r \geq 0} (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} p+r \\ p \end{bmatrix}_q z^{p+1+r} a_{p+1+r}. \quad (438)$$

Over a field, distinct geometric nodes supply the Lagrange specialization and its shifted-scale form  $z = q^{\text{start}}$ . A zero-weight node imposes no convergence condition: its weighted series is identically zero.

The definitions are `finiteAnalyticSeriesFilter` and `geometricAnalyticSeriesFilter` in `AnalyticSeriesFilter.lean`. The generic diagonal, exact-head, and zero-target declarations are `summable_finiteAnalyticSeriesFilter_diagonal`, `finiteAnalyticSeriesFilter_eq_tsum`, `finiteAnalyticSeriesFilter_eq_head_add_tail_of_exact`, and `finiteAnalyticSeriesFilter_eq_constant_add_tail_of_exact_zero`. The Gaussian, Lagrange, and shifted-scale tail declarations are `geometricAnalyticSeriesFilter_eq_constant_add_gaussian_tsum`, `geometricLagrangeAnalyticSeriesFilter_eq_constant_add_gaussian_tsum`, and `geometricLagrangeAnalyticSeriesFilter_shifted`. Thus this module has exactly two definitions and seven public theorems. These results exclude series converging only in their natural order at a boundary point. They also assert no formal-to-analytic power-series evaluation bridge, radius or uniform-convergence theorem, norm/sign/error bound, asymptotic estimate, or positivity statement. The actual Rvachev sinc-product instantiation is a separate theorem, so that the generic module retains precisely this abstract scope.

*Theorem 10.10* (Exact Gaussian filter of the Rvachev sinc product). *Let  $c, z \in \mathbb{C}$ , let  $p \in \mathbb{N}$ , put  $q = c^2$ , and assume only that*

$$j \mapsto q^j \quad \text{is injective on } \{0, \dots, p\}.$$

*Writing  $\Phi = \text{rvachevFourierProduct} = \widehat{\text{up}}$ , one has the exact entire identity*

$$\sum_{j=0}^p \lambda_{p,j}^{(q)} \Phi(c^j z) = 1 + \sum_{r=0}^{\infty} (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} p+r \\ p \end{bmatrix}_q A_{p+1+r}(z). \quad (439)$$

*There is no contraction, realness, positivity, or global nonvanishing hypothesis on  $c$ . At  $c = 1/2$ , hence  $q = 1/4$ , the finite node condition is proved internally, so (439) holds for every complex  $z$  and every natural  $p$  with no hypothesis. The same generic formula is exported for the Fourier transform of every bounded  $F$  satisfying *IsFabius*  $F$ .*

*Formal proof.* The named moment series sums to the standalone product by `hasSum_rvachevFourierMomentTerm_product`, and its iterated scaling law is `rvachevFourierMomentTerm_pow_scale`. Passing the finite Lagrange row through that unconditional sum and inserting the Gaussian residual gives `geometricLagrange_rvachevFourierProduct_eq_gaussian_tsum`. Its assumption-free  $c = 1/2$ ,  $q = 1/4$  specialization is `quarterLagrange_rvachevFourierProduct_eq_gaussian_tsum`; the model-generic Fourier-transform form is `geometricLagrange_rvachevFourier_eq_gaussian_tsum`. These five declarations exhaust `RvachevQBinomialFilter.lean`.  $\square$

The theorem closes the coefficient bridge for the *infinite* Rvachev product. It does not identify the frontier report's finite products  $P_{b,n}$ , their quotient  $P_{b,n}/P_b$ , or their Bell coefficients with this series. Nor does it prove a conditionally convergent boundary theorem, an analytic sign or bracketing/error bound, uniform or derivative convergence, or an asymptotic acceleration estimate. The algebraic  $(-1)^p$  in (439) should not be promoted to one of those analytic sign claims.

For the present half-geometric grid, write  $\lambda_{n,k}^{(1/2)}$  for the evaluation-at-zero weight belonging to the node  $2^{-k}$ . The Toeplitz row is exactly the same row in reverse order:

$$\boxed{\omega_{n,j} = \lambda_{n,n-j}^{(1/2)}} \quad (0 \leq j \leq n).$$

Indeed, for every  $d \geq 0$  its complete reversed moment is

$$\sum_{k=0}^n \omega_{n,n-k} 2^{-kd} = 2^{-nd} \frac{(2^d/2; 1/2)_n}{(1/2; 1/2)_n},$$

and for  $0 \leq d \leq n$  this is  $0^d$ , with  $0^0 = 1$ . The pointwise bridge and these two moment forms are `discreteLimitWeight_eq_geometricLagrangeWeight`, `geometricLagrangeWeight_half_eq_discreteLimitWeight`, `sum_range_discreteLimitWeight_reverse_mul_half_pow`, and `sum_range_discreteLimitWeight_reverse_moment`. Their finite-sum form is `sum_range_discreteLimitWeight_smul_eq_geometricLagrangeWeight`.

For every  $d > 0$ , the same reversed row has the sharper residual form

$$\sum_{k=0}^n \omega_{n,n-k} ((1/2)^k)^d = (-1)^n (1/2)^{n(n+1)/2} \begin{bmatrix} d-1 \\ n \end{bmatrix}_{1/2}. \quad (440)$$

This is `sum_range_discreteLimitWeight_reverse_moment_eq_halfQBinomial` in `FabiusDiscreteLimitToeplitz.lean`. Thus one Gaussian coefficient records every positive residual degree: its above-diagonal boundary gives the cancelled range, while its diagonal boundary gives the first surviving moment. The corresponding boundary declarations are `gaussianBinomial_eq_zero_of_lt` and `gaussianBinomial_self` in `FiniteQBinomialCore.lean`.

Consequently the global model in the proposition can be weakened to one finite block. For any  $s \in \mathbb{N}$ , if only the samples  $0 \leq k \leq n$  satisfy

$$u_{s+k} = L + \sum_{r=1}^n 2^{-r(s+k)} \cdot a_r,$$

then

$$\sum_{j=0}^n \omega_{n,j} \cdot u_{s+n-j} = L.$$

This is `discreteLimitWeight_qRichardson_exact_module_at_of_eq` in `FabiusDiscreteLimitToeplitz.lean`. At  $n = 0$  its hypothesis is simply  $u_s = L$ , and  $s = 0$  needs no separate convention. The grid ratio is  $1/2$ . A mode with ratio  $1/4$  and index  $r$  has half-grid degree  $2r$ , so this row cancels it only when  $2r \leq n$ ; this is not an  $n$ -mode theorem on a  $1/4$  grid.

Applying it with the opposite sign also gives the uniform variation bound

$$\sum_{j=0}^n |\omega_{n,j}| = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n} \leq 16. \quad (441)$$

Finally, every sampled degree satisfies  $2n - j \geq n$ . A finite-row Toeplitz lemma therefore carries the convergence of  $S_p^{(Q)}(x)$  to the weighted rows (416). Explicitly, given  $\epsilon > 0$ , choose  $P$  so that  $|S_p^{(Q)}(x) - \mathcal{F}(x)| < \epsilon/16$  for  $p \geq P$ . If  $n \geq P$ , then every  $2n - j \geq n \geq P$ , and the row-sum identity together with (441) gives

$$\left| \sum_{j=0}^n \omega_{n,j} S_{2n-j}^{(Q)}(x) - \mathcal{F}(x) \right| \leq \sum_{j=0}^n |\omega_{n,j}| |S_{2n-j}^{(Q)}(x) - \mathcal{F}(x)| < \epsilon.$$

This proves (406). □

Two distinctions prevent common misreadings of this theorem. First, unlike the complete finite dyadic numerator and each summand of the global  $q$ -series, a finite value  $\mathcal{D}_n^{(Q)}(x)$  can depend on  $Q$ ; it is the limit that is independent of a fixed shift. Second, the Toeplitz row (416) is not a finite binary telescope or a partial sum of (402). These finite expressions share a limit but are not equal term by term.

The two properties of the reindexed row (417) used above are both evaluations of the finite  $q$ -binomial theorem (349), at  $z = 1/2$  and at  $z = -1/2$  respectively. Both use the exponent identity  $\binom{j}{2} + j = \binom{j+1}{2}$ .

For the row sum, evaluate (349) at  $z = 1/2$ . The left side is  $(1/2; 1/2)_n$  by definition, and the right side is

$$(1/2; 1/2)_n = \sum_{j=0}^n (-1)^j 2^{-\binom{j}{2}} \left[ \begin{matrix} n \\ j \end{matrix} \right]_{1/2} \left( \frac{1}{2} \right)^j = \sum_{j=0}^n (-1)^j \left[ \begin{matrix} n \\ j \end{matrix} \right]_{1/2} 2^{-(j+1)}. \quad (442)$$

Dividing by  $(1/2; 1/2)_n$  gives exactly  $\sum_{j=0}^n \omega_{n,j} = 1$ , which is (418).

For the total variation, evaluate (349) at  $z = -1/2$  instead. Each factor  $(-1)^j (-1/2)^j = 2^{-j}$  is now positive, so the alternating signs disappear:

$$(-1/2; 1/2)_n = \sum_{j=0}^n (-1)^j 2^{-\binom{j}{2}} \left[ \begin{matrix} n \\ j \end{matrix} \right]_{1/2} \left( -\frac{1}{2} \right)^j = \sum_{j=0}^n \left[ \begin{matrix} n \\ j \end{matrix} \right]_{1/2} 2^{-(j+1)}. \quad (443)$$

Since  $\left[ \begin{matrix} n \\ j \end{matrix} \right]_{1/2} > 0$ , the summands of (443) are the absolute values of those of (442). Dividing by  $(1/2; 1/2)_n$  gives the exact evaluation

$$\sum_{j=0}^n |\omega_{n,j}| = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n}, \quad (444)$$

which is the first equality in (441). The formally proved finite-product estimate supplies the bound by 16 used in the Toeplitz argument above.

## 11 Iterated prefix sums of the signed Thue–Morse sequence

The signed Thue–Morse sequence  $\varepsilon_n = (-1)^{\text{wt}_2(n)}$  has already appeared twice in this article: once as the coefficient sequence of the locally finite global expansion (11), and once as the sign pattern that makes the finite dyadic sums of section 8 collapse. There is a third, entirely discrete, way in which the same signs produce the Fabius function. Sum the sequence, sum the result, and iterate. After  $k$  summations the sequence has been convolved with an explicit binomial kernel, and after a dyadic normalization the resulting integer grid reproduces, in the limit, the Rvachev bump and hence  $F$ .

This section develops that construction: the convolution law and its explicit binomial kernel, the signed dyadic reflection preceding each terminal zero run, the zero runs themselves and their exact boundary values, the two generating-series identities, their analytic realization and natural-boundary theorem, the discrete functional equation satisfied by the normalized grid, and the precise sense in which the normalized grid converges. The convergence statement requires one shift of the prefix order and the centered reading of the grid cells; the literal normalization does not converge, and that failure is recorded separately in section 11.6. Two further pieces of foundational material are attached at the end: the lower branch of Lambert’s  $W$ -function, which the asymptotic sections of this article use but never construct, and an exact binomial–logarithm formula for the Thue–Morse bit.

The two binary recurrences used throughout this section are

$$\varepsilon_{2m} = \varepsilon_m, \quad \varepsilon_{2m+1} = -\varepsilon_m, \quad (445)$$

$$\text{wt}_2(2m) = \text{wt}_2(m), \quad \text{wt}_2(2m+1) = \text{wt}_2(m) + 1. \quad (446)$$

They are exactly `thueMorseSign_two_mul`, `thueMorseSign_two_mul_add_one`, `binaryWeight_two_mul`, and `binaryWeight_two_mul_add_one` in `DyadicClosedForm.lean`.

### 11.1 Iterated prefix sums and the binomial convolution law

**Definition 11.1** (Iterated prefix sums). Set  $\sigma_0(n) = \varepsilon_n$  and, for  $k \geq 0$ ,

$$\sigma_{k+1}(n) = \sum_{j=0}^n \sigma_k(j). \quad (447)$$

All sums are *inclusive*: the index  $n$  itself is summed.

Every  $\sigma_k(n)$  is an integer, and  $\sigma_k(0) = 1$  for all  $k$ . The inclusive convention has one consequence that must be kept in view throughout, because it is the source of an index shift later:

$$\sigma_{k+1}(n+1) - \sigma_{k+1}(n) = \sigma_k(n+1). \quad (448)$$

The increment of a prefix sum is the lower-order value at the *new* index, not at the old one.

The first prefix sum nevertheless vanishes at every odd index:

$$\sigma_1(2m+1) = 0 \quad (m \in \mathbb{N}).$$

This is exactly `iteratedPrefix_one_two_mul_add_one` in `ThueMorsePrefix.lean`.

The general structure is a discrete convolution. Write

$$\mathcal{B}_k(i) = \binom{i+k-1}{k-1} \quad (k \geq 1, i \geq 0) \quad (449)$$

which is `iteratedPrefixKernel`. Because the argument  $i = n - m$  is nonnegative in the sums below, no separate indicator is needed.

**Theorem 11.2** (Binomial convolution law). For every  $k \geq 1$  and every  $n \geq 0$ ,

$$\sigma_k(n) = \sum_{m=0}^n \mathcal{B}_k(n-m) \varepsilon_m = \sum_{m=0}^n \binom{n-m+k-1}{k-1} \varepsilon_m. \quad (450)$$

Equivalently, in a form valid for every  $k \geq 0$ ,

$$\sigma_{k+1}(n) = \sum_{m=0}^n \binom{n-m+k}{k} \varepsilon_m. \quad (451)$$

*Proof.* Induct on  $k$  in the form (451). For  $k = 0$  the binomial coefficient is 1 and the assertion is the definition of  $\sigma_1$ . Assume (451) for  $k$ . Then

$$\begin{aligned} \sigma_{k+2}(n) &= \sum_{j=0}^n \sigma_{k+1}(j) = \sum_{j=0}^n \sum_{m=0}^j \binom{j-m+k}{k} \varepsilon_m \\ &= \sum_{m=0}^n \varepsilon_m \sum_{i=0}^{n-m} \binom{i+k}{k} = \sum_{m=0}^n \binom{n-m+k+1}{k+1} \varepsilon_m, \end{aligned}$$

where the interchange is the triangular reindexing  $i = j - m$  and the inner evaluation is the hockey-stick identity  $\sum_{i=0}^N \binom{i+k}{k} = \binom{N+k+1}{k+1}$ .  $\square$

*Remark 11.3.* The kernel in (450) is exactly the coefficient sequence of  $(1-z)^{-k}$ . [Theorem 11.2](#) is therefore the coefficient form of the series identity proved in [theorem 11.11](#) below, and either statement may be taken as the definition of the other. Both are recorded here because the coefficient form is what one actually evaluates, while the series form is what one manipulates.

## 11.2 Prouhet annihilation, signed dyadic reciprocity, and exact zero runs

Equation (325) already records that a complete dyadic block of Thue–Morse signs annihilates every polynomial of degree below the block exponent. For the prefix sums we also need the sharp boundary case, which identifies the first power that survives.

**Lemma 11.4** (Sharp Prouhet extraction). *Write  $S_{r,d} = \sum_{h < 2^r} \varepsilon_h h^d$ . Then  $S_{r,d} = 0$  for  $0 \leq d < r$ , and*

$$S_{r,r} = (-1)^r 2^{\binom{r}{2}} r!. \quad (452)$$

Consequently, for all rational numbers  $a_0, \dots, a_r$ ,

$$\sum_{h < 2^r} \varepsilon_h \sum_{d=0}^r a_d h^d = a_r (-1)^r 2^{\binom{r}{2}} r!, \quad (453)$$

so summation against a complete dyadic block of Thue–Morse signs annihilates every polynomial of degree below  $r$  and, on polynomials of degree at most  $r$ , extracts the leading coefficient up to the explicit nonzero factor  $(-1)^r 2^{\binom{r}{2}} r!$ .

*Proof.* The vanishing statement is (325) with  $X = 0, Y = 1$ . For the boundary case, split the block by parity rather than by halves. Since  $\varepsilon_{2j} = \varepsilon_j$  and  $\varepsilon_{2j+1} = -\varepsilon_j$  by (445),

$$S_{r+1,d} = \sum_{j < 2^r} \varepsilon_j [(2j)^d - (2j+1)^d] = - \sum_{c=0}^{d-1} \binom{d}{c} 2^c S_{r,c},$$

because  $(2j+1)^d - (2j)^d = \sum_{c < d} \binom{d}{c} (2j)^c$ . Take  $d = r+1$ . All terms with  $c < r$  vanish by the first assertion, leaving only  $c = r$ :

$$S_{r+1,r+1} = - \binom{r+1}{r} 2^r S_{r,r} = -(r+1) 2^r S_{r,r}.$$

Since  $S_{0,0} = \varepsilon_0 = 1$ , induction gives  $S_{r,r} = (-1)^r r! 2^{0+1+\dots+(r-1)} = (-1)^r r! 2^{\binom{r}{2}}$ , which is (452). Formula (453) follows by exchanging the two sums: every index  $d < r$  contributes  $a_d S_{r,d} = 0$ , and the index  $d = r$  contributes  $a_r S_{r,r}$ .  $\square$

**Corollary 11.5** (Affine sharp case). *For all  $X, Y \in \mathbb{Q}$ ,*

$$\sum_{h < 2^r} \varepsilon_h (X + Yh)^r = (-1)^r Y^r 2^{\binom{r}{2}} r!.$$

*In particular the sum does not depend on the translation  $X$ .*

*Proof.* Expand  $(X + Yh)^r$  by the binomial theorem and apply (453): the coefficient of  $h^r$  is  $Y^r$ .  $\square$

The sharp extractor also has a direct commutative-ring form. If  $R$  is any commutative ring and  $p \in R[X]$  satisfies  $\deg p \leq r$ , then, with the signs and natural numbers mapped into  $R$ ,

$$\sum_{h < 2^r} \varepsilon_h p(h) = [X^r]p (-1)^r 2^{\binom{r}{2}} r!. \quad (454)$$

This is `thueMorse_polynomial_sum_eq_coeff_ring_of_degree_le`; its strict-degree cancellation form is `thueMorse_polynomial_sum_eq_zero_ring_of_degree_lt`, both in `ThueMorsePrefix.lean`. The word “extractor” here describes the functional identity, not division by its sharp factor: in positive characteristic the image of  $2^{\binom{r}{2}} r!$  can vanish, so the displayed equality need not recover  $[X^r]p$  from the sum. The degree-valued hypothesis also treats the zero polynomial

correctly; at  $r = 0$  the extraction formula includes constant polynomials and the zero polynomial, whereas the strict-degree condition admits only the latter.

Likewise the affine sharp identity holds directly in  $R$ :

$$\sum_{h < 2^r} \varepsilon_h(x + yh)^r = (-1)^r y^r 2^{\binom{r}{2}} r!, \quad (455)$$

formalized as `thueMorse_affine_power_sum_self_ring`, with the unified range  $d \leq r$  given by `thueMorse_affine_power_sum_of_le_ring`. For  $r > 0$  the sharp sum vanishes at  $y = 0$ . At the joint boundary  $r = 0, y = 0$ , Lean’s convention  $0^0 = 1$  instead gives the correct one-term value 1 on both sides; no positive-degree reading should be imposed on that case.

The binary-support parametrization used next has a canonical global foundation. Write  $\text{bits}(n) = \text{bitSupport}(n)$  and  $E(T) = \sum_{j \in T} 2^j$  for a finite set  $T \subseteq \mathbb{N}$ . This support is exactly Mathlib’s finite set of binary indices, and

$$j \in \text{bits}(n) \iff n.\text{testBit}(j) = \text{true}, \quad E(\text{bits}(n)) = n, \quad \text{bits}(E(T)) = T. \quad (456)$$

Consequently

$$\text{wt}_2(E(T)) = |T|, \quad \varepsilon_{E(T)} = (-1)^{|T|}. \quad (457)$$

The exact declarations are `bitSupport_eq_toFinset_bitIndices`, `mem_bitSupport`, `sum_two_pow_bitSupport`, `bitSupport_sum_two_pow`, `binaryWeight_sum_two_pow_eq_card`, and `thueMorseSign_sum_two_pow` in `ThueMorseBitSupport.lean`.

Restricting the same inverse pair gives the canonical equivalence

$$\{T : T \subseteq \{j \in \mathbb{N} : j < m\}\} \simeq \{h \in \mathbb{N} : h < 2^m\}, \quad T \mapsto E(T),$$

and hence, for every function into an additive commutative monoid,

$$\sum_{T \subseteq \{j \in \mathbb{N} : j < m\}} f(E(T)) = \sum_{h=0}^{2^m-1} f(h). \quad (458)$$

These are `powersetRangeEquivFin` and `sum_powerset_two_pow`; the exact image statement is `image_sum_two_pow_powerset_range`. The singleton boundary at  $m = 0$  is included explicitly by `powersetRangeEquivFin_zero_apply`, all in `ThueMorseBitSupport.lean`.

The public Boolean-cube theorem uses the genuine degree  $\deg p \in \mathbb{N} \cup \{-\infty\}$ , rather than the natural degree that assigns the zero polynomial degree zero. Over any commutative ring, for a finite set  $S$ , weights  $w_j$ , and a polynomial satisfying  $\deg p < |S|$ , it states

$$\sum_{T \subseteq S} (-1)^{|T|} p\left(x + \sum_{j \in T} w_j\right) = 0. \quad (459)$$

This is `sum_powerset_neg_one_pow_eval_of_degree_lt` in `ThueMorseSparseProuhet.lean`. The degree convention matters at the boundary  $S = \emptyset$ : the hypothesis then forces  $p = 0$ , so the theorem correctly includes the empty-support case instead of declaring it vacuous. With  $S$  the bit support of  $n$  and  $w_j = 2^j h$ , the corresponding sparse submask parametrization is

$$\sum_{T \subseteq \text{bits}(n)} (-1)^{|T|} p\left(x + h \sum_{j \in T} 2^j\right) = 0 \quad \text{when } \deg p < \text{wt}_2(n), \quad (460)$$

formalized as `sum_submask_neg_one_pow_eval_of_degree_lt`. It is valid also at  $n = 0$ , where again the only polynomial satisfying the strict degree bound is zero.

At the level of the centered formal exponential series, the corresponding zero has exact order  $r$ ; this is `order_thueMorseCenteredPowerSeries`. For the binomial kernel (449), the directly applicable formally proved form is instead the coefficient-extraction theorem `thueMorse_polynomial_sum_eq_coeff`, displayed in (453).



**Theorem 11.6** (Signed dyadic reciprocity). *Let  $k, r, d \in \mathbb{N}$  satisfy  $k \leq r$  and  $k + d < 2^r$ . Then the part of the order- $k$  prefix row immediately preceding its final  $k$  positions is the initial part of that row, read backwards and multiplied by one fixed sign:*

$$\sigma_k(2^r - k - 1 - d) = (-1)^{r-k} \sigma_k(d). \quad (461)$$

Consequently reflection preserves, rather than excludes, its zero locus:

$$\sigma_k(2^r - k - 1 - d) = 0 \iff \sigma_k(d) = 0. \quad (462)$$

The identity (461) and equivalence (462) are formalized respectively as `iteratedPrefix_dyadic_reverse_window` and `iteratedPrefix_dyadic_reverse_window_eq_zero_iff`. At  $k = 0$  the first identity is precisely the dyadic bit-complement law `thueMorseSign_dyadic_complement`. Its proof below uses the independently established left boundary in [theorem 11.7](#) as the seed for the prefix recurrence.

**Theorem 11.7** (Exact dyadic zero runs). *Let  $1 \leq k \leq r$ . Then*

$$\sigma_k(n) = 0 \quad \text{for } 2^r - k \leq n \leq 2^r - 1, \quad (463)$$

$$\sigma_k(2^r - k - 1) = (-1)^{r-k}, \quad (464)$$

$$\sigma_k(2^r) = -1. \quad (465)$$

Formula (464) also holds for  $k = 0$ , where it reduces to  $\varepsilon_{2^r-1} = (-1)^r$ , and (465) also holds for  $k = 0$ . Hence the zero run terminating at  $2^r - 1$  has length exactly  $k$ : it is bounded on the left by a value of modulus one and on the right by the value  $-1$ .

*Formal proof and Lean correspondence.* The terminal block is formalized by `iteratedPrefix_dyadic_zero_run`; its endpoint and reverse-indexed forms are `iteratedPrefix_dyadic_endpoint` and `iteratedPrefix_dyadic_zero_reverse`. Its left and right boundaries are `iteratedPrefix_before_dyadic_run` and `iteratedPrefix_at_dyadic`. All five declarations are in `ThueMorsePrefix.lean`. The mathematical argument behind them is as follows.

*Endpoint.* Put  $q = k - 1$ , so  $0 \leq q < r$ . By (450),

$$\sigma_k(2^r - 1) = \sum_{m=0}^{2^r-1} \binom{2^r-1-m+q}{q} \varepsilon_m = \frac{1}{q!} \sum_{m < 2^r} \varepsilon_m \prod_{i=1}^q (2^r - 1 - m + i).$$

The product is a polynomial in  $m$  of degree  $q < r$ , so the sum vanishes by (453).

*Rest of the run.* We prove, by induction on  $d$ , that  $\sigma_k(2^r - 1 - d) = 0$  whenever  $1 \leq k \leq r$  and  $0 \leq d < k$ . The case  $d = 0$  is the endpoint. Suppose the claim holds for  $d$  and that  $d + 1 < k$ , so  $k \geq 2$  and  $1 \leq k - 1 \leq r$ . The induction hypothesis applied at order  $k$  gives  $\sigma_k(2^r - 1 - d) = 0$ , and applied at order  $k - 1$ , which is legitimate because  $d < k - 1$ , gives  $\sigma_{k-1}(2^r - 1 - d) = 0$ . Now use (448) with  $n = 2^r - 1 - (d + 1)$ , so that  $n + 1 = 2^r - 1 - d$ :

$$0 - \sigma_k(n) = \sigma_k(n + 1) - \sigma_k(n) = \sigma_{k-1}(n + 1) = 0.$$

This proves (463), since the indices  $2^r - k, \dots, 2^r - 1$  are exactly the  $2^r - 1 - d$  with  $0 \leq d < k$ .

*Left boundary.* Induct on  $k$ . For  $k = 0$  the number  $2^r - 1$  has  $r$  binary digits, all equal to one, so  $\varepsilon_{2^r-1} = (-1)^r$ . Assume (464) at order  $k$  and let  $k + 1 \leq r$ . Put  $n = 2^r - (k + 1) - 1$ , so  $n + 1 = 2^r - k - 1$ . By the induction hypothesis  $\sigma_k(n + 1) = (-1)^{r-k}$ , and by (463) at order  $k + 1$  we have  $\sigma_{k+1}(n + 1) = \sigma_{k+1}(2^r - (k + 1)) = 0$ . Then (448) gives  $-\sigma_{k+1}(n) = (-1)^{r-k}$ , that is,  $\sigma_{k+1}(n) = (-1)^{r-(k+1)}$ .

*Right boundary.* Induct on  $k$ . For  $k = 0$ ,  $\text{wt}_2(2^r) = 1$ , so  $\varepsilon_{2^r} = -1$ . If (465) holds at order  $k$  and  $k + 1 \leq r$ , then by (448) and (463),  $\sigma_{k+1}(2^r) = \sigma_{k+1}(2^r - 1) + \sigma_k(2^r) = 0 - 1 = -1$ .  $\square$

*Proof of theorem 11.6.* Induct first on the prefix order  $k$ . At order zero,  $\sigma_0 = \varepsilon$ , and reflecting  $d$  in the block  $0, \dots, 2^r - 1$  complements its first  $r$  binary digits. Thus  $\varepsilon_{2^r-1-d} = (-1)^r \varepsilon_d$ , which is (461) for  $k = 0$ .

Now suppose the identity is known at order  $k$  and prove it at order  $k + 1$  by induction on  $d$ . For  $d = 0$ , equation (464) and  $\sigma_{k+1}(0) = 1$  give exactly the required identity. For the induction step, put

$$n = 2^r - (k + 1) - 1 - (d + 1), \quad s = (-1)^{r-(k+1)}.$$

The same-order induction hypothesis at offset  $d$  and the lower-order hypothesis at offset  $d + 1$  have the common reflected index  $n + 1$  and give

$$\sigma_{k+1}(n + 1) = s\sigma_{k+1}(d), \quad \sigma_k(n + 1) = (-1)^{r-k}\sigma_k(d + 1) = -s\sigma_k(d + 1).$$

Applying the forward-difference law (448) first at  $n$  and then at  $d$  therefore yields

$$\begin{aligned} \sigma_{k+1}(n) &= \sigma_{k+1}(n + 1) - \sigma_k(n + 1) \\ &= s(\sigma_{k+1}(d) + \sigma_k(d + 1)) = s\sigma_{k+1}(d + 1), \end{aligned}$$

which closes both inductions. Finally, the multiplier  $(-1)^{r-k}$  is nonzero, so (461) is equivalent to (462).  $\square$

*Remark 11.8.* At the level of statements, setting  $d = 0$  in (461) and using `iteratedPrefix_at_zero`, namely  $\sigma_k(0) = 1$ , recovers the left boundary (464). Reciprocity pairs any zeros that occur in the pre-run window; it does not claim that window is zero-free. Nor does it imply the terminal zero run (463) or the separate right boundary (465), both of which were proved independently above.

*Remark 11.9.* The right boundary value is not  $+1$ . Every iterated prefix sum, at every order  $k \leq r$ , takes the value  $-1$  at the index  $2^r$ . This single sign is what makes the literal normalization of section 11.5 fail at the right endpoint; see section 11.6.

### 11.3 Two generating-series identities

All series in this subsection are formal power series over  $\mathbb{Z}$ ; no convergence hypothesis on a complex variable is used or needed.

Let

$$\mathcal{K}_r(z) = \sum_{n < 2^r} \varepsilon_n z^n, \quad \mathcal{K}(z) = \sum_{n \geq 0} \varepsilon_n z^n, \quad \mathcal{S}_k(z) = \sum_{n \geq 0} \sigma_k(n) z^n. \quad (466)$$

**Proposition 11.10** (Block product and coefficient stabilization). *For every  $r \geq 0$ ,*

$$\mathcal{K}_r(z) = \prod_{j=0}^{r-1} (1 - z^{2^j}), \quad (467)$$

*and, whenever  $n < 2^r$ , the coefficient of  $z^n$  in this finite product is  $\varepsilon_n$ . Hence each coefficient stabilizes once the finite product reaches a large enough dyadic block.*

*Formal proof.* The finite product is `thueMorseBlockPolynomial_eq_product`; the stable coefficient identity is `coeff_finite_thueMorse_product`, both in `ThueMorseGenerating.le` an.  $\square$

**Theorem 11.11** (Prefix generating series). *For every  $k \geq 0$ ,*

$$\mathcal{S}_k(z) = \frac{\mathcal{K}(z)}{(1 - z)^k}, \quad (468)$$

*or equivalently*

$$(1 - z)^k \mathcal{S}_k(z) = \mathcal{K}(z). \quad (469)$$

*For each fixed coefficient index, either side can be computed from a sufficiently long finite product (467).*



*Formal proof.* The two identities are `iteratedPrefixSeries_eq` and `one_sub_X_pow_mul_iteratedPrefixSeries`; their finite coefficient bridge is `coeff_eq6_finite` in `ThueMorseGenerating.lean`.  $\square$

We use  $\prod_{j \geq 0} (1 - z^{2^j})$  only as shorthand for the coefficientwise stabilization of the finite products (467); no separate infinite-product object is asserted.

The two identities of [theorem 11.11](#) connect the prefix sums to the polynomial approximants used in [section 11.7](#). These are the combinatorial step polynomials already introduced in (85); their coefficient interpretation and exact degree were proved there. In the index convention of this section,

$$\mathcal{P}_n(z) = \prod_{i=1}^n (1 + z + \cdots + z^{2^i-1}), \quad g_n = \deg \mathcal{P}_n = \sum_{i=1}^n (2^i - 1) = 2^{n+1} - n - 2. \quad (470)$$

Since  $1 - z^{2^j} = (1 - z)(1 + z + \cdots + z^{2^j-1})$  and the factor with  $j = 0$  is trivial, (467) factors as

$$\mathcal{K}_{n+1}(z) = (1 - z)^{n+1} \mathcal{P}_n(z). \quad (471)$$

Comparing (471) with (469) at order  $k = n + 1$  and using stabilization below  $2^{n+1}$  gives the exact coefficient identification

$$\sigma_{n+1}(m) = [z^m] \mathcal{P}_n(z) \quad (0 \leq m < 2^{n+1}). \quad (472)$$

Note that (472) must fail at  $m = 2^{n+1}$ , because  $g_n < 2^{n+1}$  makes the right side zero while the left side is  $-1$  by (465). That single index is the exceptional right endpoint of [theorem 11.19](#).

## 11.4 The analytic Thue–Morse series and its natural boundary

The series  $\mathcal{K}$  in (466) was interpreted there as a formal power series. On the open unit disc the same coefficients define an analytic function, and we retain the notation  $\mathcal{K}$ :

$$\mathcal{K}(z) = \sum_{n=0}^{\infty} \varepsilon_n z^n, \quad |z| < 1. \quad (473)$$

**Proposition 11.12** (Analytic product and Mahler equation). *For every  $z \in \mathbb{C}$  with  $|z| < 1$ , the series (473) converges and*

$$\mathcal{K}(z) = \prod_{j=0}^{\infty} (1 - z^{2^j}). \quad (474)$$

*The function  $\mathcal{K}$  is holomorphic on  $|z| < 1$  and satisfies*

$$\mathcal{K}(z) = (1 - z) \mathcal{K}(z^2), \quad (475)$$

$$\mathcal{K}(z) = \left( \prod_{j=0}^{k-1} (1 - z^{2^j}) \right) \mathcal{K}(z^{2^k}) \quad (k \in \mathbb{N}). \quad (476)$$

*On the real radius,*

$$|\mathcal{K}(x)| \leq 1 - x \quad (0 \leq x < 1), \quad (477)$$

*and  $\mathcal{K}(0) = 1$ .*

*Formal proof.* The analytic function is `thueMorseDiscSeries`. Convergence, the product identity, and holomorphy are respectively `summable_thueMorseDiscSeries`, `thueMorseDiscSeries_eq_tprod`, and `thueMorseDiscSeries_differentiableOn`. Convergence of the product itself is `multipliable_one_sub_pow_two_pow_complex` in `ThueMorseInfiniteProduct.lean`.

Equations (475) and (476) are `thueMorseDiscSeries_mahler` and `thueMorseDiscSeries_iterate`; (477) and the value at zero are `thueMorseDiscSeries_real_le` and `thueMorseDiscSeries_zero`. Except for the product-convergence declaration just cited, all these declarations are in `ThueMorseDiscSeries.lean`.  $\square$

**Theorem 11.13** (Dyadic radial zeros and the natural boundary). *If  $k \in \mathbb{N}$  and  $\zeta \in \mathbb{C}$  satisfy  $\zeta^{2^k} = 1$ , then*

$$\lim_{\rho \rightarrow 1^-} \mathcal{K}(\rho\zeta) = 0. \quad (478)$$

Moreover, for every  $z_0 \in \mathbb{C}$  with  $|z_0| = 1$  and every  $R > 0$ , there is no function  $g : \mathbb{C} \rightarrow \mathbb{C}$  which is holomorphic on  $\{z : |z - z_0| < R\}$  and satisfies

$$g(z) = \mathcal{K}(z) \quad \text{whenever} \quad |z - z_0| < R \quad \text{and} \quad |z| < 1. \quad (479)$$

Thus, in precisely this local-disc sense, every point of the unit circle is singular and the unit circle is a natural boundary for  $\mathcal{K}$ .

*Formal proof.* The radial limit is `tendsto_thueMorseDiscSeries_radial_zero`. More generally, if  $n \in \mathbb{N}$  is nonzero and  $\zeta^n = 1$ , then  $|\zeta| = 1$ ; this is `norm_eq_one_of_pow_eq_one`. The exact quantified nonextension statement (479), and hence the natural-boundary conclusion in the stated local-disc sense, is `thueMorse_natural_boundary`. All three declarations are in `ThueMorseNaturalBoundary.lean`.  $\square$

## 11.5 The functional equation of the normalized prefix grid

The natural dyadic normalization of the prefix sums is

$$\mathcal{V}_k(j) = \frac{\sigma_k(j)}{2^{\binom{k}{2}}}, \quad x_{k,j} = \frac{j}{2^k}, \quad (480)$$

so that  $\mathcal{V}_k$  is a rational-valued function on the level- $k$  dyadic grid. The exponent  $\binom{k}{2}$  is forced: it is the exponent that appears in the sharp Prouhet value (452), in the centered spline (95), and in the derivative scaling (195).

**Theorem 11.14** (Discrete functional equation). *Let  $q \geq 0$  and  $0 \leq j < 2^q$ , and put  $h = 2^{-(q+1)}$  and  $x = x_{q+1,j}$ . Then*

$$\frac{\mathcal{V}_{q+1}(j+1) - \mathcal{V}_{q+1}(j)}{h} = 2 \mathcal{V}_q(j+1), \quad (481)$$

and the abscissa attached to the right-hand value is

$$x_{q,j+1} = 2(x + h) = 2x_{q+1,j+1} \in [0, 1]. \quad (482)$$

The same identity holds verbatim for the shifted grid  $\tilde{\mathcal{V}}_k(j) = \sigma_{k+1}(j)/2^{\binom{k}{2}}$  of section 11.7.

*Proof.* Because  $\binom{q+1}{2} = \binom{q}{2} + q$ , the forward difference (448) gives

$$\mathcal{V}_{q+1}(j+1) - \mathcal{V}_{q+1}(j) = \frac{\sigma_{q+1}(j+1) - \sigma_{q+1}(j)}{2^{\binom{q}{2}+q}} = \frac{\sigma_q(j+1)}{2^{\binom{q}{2}}} \cdot \frac{1}{2^q} = \frac{\mathcal{V}_q(j+1)}{2^q}.$$

Dividing by  $h = 2^{-(q+1)}$  multiplies by  $2^{q+1}$  and gives (481). For (482),  $2(x + h) = 2(j+1)/2^{q+1} = (j+1)/2^q = x_{q,j+1}$ , and the hypothesis  $j < 2^q$  is exactly the condition  $j+1 \leq 2^q$  that keeps this abscissa in the unit interval. The shifted grid satisfies the same computation with  $\sigma$  replaced by  $\sigma_{\bullet+1}$  throughout.  $\square$

Equation (481) is the discrete counterpart of the global differential equation (194): a difference quotient at level  $q + 1$  equals twice a level- $q$  value at the doubled abscissa. Two features of the discrete statement have no continuous analogue and are easy to lose. First, the right-hand side is evaluated at the *forward* node  $2(x + h)$  rather than at  $2x$ ; this is the inclusive-prefix convention (448). Second, the identity is an assertion about a level- $q$  value, so it is meaningful as a statement about  $F$  or up only when the doubled abscissa lies in the relevant interval, that is, only when  $j < 2^q$ . Both features are used in section 11.7.

## 11.6 The literal right endpoint does not converge

The unshifted normalization printed in the source formula cannot converge to a function whose right-endpoint value is 1: the sequence of literal values at the index  $2^k$  does not tend to 1. This exact negative statement is `paperPrefixGridValue_endpoint_not_tendsto_one` in `ThueMorseGenerating.lean`. The corrected normalization below is therefore a mathematical requirement, not a cosmetic reindexing.

## 11.7 Corrected normalization and pointwise convergence

We now identify the normalized prefix grid with the histogram approximants of the probabilistic construction, and deduce convergence. Two adjustments to the naive reading are required, and both are forced by identities already proved: the prefix order must be shifted by one, because (472) matches  $\sigma_{n+1}$  with  $\mathcal{P}_n$ ; and the grid index  $m$  must be read as labelling the *center* of a cell of width  $2^{-n}$ , not the left endpoint of a cell of width  $2^{-(n+1)}$ .

For the polynomial  $\mathcal{P}_n$  of (470), place the coefficient of the monomial  $z^m$  at

$$\xi_{n,m} = \frac{2m - g_n}{2^{n+1}}, \quad (483)$$

and spread it uniformly over the cell  $I_{n,m}$  of width  $2^{-n}$  centered at  $\xi_{n,m}$ . The consecutive centers are spaced  $2^{-n}$  apart, so these finitely many cells tile their support interval

$$\left[ -1 + \frac{n+1}{2^{n+1}}, 1 - \frac{n+1}{2^{n+1}} \right].$$

The resulting step approximant is

$$\mathcal{W}_n(x) = \frac{2^n}{2^{\binom{n+1}{2}}} \sum_{m=0}^{g_n} [z^m] \mathcal{P}_n(z) \cdot \mathbf{1}_{I_{n,m}}(x), \quad (484)$$

where a point shared by two adjacent cells is given the weight  $\frac{1}{2}$  in each, so that the two contributions add to the single value the formula is intended to have. This endpoint convention is not cosmetic: with ordinary closed-interval indicators the value at  $x = 0$  would be counted twice already for  $n = 1$ .

**Theorem 11.15** (Histogram approximants). *For every  $n \geq 0$  the function  $\mathcal{W}_n$  satisfies  $0 \leq \mathcal{W}_n \leq 1$ , is nondecreasing on  $(-\infty, 0)$ , is nonincreasing on  $(0, \infty)$ , and satisfies  $\mathcal{W}_n(0) = 1$ . Moreover  $\mathcal{W}_n(x) \rightarrow \text{up}(x)$  for every real  $x$ .*

*Proof.* For  $n \geq 1$ , this is exactly theorem 4.12, since (470) and (85) define the same polynomial, while (483) and (484) give the same cells and normalization as (88) and (89). At  $n = 0$  the empty product is  $\mathcal{P}_0 = 1$ ,  $g_0 = 0$ , and  $\mathcal{W}_0$  is the half-weighted indicator of  $[-1/2, 1/2]$ , so the stated order and normalization properties are immediate. The pointwise convergence of the sequence is already the final assertion of theorem 4.12.  $\square$

**Theorem 11.16** (Exact identification of the corrected grid). *Put  $\tilde{\mathcal{V}}_n(m) = \sigma_{n+1}(m)/2^{\binom{n}{2}}$ . For every  $n \geq 0$  and every  $m$  with  $0 \leq m < 2^{n+1}$ ,*

$$\tilde{\mathcal{V}}_n(m) = \mathcal{W}_n(\xi_{n,m}). \quad (485)$$

*Proof.* If  $m \leq g_n$ , the point  $\xi_{n,m}$  is the center of  $I_{n,m}$  and lies in no other cell, so (484) evaluates there to

$$\frac{2^n}{2^{\binom{n+1}{2}}} [z^m] \mathcal{P}_n(z).$$

Since  $\binom{n+1}{2} = \binom{n}{2} + n$ , this is  $[z^m] \mathcal{P}_n / 2^{\binom{n}{2}} = \tilde{\mathcal{V}}_n(m)$  by (472). If  $g_n < m < 2^{n+1}$ , then  $\xi_{n,m}$  lies strictly to the right of the finite support interval above, so  $\mathcal{W}_n(\xi_{n,m}) = 0$ ; meanwhile  $[z^m] \mathcal{P}_n = \sigma_{n+1}(m) = 0$  because  $\deg \mathcal{P}_n = g_n$ . Thus both sides again agree.  $\square$

Sampling at the dyadic floor now gives an approximation scheme. For  $x \in \mathbb{R}$  set

$$j_n(x) = \max(0, \lfloor 2^{n+1}x \rfloor), \quad \mathcal{J}_n(x) = \tilde{\mathcal{V}}_n(j_n(x)) = \frac{\sigma_{n+1}(j_n(x))}{2^{\binom{n}{2}}}. \quad (486)$$

**Lemma 11.17** (Moving cell centers). *For every  $x \in [0, 1]$ ,  $\xi_{n,j_n(x)} \rightarrow 2x - 1$  as  $n \rightarrow \infty$ .*

*Proof.* By (483),  $\xi_{n,j_n(x)} = 2 \cdot j_n(x) 2^{-(n+1)} - g_n 2^{-(n+1)}$ . The first term tends to  $2x$  because  $0 \leq 2^{n+1}x - j_n(x) < 1$ , and  $g_n 2^{-(n+1)} = 1 - (n+2)2^{-(n+1)} \rightarrow 1$ .  $\square$

**Lemma 11.18** (Stability under moving arguments). *If  $u_n \rightarrow y$  in  $\mathbb{R}$ , then  $\mathcal{W}_n(u_n) \rightarrow \text{up}(y)$ .*

*Proof.* Let  $\varepsilon > 0$ . Suppose first  $y > 0$ . By continuity of  $\text{up}$  choose  $d \in (0, y/2]$  with  $|\text{up}(y \pm d) - \text{up}(y)| < \varepsilon/2$ . For all large  $n$  we have  $u_n \in [y - d, y + d] \subset (0, \infty)$ , so the monotonicity of theorem 11.15 on  $(0, \infty)$  gives

$$\mathcal{W}_n(y + d) \leq \mathcal{W}_n(u_n) \leq \mathcal{W}_n(y - d).$$

By pointwise convergence at the two fixed points  $y \pm d$ , for all large  $n$  both outer terms lie within  $\varepsilon/2$  of  $\text{up}(y \pm d)$ , hence within  $\varepsilon$  of  $\text{up}(y)$ . The case  $y < 0$  is identical using monotonicity on  $(-\infty, 0)$ . For  $y = 0$  the upper bound is  $\mathcal{W}_n(u_n) \leq 1 = \text{up}(0)$ , and the lower bound is  $\mathcal{W}_n(-d)$ ,  $\mathcal{W}_n(0) = 1$ , or  $\mathcal{W}_n(d)$  according to the sign of  $u_n$ ; each of the three converges to a value within  $\varepsilon/2$  of  $\text{up}(0)$ .  $\square$

**Theorem 11.19** (Corrected prefix approximation). *For every  $x \leq 1$ ,*

$$\mathcal{J}_n(x) \longrightarrow \text{up}(2x - 1), \quad (487)$$

*and on the whole nonpositive half-line the finite-stage values are exactly*

$$\mathcal{J}_n(x) = \mathcal{J}_n(0) = 2^{-\binom{n}{2}} \quad (x \leq 0). \quad (488)$$

*and consequently, for every  $x \in [0, 1]$ ,*

$$\mathcal{J}_n\left(\frac{x}{2}\right) \longrightarrow F(x). \quad (489)$$

*Proof.* For  $x \leq 0$ , the natural-floor clamp is exact: `correctedPrefixGridSample_eq_at_zero_of_nonpos` and `correctedPrefixGridSample_zero` give (488), while `correctedPrefixGridSample_tendsto_of_nonpos` gives the asserted limit. Now let  $0 < x < 1$ . Then  $2^{n+1}x < 2^{n+1}$ , so  $j_n(x) < 2^{n+1}$  and theorem 11.16 applies:  $\mathcal{J}_n(x) = \mathcal{W}_n(\xi_{n,j_n(x)})$ . Combining lemma 11.17 with lemma 11.18 gives  $\mathcal{J}_n(x) \rightarrow \text{up}(2x - 1)$ .

At  $x = 1$  the identification fails, because  $j_n(1) = 2^{n+1}$  is exactly the first index not covered by (472). Here the value is computed instead from (465) with  $k = r = n + 1$ :

$$\mathcal{J}_n(1) = \frac{\sigma_{n+1}(2^{n+1})}{2^{\binom{n}{2}}} = -\frac{1}{2^{\binom{n}{2}}} \rightarrow 0 = \text{up}(1),$$

since  $\binom{n}{2} \rightarrow \infty$  and  $\text{up}(1) = F(0) = 0$ . Altogether this is the public theorem `correctedPrefixGridSample_tendsto_rvachevUp_of_le_one` in `ThueMorseApproximation.lean`. For (489), apply it at  $x/2 \in [0, 1]$  and use  $2(x/2) - 1 = x - 1 \leq 0$ , so that  $\text{up}(x - 1) = F(x)$  by (7).  $\square$

*Warning 11.20.* [Theorem 11.19](#) is a pointwise statement and nothing more. No uniform rate is asserted, and none is proved. The proof passes through [lemma 11.18](#), whose only quantitative inputs are the shared monotonicity of the  $\mathcal{W}_n$  and the modulus of continuity of the limit  $\text{up}$  at the single point  $2x - 1$ ; the sandwich it uses degrades as that modulus degrades. A uniform bound of the kind supplied for the centered splines by (101) is available from [section 4.6](#), but it is a different approximation scheme, and the two should not be conflated.

*Remark 11.21.* The two corrections are independent of one another and both are necessary. Shifting the prefix order from  $k$  to  $k + 1$  is what makes (472) available; reading the index  $m$  as the center (483) of a cell of width  $2^{-n}$  rather than as an abscissa  $m2^{-k}$  is what supplies the affine change of variable  $x \mapsto 2x - 1$  between the unit interval and the support of  $\text{up}$ . In particular the corrected grid is a sample of the compactly supported bump, not of the distribution function; this is why its right-endpoint limit is 0 and not 1.

## 11.8 The lower Lambert branch

The asymptotic analysis of [section 12.5](#) is organized around the coordinate (511), where  $W_{-1}$  denotes the lower real branch of Lambert’s  $W$ -function [11]. That section, and the elementary expansions derived from it in [section 12.10](#), use three facts about  $W_{-1}$  without proving any of them: that the branch exists on  $(-e^{-1}, 0)$ , that it is characterized by being the solution below  $-1$ , and that it has the two-term expansion recorded below. We supply them here, since they are the foundational input for (513) and for every subsequent estimate in the Lambert variable.

**Proposition 11.22** (Existence, defining equation, and uniqueness). *For  $z \in (-e^{-1}, 0)$ , let  $W_{-1}(z)$  denote the totalized lower branch `lowerLambertW`. Then*

$$W_{-1}(z) < -1, \quad W_{-1}(z) e^{W_{-1}(z)} = z, \quad (490)$$

and  $W_{-1}(z)$  is the unique real  $w < -1$  with  $we^w = z$ .

*Formal proof.* These are respectively `lowerLambertW_lt_neg_one`, `lowerLambertW_mul_exp`, and `lowerLambertW_unique` in `LowerLambertW.lean`.  $\square$

On its natural domain,  $W_{-1}$  is continuous and strictly decreasing, and its image is exactly  $(-\infty, -1)$ . At every  $z \in (-e^{-1}, 0)$ , it is differentiable with

$$\frac{d}{dz} W_{-1}(z) = \frac{W_{-1}(z)}{z(1 + W_{-1}(z))} < 0. \quad (491)$$

These four assertions are `lowerLambertW_continuousOn`, `lowerLambertW_strictAntiOn`, `lowerLambertW_image`, `deriv_lowerLambertW`, and `deriv_lowerLambertW_neg`; the underlying derivative witness is `lowerLambertW_hasDerivAt`, all in `LowerLambertW.lean`.

**Corollary 11.23** (Solution of the saddle equation). *Let  $x > 0$  with  $Lx < e^{-1}$ , where  $L = \log 2$  as in (496), and put  $\lambda = -W_{-1}(-Lx)/L$  as in (512). Then  $\lambda 2^{-\lambda} = x$ , which is (513).*

*Formal proof.* The saddle equation is `paperLambertN_eq9` in `LowerLambertW.lean`.  $\square$

**Proposition 11.24** (Two-term expansion). *As  $\epsilon \downarrow 0$ ,*

$$W_{-1}(-\epsilon) - (\log \epsilon - \log |\log \epsilon|) \longrightarrow 0. \quad (492)$$

*Formal proof.* This is `tendsto_lowerLambertW_expansion` in `LowerLambertW.lean`.  $\square$

### 11.9 A binomial–logarithm formula for the Thue–Morse bit

The zero-one Thue–Morse bit is  $\tau(n) = \text{wt}_2(n) \bmod 2$ , so that  $\varepsilon_n = 1 - 2\tau(n)$ . Its primitive-recursive evaluator is `tmBitPR`; the theorems `tmBitPR_of_pos` and `tmBitPR_eq_thueMorseBit` prove its agreement with the binary-weight definition. The bit also admits a closed formula in which the binary structure is hidden inside a single row of Pascal’s triangle. Set

$$\mathcal{X}(n) = \sum_{k=0}^n (-1)^{\binom{n}{k}} \in \mathbb{Z}. \quad (493)$$

**Lemma 11.25** (Glaisher count). *The number of odd entries in the  $n$ th row of Pascal’s triangle is  $2^{\text{wt}_2(n)}$ .*

*Proof.* Let  $\nu(n)$  denote that number, so that  $\nu(n)$  is the number of nonzero coefficients of  $(1+z)^n$  in  $\mathbb{F}_2[z]$ . There  $(1+z)^2 = 1+z^2$ , hence

$$(1+z)^{2n} = (1+z^2)^n, \quad (1+z)^{2n+1} = (1+z^2)^n(1+z).$$

The substitution  $z \mapsto z^2$  does not change the number of nonzero coefficients, so  $\nu(2n) = \nu(n)$ ; and multiplying a polynomial in  $z^2$  by  $1+z$  produces two summands with disjoint supports, one even and one odd, so  $\nu(2n+1) = 2\nu(n)$ . Since  $\text{wt}_2(2n) = \text{wt}_2(n)$  and  $\text{wt}_2(2n+1) = \text{wt}_2(n) + 1$  by (446), and  $\nu(0) = 1 = 2^{\text{wt}_2(0)}$ , induction on the binary representation gives  $\nu(n) = 2^{\text{wt}_2(n)}$ .  $\square$

**Proposition 11.26** (The logarithm is exact). *For every  $n \geq 0$ ,*

$$n+1 - \mathcal{X}(n) = 2^{\text{wt}_2(n)+1}. \quad (494)$$

*In particular  $n+1 - \mathcal{X}(n)$  is a positive power of two, and  $\log_2(n+1 - \mathcal{X}(n)) = \text{wt}_2(n) + 1$  is an integer.*

*Proof.* Each summand of (493) is  $-1$  at an odd entry and  $+1$  at an even entry, and the row has  $n+1$  entries. Hence  $\mathcal{X}(n) = (n+1) - 2\nu(n)$  with  $\nu$  as in lemma 11.25, so  $n+1 - \mathcal{X}(n) = 2\nu(n) = 2^{\text{wt}_2(n)+1}$ .  $\square$

**Theorem 11.27** (Binomial–logarithm formula). *For every  $n \geq 0$ ,*

$$\tau(n) = \frac{1 + (-1)^{\log_2(n+1 - \sum_{k=0}^n (-1)^{\binom{n}{k}})}}{2}, \quad \varepsilon_n = 1 - 2\tau(n). \quad (495)$$

*Proof.* By proposition 11.26 the exponent equals  $\text{wt}_2(n) + 1$ . If  $\text{wt}_2(n)$  is even then  $\tau(n) = 0$  and  $(-1)^{\text{wt}_2(n)+1} = -1$ , so the right side is 0; if  $\text{wt}_2(n)$  is odd then  $\tau(n) = 1$  and  $(-1)^{\text{wt}_2(n)+1} = 1$ , so the right side is 1. The second identity is  $\varepsilon_n = (-1)^{\text{wt}_2(n)} = 1 - 2(\text{wt}_2(n) \bmod 2)$ .  $\square$

*Remark 11.28.* The order of the two steps matters. As printed, (495) raises  $-1$  to a logarithm, an operation that is meaningful only because the argument of that logarithm has first been proved to be exactly a power of two. The formal development therefore proves (494) before it interprets the outer expression at all, evaluates the logarithm on the natural numbers, and reads the final division in  $\mathbb{Q}$ . A floating-point evaluation of the same display, or an interpretation of the logarithm as a real-valued function whose output is then rounded, is an approximation to a formula rather than the formula.

## 12 Endpoint asymptotics: the exact formal hierarchy

The behavior of  $F(x)$  as  $x \downarrow 0$  is governed by three nested structures: an exact dyadic product on the negative Laplace axis, a genuinely nonconstant periodic correction on logarithmic scale, and a lower-Lambert saddle coordinate. This section states the hierarchy in the form proved in Lean. Longer heuristic derivations, numerical experiments, and proposed refinements without exact theorem counterparts are collected in `docs/semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex`.

Throughout, put

$$L = \log 2. \quad (496)$$

### 12.1 The negative-Laplace product and its exact decomposition

For  $s > 0$ , the entire generating function  $G$  from (281) satisfies

$$G(-s) = \prod_{j=1}^{\infty} \frac{1 - \exp(-s/2^j)}{s/2^j}. \quad (497)$$

This is the real specialization of `complexGeneratingFunction_neg_eq_tprod` in `NegativeLaplaceVertical.lean`. If

$$\kappa(u) = \log\left(\frac{1 - \exp(-u)}{u}\right), \quad \Lambda(s) = \log G(-s) = \sum_{j=1}^{\infty} \kappa(s/2^j),$$

then the series is absolutely summable for every  $s > 0$ ; the precise declarations are `summable_negativeLaplaceTerm`, `exp_negativeLaplaceLog_eq_generatingFunction_neg`, and `negativeLaplaceLog_eq_log_laplaceMoment`.

Let  $P$  denote `negativeLaplacePeriodicCorrection` and let  $E$  denote `negativeLaplaceTailError`. The exact decomposition is

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + P(\log_2 s) + E(s) \quad (s > 0). \quad (498)$$

Moreover,

$$0 \leq E(s) \leq \frac{\exp(-s)}{(1 - \exp(-s))^2}, \quad E(s) \leq 4 \exp(-s) \quad (s \geq L). \quad (499)$$

These are exactly `negativeLaplaceLog_exact_periodic_decomposition`, `negativeLaplaceTailError_nonneg`, `abs_negativeLaplaceTailError_le`, and `abs_negativeLaplaceTailError_le_four_exp` in `NegativeLaplace.lean`. The correction  $P$  is  $C^\infty$  and one-periodic by `contDiff_infty_negativeLaplacePeriodicCorrection` and `negativeLaplacePeriodicCorrection_add_one`.

Write

$$\overline{P} = \int_0^1 P(u) \, du, \quad \Psi(t) = P(t) - \overline{P}. \quad (500)$$

Then  $\Psi$  is  $C^\infty$ , one-periodic, and has mean zero:

$$\Psi(t+1) = \Psi(t), \quad \int_0^1 \Psi(t) \, dt = 0.$$

The formal counterparts are `contDiff_infty_negativeLaplacePsi`, `negativeLaplacePsi_add_one`, and `integral_negativeLaplacePsi_zero`.



The first logarithmic derivative has a correspondingly sharp full-ray expansion. As  $s \rightarrow +\infty$ ,

$$\Lambda'(s) = \frac{-\log s/L + 1/2 + \Psi'(\log_2 s)/L}{s} + \mathcal{O}(s^{-2}). \quad (501)$$

Here  $\Lambda'$  is the formal `negativeLaplaceLogFirst`, identified with the derivative of  $\Lambda$  by `negativeLaplaceLog_hasDerivAt`. The exact  $\mathcal{O}(s^{-2})$  statement is `negativeLaplaceLogFirst_sub_periodic_main_isBigO_inv_sq_real` in `LaplacePeriodicSecondOrder.lean`. This promotes the proved derivative estimate without importing the longer unformalized contour and Mellin derivations from the frontier volume.

## 12.2 The Gamma–zeta Fourier series

For  $k \in \mathbb{Z}$ , use the Fourier convention

$$\widehat{\Psi}(k) = \int_0^1 \Psi(t) \exp(-2\pi i k t) dt, \quad \chi_k = \frac{2\pi i k}{L}. \quad (502)$$

The zero mode vanishes, and every nonzero mode has the closed form

$$\widehat{\Psi}(0) = 0, \quad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L} \quad (k \neq 0). \quad (503)$$

Every coefficient with  $k \neq 0$  is nonzero, so  $\Psi$  is genuinely nonconstant. The coefficient sequence is absolutely summable and reconstructs  $\Psi$  pointwise:

$$\Psi(t) = \sum_{k \in \mathbb{Z}} \widehat{\Psi}(k) \exp(2\pi i k t). \quad (504)$$

These assertions are `negativeLaplacePsiFourierCoeff_zero`, `negativeLaplacePsiFourierCoeff_eq_neg_gamma_zeta`, `negativeLaplacePsiFourierCoeff_ne_zero`, `negativeLaplacePsi_not_constant`, `summable_negativeLaplacePsiGammaZetaFourierCoeff`, and `hasSum_negativeLaplacePsi_gammaZeta_fourierSeries` in `PeriodicFourier.lean`. No numerical truncation or stronger coefficient-decay claim is needed here.

## 12.3 The evaluated constant

Let  $\gamma$  be Euler’s constant and let  $\gamma_1$  denote the formally defined first Stieltjes constant `firstStieltjesConstant`. The regularized Gamma–zeta combination

$$A_0 = \frac{\gamma^2}{2} + \gamma_1 - \frac{\pi^2}{12}$$

is `gammaZetaConstant`; its equivalent common-denominator form is proved by `gammaZetaConstant_eq_div_twelve`. The split, convergent finite-part Bose integral is evaluated as this same constant by `boseFinitePartIntegral_eq_gammaZetaConstant`.

The formal finite part is independent of where the convergent integral is split. More precisely, put

$$\kappa_0(x) = \frac{1}{x} \log\left(\frac{1 - \exp(-x)}{x}\right), \quad \kappa_\infty(x) = \frac{1}{x} \log(1 - \exp(-x)).$$

For every  $c > 0$ , both integrals below are convergent and

$$\boxed{\int_{(0,c]} \kappa_0(x) dx + \int_{(c,\infty)} \kappa_\infty(x) dx + \frac{(\log c)^2}{2} = A_0.} \quad (505)$$



If  $0 < a \leq b$ , changing the splitting point is governed by the exact transition identity

$$\int_{(a,b]} (\kappa_\infty(x) - \kappa_0(x)) dx = \frac{(\log b)^2 - (\log a)^2}{2}. \quad (506)$$

Here the two kernels are `boseFinitePartSmallKernel` and `boseFinitePartLargeKernel`; their integrability is `integrableOn_boseFinitePartSmallKernel_Ioc` and `integrableOn_boseFinitePartLargeKernel_Ioi_of_pos`. Equations (505) and (506) are respectively `boseFinitePartIntegral_eq_gammaZetaConstant_of_pos` and `integral_boseFinitePartLargeKernel_sub_small_Ioc` in `BoseFinitePartIntegral.lean`. Thus the constant is represented by ordinary convergent real integrals, without assigning a value to a divergent integral.

In turn, `negativeLaplacePeriodicMean_eq` gives

$$\bar{P} = \frac{A_0}{L} - \frac{L}{12}.$$

After the Gaussian normalization, the sharp Fabius constant is

$$C_F = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi. \quad (507)$$

This is the exact identity `fabiusSharpAsymptoticConstant_eq` in `FabiusSharpConstant.lean`.

## 12.4 A fully effective dyadic bound

The dyadic specialization admits a proved numerical threshold and error constant. For  $n \geq 1$ , define

$$\begin{aligned} D_n = & -\frac{L}{2}n^2 - n \log n + \left(1 + \frac{L}{2}\right)n - \frac{(\log n)^2}{2L} + C_F + \Psi(\log_2 n) \\ & - \frac{n}{2}(\Lambda''(n) + \Lambda'(n)^2). \end{aligned} \quad (508)$$

Here the final term is exactly `dyadicEndpointSecondOrder`; the derivative identifications are `negativeLaplaceLog_hasDerivAt` and `negativeLaplaceLogFirst_hasDerivAt`. Then, for every  $n \geq 224043$ ,

$$\boxed{|\log F(2^{-n}) - D_n| \leq \frac{2512945}{12n}}. \quad (509)$$

This is `abs_log_fabius_dyadic_sub_explicitCumulantMain_le` in `FabiusDyadicSharpCumulant.lean`. The threshold is a sufficient certified threshold, not an optimality claim. Its proof uses the pointwise bounds `dyadicEndpointSecondOrder_sq_le`, `dyadicHigherLaplaceMoments_le`, and `abs_dyadicEndpointLaplaceLogError_add_secondOrder_le_unconditional` in `LaplaceMomentBounds.lean`, with respective constants  $256/n$ ,  $104448/n$ , and  $209408/n$  on their stated ranges.

## 12.5 The lower-Lambert saddle and exact reduction

On the endpoint-inclusive lower-branch domain

$$\mathcal{D}_L = \left(0, \frac{e^{-1}}{L}\right], \quad (510)$$

let

$$W = W_{-1}(-Lx), \quad (511)$$

and define

$$\lambda = \lambda(x) = -\frac{W}{L}, \quad r = r(x) = 2^\lambda. \quad (512)$$

The global Lean definitions are `fabiusLambertPhase` and `fabiusLambertRadius`; here we use them on  $\mathcal{D}_L$ . For every  $x \in \mathcal{D}_L$ , their exact identities are

$$\lambda 2^{-\lambda} = x, \quad \frac{\lambda}{r} = x, \quad rx = \lambda, \quad \log r = L\lambda. \quad (513)$$

The endpoint-inclusive quotient and product identities are `fabiusLambertPhase_div_radiu s_of_le` and `fabiusLambertRadius_mul_argument_of_le`; the first equality in (513) is the quotient identity rewritten using  $r = 2^\lambda$ . The logarithmic identity in fact holds for every real  $x$ , not only on  $\mathcal{D}_L$ ; this is `log_fabiusLambertRadius`. The corresponding strict-interior wrappers are `fabiusLambertPhase_div_radius` and `fabiusLambertRadius_mul_argument`. All five declarations are in `FabiusLambertSaddle.lean`.

This lower phase is also the exact specialization at  $(m, A, \beta) = (1, 1, L)$  of the generic scaled power–exponential inverse; this is `lowerPowerExponentialPhase_one_one_log_two` in `PowerExponentialLambertFabius.lean`. More generally, for  $m \in \mathbb{N} \setminus \{0\}$ ,  $A > 0$ , and  $\beta > 0$ , put

$$S_{m,A,\beta}(t) = At^m e^{-\beta t}, \quad P_{m,A,\beta} = A \left( \frac{m}{\beta} e^{-1} \right)^m.$$

Writing  $\lambda_0$  and  $\lambda_{-1}$  for the principal and lower Lambert phases, their exact images are

$$\lambda_0([0, P_{m,A,\beta}]) = \left[ 0, \frac{m}{\beta} \right], \quad \lambda_{-1}((0, P_{m,A,\beta})) = \left[ \frac{m}{\beta}, \infty \right].$$

These are `principalPowerExponentialPhase_image_Icc` and `lowerPowerExponentialPhase_image_Ioc`; the companion declarations `principalPowerExponentialPhase_invOn` and `lowerPowerExponentialPhase_invOn` make each phase a two-sided `InvOn` inverse of  $S_{m,A,\beta}$  on the corresponding displayed interval pair. They are proved in `PowerExponentialLambertInverse.lean`. On the principal branch, the closed endpoint step uses the exact restriction  $W_0([-e^{-1}, 0]) = [-1, 0]$ , namely `principalLambertW_image_Icc` in `PrincipalLambertW.lean`.

On the same closed-upper domain, the phase has the sharp lower bound and endpoint characterization

$$\lambda(x) \geq \frac{1}{L}, \quad \lambda(x) = \frac{1}{L} \iff x = \frac{e^{-1}}{L}. \quad (514)$$

In particular,

$$\lambda\left(\frac{e^{-1}}{L}\right) = \frac{1}{L}. \quad (515)$$

Moreover,  $\lambda$  is strictly decreasing on  $\mathcal{D}_L$ , and its exact image is

$$\lambda(\mathcal{D}_L) = \left[ \frac{1}{L}, \infty \right). \quad (516)$$

The bound, equality characterization, endpoint value, strict antitonicity, and image are respectively `fabiusLambertPhase_ge_inv_log_two`, `fabiusLambertPhase_eq_inv_log_two_iff`, `fabiusLambertPhase_branchPoint`, `fabiusLambertPhase_strictAntiOn_Ioc`, and `fabiusLambertPhase_image_Ioc` in `FabiusLambertSaddle.lean`.

Equivalently, under the hypotheses  $x > 0$  and  $Lx \leq e^{-1}$ , the same lower bound and the strict positivity  $0 < \lambda(x)$  are `one_div_log_two_le_fabiusLambertPhase` and `fabiusLambertPhase_pos_of_le`. Thus the algebraic saddle identities extend to the finite branch point even though the smooth formulas below do not.

The smooth API is deliberately restricted to the open interior  $0 < x < e^{-1}/L$ . There  $\lambda$  is continuous and strictly decreasing, its image is  $(1/L, \infty)$ , and it is differentiable with

$$\lambda'(x) = \frac{\lambda(x)}{x(1 - L\lambda(x))} < 0. \quad (517)$$

These are `fabiusLambertPhase_continuousOn`, `fabiusLambertPhase_strictAntiOn`, `fabiusLambertPhase_image`, `deriv_fabiusLambertPhase`, and `deriv_fabiusLambertPhase_neg` in `FabiusLambertSaddle.lean`; the derivative witness is `fabiusLambertPhase_hasDerivAt`, and pointwise continuity is `fabiusLambertPhase_continuousAt`. At the small-argument endpoint  $x = 0$ , the exact phase and its reciprocal have the limits

$$\lambda(x) \longrightarrow +\infty, \quad \lambda(x)^{-1} \longrightarrow 0 \quad (x \downarrow 0). \quad (518)$$

These are `tendsto_fabiusLambertPhase_nhdsGT_zero_atTop` and `tendsto_inv_fabiusLambertPhase_nhdsGT_zero` in `FabiusLambertRates.lean`.

The corrected main term is

$$\mathcal{M}(x) = -\frac{L}{2}\lambda(x)^2 + \left(1 + \frac{L}{2}\right)\lambda(x) - \frac{1}{2}\log \lambda(x) + C_F + \Psi(\lambda(x)). \quad (519)$$

This is definition `fabiusSharpLambertMain`. The exact saddle action itself is endpoint-inclusive: whenever  $x > 0$  and  $Lx \leq e^{-1}$ ,

$$r(x)x + \Lambda(r(x)) - \frac{1}{2}\log(2\pi\lambda(x)) = \mathcal{M}(x) + E(r(x)). \quad (520)$$

This is `fabiusLambertSaddleAction_eq_of_le` in `FabiusSharpLambertMain.lean`. If  $\mathcal{R}(x)$  denotes the normalized saddle ratio `fabiusSaddleRatio` at  $r(x)$  and  $\lambda(x)$ , then Lean proves the exact error separation

$$\log F(x) - \mathcal{M}(x) = \log \mathcal{R}(x) + E(r(x)). \quad (521)$$

The declaration is `log_fabius_sub_sharpLambertMain_eq_log_ratio_add_tail` in `FabiusSharpExactReduction.lean`.

The forward product tail in this separation is negligible at every algebraic order of the exact phase. Along  $x = 2^{-t}$ , for each  $N \in \mathbb{N}$ ,

$$E(r(2^{-t})) = o(\lambda(2^{-t})^{-N}) \quad (t \rightarrow +\infty). \quad (522)$$

This is `negativeLaplaceTailError_dyadicLambert_isLittleO_inv_pow` in `FabiusSharpLambertTransfer.lean`. At  $N = 0$  it says directly that the tail tends to zero; larger  $N$  show that it contributes to none of the inverse-phase coefficients.

The analytic input to that reduction is an exact Bromwich formula. For every  $r > 0$  and  $x \in \mathbb{R}$ , as an identity in  $\mathbb{C}$ ,

$$F(x) = \exp(rx) \int_{\mathbb{R}} \exp(2\pi i \xi x) \frac{G(-(r + 2\pi i \xi))}{r + 2\pi i \xi} d\xi. \quad (523)$$

This is `fabius_bromwich`; the dimensionless, Gaussian-normalized version used by the saddle machinery is `fabius_bromwich_scaled`. The supporting Laplace-transform identity is `generatingFunction_neg_div_eq_laplace`. All three occur in `LaplaceTransform.lean` and `FabiusBromwichInput.lean` with the positivity and integrability hypotheses shown there.

## 12.6 The compact online formula and its missing periodic factor

A January 2022 comment in a Mathematics Stack Exchange discussion asked whether the preceding calculation implied a particularly compact lower-Lambert formula for the endpoint behavior [20]. With the notation of (511), its logarithmic main term is

$$\mathcal{W}_{\text{MSE}}(x) = -\frac{7L}{12} - \frac{1}{2} \log(\pi x) + \frac{1}{L} \left( \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12} - W - \frac{W^2}{2} \right). \quad (524)$$

This is the human-readable form of definition `fabiusWikipediaLambertMain` in `FabiusWikipediaMain.lean`. Written in ordinary mathematical notation, the exponential of this logarithmic encoding is the expression printed in the discussion:

$$A_{\text{MSE}}(x) = \frac{1}{2^{7/12} \sqrt{\pi x}} \exp \left[ \frac{1}{L} \left( \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12} - W - \frac{W^2}{2} \right) \right]. \quad (525)$$

For  $x > 0$ , this factor is strictly positive. Lean names it `fabiusWikipediaLambertFactor`; `exp_fabiusWikipediaLambertMain_eq_WikipediaLambertFactor` identifies it with  $\exp(\mathcal{W}_{\text{MSE}}(x))$ , and `fabiusWikipediaLambertFactor_pos` proves its positivity. All three declarations occur in `FabiusWikipediaMain.lean`. The proposed symbol  $F(x) \sim A_{\text{MSE}}(x)$  means that the ratio should tend to 1 as  $x \downarrow 0$ .

That ratio-one claim is false. The answer on the cited page first derives a Gamma–zeta periodic term, then explicitly omits it from the subsequent approximation while noting that it must be retained for more precise asymptotics. The exact formal decomposition shows what is lost: on the positive lower-Lambert domain,

$$\mathcal{M}(x) = \mathcal{W}_{\text{MSE}}(x) + \Psi(\lambda(x)). \quad (526)$$

Here `fabiusCorrectedWikipediaMain_eq_WikipediaLambertMain_add` is the all-real identity between the two named compact definitions, while `fabiusCorrectedWikipediaMain_eq_sharpLambertMain` identifies the corrected one with  $\mathcal{M}$  on the stated domain. Consequently the proved ratio-one asymptotic is

$$\boxed{F(x) \sim \frac{1}{2^{7/12} \sqrt{\pi x}} \exp \left[ \frac{1}{L} \left( \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12} - W - \frac{W^2}{2} \right) \right] \times \exp(\Psi(-W/L))} \quad (x \downarrow 0). \quad (527)$$

This is the direct theorem `fabius_isEquivalentent_WikipediaLambertFactor_mul_periodic` in `FabiusSharpAsymptotic.lean`; the underlying exact positive-side factorization is `exp_fabiusCorrectedWikipediaMain_eq_WikipediaLambertFactor_mul_periodic` in `FabiusWikipediaMain.lean`. The plus sign in (526) is essential: the missing multiplier is  $\exp(\Psi(\lambda(x)))$ , with  $\lambda(x) = -W/L$ .

The logical bridge from logarithmic error to ratio-one equivalence is itself formalized. Along any filter, if  $f$  is eventually positive, then

$$f \sim \exp(g) \iff \log f - g \longrightarrow 0. \quad (528)$$

This is `isEquivalentent_exp_iff_tendsto_log_sub`. In the present case Lean proves both

$$\log F(x) - \mathcal{W}_{\text{MSE}}(x) \not\longrightarrow 0 \quad (x \downarrow 0) \quad (529)$$

and that the same residual is not  $\mathcal{O}((-\log x)^{-1})$ . These are respectively `log_fabius_sub_WikipediaLambertMain_not_tendsto_zero` and `log_fabius_sub_WikipediaLambertMain_not_isBigO`. Applying (528) gives the direct theorem

$$F \not\sim A_{\text{MSE}}, \quad (530)$$

namely `fabius_not_isEquivalent_exp_WikipediaLambertMain`; its direct literal-factor form is `fabius_not_isEquivalent_WikipediaLambertFactor`. The criterion and all four obstruction declarations in this paragraph occur in `FabiusSharpAsymptotic.lean`. Thus the nonconvergence of the sampled periodic correction, rather than its numerical size, is decisive for ratio-one equivalence.

**Theorem 12.1** (Exact cluster interval for the online quotient). *For  $x > 0$ , put*

$$R_{\text{MSE}}(x) := \frac{F(x)}{A_{\text{MSE}}(x)}, \quad m_{\Psi} := \min_{0 \leq t \leq 1} \Psi(t), \quad M_{\Psi} := \max_{0 \leq t \leq 1} \Psi(t).$$

*Then  $m_{\Psi} < M_{\Psi}$ , and the quotient has the sharper additive asymptotic model*

$$R_{\text{MSE}}(x) - \exp(\Psi(\lambda(x))) \longrightarrow 0 \quad (x \downarrow 0). \quad (531)$$

*Write  $\text{Cl}_{0+}(R_{\text{MSE}})$  for the set of all  $z \in \mathbb{R}$  for which there is a sequence  $x_n \rightarrow 0$ , eventually positive, such that  $R_{\text{MSE}}(x_n) \rightarrow z$ . Its complete cluster set is*

$$\text{Cl}_{0+}(R_{\text{MSE}}) = [\exp(m_{\Psi}), \exp(M_{\Psi})]. \quad (532)$$

*In particular,*

$$\liminf_{x \downarrow 0} R_{\text{MSE}}(x) = \exp(m_{\Psi}) < \exp(M_{\Psi}) = \limsup_{x \downarrow 0} R_{\text{MSE}}(x). \quad (533)$$

*Consequently no real constant repairs the omitted factor: in Mathlib’s filter-theoretic asymptotic-equivalence relation,*

$$\forall c \in \mathbb{R}, \quad F \not\sim c A_{\text{MSE}} \quad (x \downarrow 0). \quad (534)$$

*For  $c \neq 0$ , the last assertion is the usual failure of  $F(x)/(c A_{\text{MSE}}(x)) \rightarrow 1$ . The formal declaration also includes  $c = 0$ , without interpreting a quotient by zero.*

The online discussion supplies  $A_{\text{MSE}}$ ; the exact cluster interval is a consequence of the corrected formal asymptotic. The endpoint descriptions  $\exp(m_{\Psi})$  and  $\exp(M_{\Psi})$  are exact variational formulas in terms of the explicit periodic correction. No finite elementary evaluation, numerical certification, sign claim for the extrema, or uniqueness claim for their phases is asserted here.

The phase-sampling mechanism is reusable. Every value of a one-periodic map  $p$  is a cluster value of  $p(\lambda(x))$  as  $x \downarrow 0$ ; if  $p$  is continuous and its codomain is Hausdorff, the complete cluster set is exactly  $p(\mathbb{R})$ . These statements are `mapClusterPt_periodic_comp_fabiusLambertPhase` and `mapClusterPt_periodic_comp_fabiusLambertPhase_iff` in `FabiusWikipediaObstruction.lean`.

For the specialization above,  $R_{\text{MSE}}$  is `fabiusWikipediaLambertRatio`, while  $m_{\Psi}$  and  $M_{\Psi}$  are `negativeLaplacePsiPeriodMin` and `negativeLaplacePsiPeriodMax`. The exact global range, pointwise extremal bounds, and endpoint order are respectively `range_negativeLaplacePsi_eq_Icc_periodMin_periodMax`, `negativeLaplacePsiPeriodMin_le_value`, `negativeLaplacePsi_value_le_periodMax`, and `negativeLaplacePsiPeriodMin_le_periodMax`; attainment and strict separation are `exists_mem_Icc_negativeLaplacePsi_eq_periodMin`, `exists_mem_Icc_negativeLaplacePsi_eq_periodMax`, and `negativeLaplacePsiPeriodMin_lt_periodMax`. Exponentiation of the exact range is `range_exp_negativeLaplacePsi_eq_Icc_periodExtrema`.

The ratio-one comparison with the sampled periodic model is `fabiusWikipediaLambertRatio_isEquivalent_periodic`, and the stronger additive statement (531) is `tendsto_fabiusWikipediaLambertRatio_sub_periodic`. The interval identity (532) is `mapClusterPt_fabiusWikipediaLambertRatio_nhdsGT_zero_iff`; the two endpoint limits are `liminf_fabiusWikipediaLambertRatio_nhdsGT_zero_eq` and `limsup_fabiusWikipediaLambertRatio_nhdsGT_zero_eq`; and the universal obstruction is `fabius_not_isEquivalent_const_mul_WikipediaLambertFactor`. These declarations all occur in `FabiusSharpAsymptotic.lean`.

## 12.7 The first two periodic saddle corrections

Define

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \quad (535)$$

This is `fabiusFirstSaddleCorrection`. It is continuous, one-periodic, and bounded by `continuous_fabiusFirstSaddleCorrection`, `fabiusFirstSaddleCorrection_periodic`, and `isBounded_range_fabiusFirstSaddleCorrection`. The first corrected expansion is

$$\boxed{\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda(x))}{\lambda(x)} + \mathcal{O}(\lambda(x)^{-2})} \quad (x \downarrow 0). \quad (536)$$

It is the  $N = 2$  specialization of `log_fabius_sub_sharpLambertExpansion_isBigO`, together with `fabiusSharpLambertExpansion_two`.

The second coefficient has a compact closed form in the bounded exponent jets. For each  $m$ , write  $J_m(u)$  for the jet `negativeLaplaceBoundedExponentJet(m, u)`, and define

$$A_2(u) = \frac{1}{24} (8J_0^3 + 12J_0^2 J_1 + 12J_0^2 + 48J_0 J_1 + 12J_0 J_2 + 6J_1^2 + 24J_1 + 20J_2 + 3J_3)(u). \quad (537)$$

This is exactly `fabiusSecondSaddleCorrection`; it is continuous, one-periodic, and bounded by `continuous_fabiusSecondSaddleCorrection`, `fabiusSecondSaddleCorrection_periodic`, and `isBounded_range_fabiusSecondSaddleCorrection`. The formally proved second-correction theorem is

$$\boxed{\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda(x))}{\lambda(x)} + \frac{A_2(\lambda(x))}{\lambda(x)^2} + \mathcal{O}(\lambda(x)^{-3})} \quad (x \downarrow 0). \quad (538)$$

Its exact declarations are `fabiusSharpLambertExpansion_three` and `log_fabius_sub_twoSaddleCorrections_isBigO` in `FabiusSecondSaddleExpansion.lean`; the companion theorem `log_fabius_sub_twoSaddleCorrections_isBigO_negLog` gives the literal remainder  $\mathcal{O}((-\log x)^{-3})$ .

## 12.8 The recursive all-orders theorem

For  $j \in \mathbb{N}$ , let  $a_j(u)$  denote `fabiusSaddleLogCoefficient(j, u)`. Every  $a_j$  is continuous, one-periodic, and bounded, by `continuous_fabiusSaddleLogCoefficient`, `fabiusSaddleLogCoefficient_periodic`, and `isBounded_range_fabiusSaddleLogCoefficient`. The initial identifications are

$$a_0 = 0, \quad a_1 = A_1, \quad a_2 = A_2,$$

proved by `fabiusSaddleLogCoefficient_zero`, `fabiusSaddleLogCoefficient_one_eq_firstSaddleCorrection`, and `fabiusSaddleLogCoefficient_two_eq_secondSaddleCorrection`.

For every  $N \in \mathbb{N}$ , the complete exact-phase expansion is

$$\boxed{\log F(x) = \mathcal{M}(x) + \sum_{j=0}^{N-1} \frac{a_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}(\lambda(x)^{-N})} \quad (x \downarrow 0). \quad (539)$$

The precise formal statement is `log_fabius_sub_sharpLambertMain_hasAsymptoticExpansion`; its finite-remainder forms are `log_fabius_sub_sharpLambertExpansion_isBigO` and

`log_fabius_sub_sharpLambertExpansion_isBigO_negLog` in `FabiusFullAsymptoticExpansion.lean`.

The coefficients are constructive rather than existential. The formal development identifies the jet weights as coefficients of

$$\prod_{k=0}^{n-1} (X - (k + 1)),$$

proves closed forms for the periodic and bounded exponent jets, and gives recursive exponent-polynomial identities. The exact declarations are `negativeLaplaceJetPolynomial_eq_prod`, `negativeLaplaceJetStirling_eq_coeff`, `negativeLaplacePeriodicJet_eq_closedForm`, `negativeLaplaceBoundedExponentJet_eq_closedForm`, `negativeLaplaceExponentCoefficient_succ_eq`, and `negativeLaplaceExponentPolynomial_succ_eq`. This coefficient-of-a-product description is the proved statement; no identification with an external Stirling-number API is asserted here.

The algebraic conversion between relative-mass and logarithmic coefficients is normalized at every index rather than only asymptotically. If `ExpCoeff` and `LogCoeff` denote the formal recurrences `expCoeff` and `logCoeff`, then

$$\begin{aligned} \text{ExpCoeff}(\text{LogCoeff}(a))_n &= \begin{cases} 1, & n = 0, \\ a_n, & n > 0, \end{cases} \\ \text{LogCoeff}(\text{ExpCoeff}(E))_n &= \begin{cases} 0, & n = 0, \\ E_n, & n > 0. \end{cases} \end{aligned} \quad (540)$$

These are `expCoeff_logCoeff_eq_ite` and `logCoeff_expCoeff_eq_ite`; the corresponding identities of formal power series are `expSeries_logCoeff_eq_massSeries_normalize_zero` and `logSeries_expCoeff_eq_exponentSeries_normalize_zero` in `SaddleLogExpansionAlgebra.lean` and `SaddleLogExpansionPowerSeries.lean`.

The recurrence-defined logarithm is also multiplicative at the level of formal power series. Let  $R$  be a commutative  $\mathbb{Q}$ -algebra and define

$$\mathfrak{L}(C) := \text{SaddleExpansion.logOf}(C), \quad C \in R[[X]],$$

by applying `logSeries` to the coefficient family of  $C$ . Rebuilding a series from that family gives  $C$  itself by `SaddleExpansion.massSeries_coeff`; this reconstruction lemma needs only that  $R$  be a commutative ring. For every  $C$ , even before a unit-constant hypothesis is imposed,

$$[X^0]\mathfrak{L}(C) = 0, \quad (541)$$

which is `SaddleExpansion.constantCoeff_logOf`. If  $[X^0]C = 1$ , then

$$C \mathfrak{L}(C)' = C', \quad (542)$$

by `SaddleExpansion.mul_derivative_logOf`; conversely, `SaddleExpansion.logOf_eq_of` says that (541) and (542) characterize  $\mathfrak{L}(C)$  uniquely.

It follows that unit-constant series satisfy

$$\mathfrak{L}(AB) = \mathfrak{L}(A) + \mathfrak{L}(B), \quad \mathfrak{L}(1) = 0, \quad (543)$$

and, for every finite set  $S$  and family with  $[X^0]F_i = 1$ ,

$$\mathfrak{L}\left(\prod_{i \in S} F_i\right) = \sum_{i \in S} \mathfrak{L}(F_i). \quad (544)$$

These are `SaddleExpansion.logOf_mul`, `SaddleExpansion.logOf_one`, and `SaddleExpansion.logOf_prod` in `LogSeriesMultiplicative.lean`. They are identities in  $R[[X]]$ : no convergence, analytic logarithm, branch choice, or passage to an infinite product is asserted.

## 12.9 Universal finite endpoint-transfer algebra

The formal coefficient stage of endpoint transfer is independent of every probability and asymptotic estimate. To distinguish the universal endpoint polynomials from the periodic correction  $P$ , write  $P_r^{\text{end}}(X) = \text{endpointTransferPolynomial}(r)$ . In  $\mathbb{Q}[X]$ , define

$$e_0(X) = 0, \quad e_{n+1}(X) = -\frac{X^{n+2}}{n+2}, \quad P_r^{\text{end}}(X) = \text{ExpCoeff}(e)_r. \quad (545)$$

Then the recursive, formal-series, and unordered-partition descriptions agree:

$$P_{n+1}^{\text{end}}(X) = -\frac{1}{n+1} \sum_{j=0}^n \frac{j+1}{j+2} X^{j+2} P_{n-j}^{\text{end}}(X), \quad (546)$$

$$\sum_{r \geq 0} P_r^{\text{end}}(X) t^r = \exp \left( \sum_{r \geq 0} e_r(X) t^r \right) = \exp \left( - \sum_{j \geq 2} \frac{X^j t^{j-1}}{j} \right), \quad (547)$$

$$P_r^{\text{end}}(X) = \sum_{f \in \mathcal{P}_r} \prod_{j=1}^r \frac{e_j(X)^{f(j)}}{f(j)!},$$

$$\mathcal{P}_r = \left\{ f : \mathbb{N} \rightarrow \mathbb{N} \mid \begin{array}{l} \text{supp}(f) \subseteq \{1, \dots, r\}, \\ \sum_{j=1}^r j f(j) = r \end{array} \right\}. \quad (548)$$

The first coefficients are

$$P_0^{\text{end}} = 1, \quad P_1^{\text{end}} = -\frac{X^2}{2}, \quad P_2^{\text{end}} = \frac{X^4}{8} - \frac{X^3}{3}, \quad P_3^{\text{end}} = -\frac{X^6}{48} + \frac{X^5}{6} - \frac{X^4}{4}. \quad (549)$$

Because  $\mathbb{Q}[X]$  is the free rational algebra on one generator, evaluating  $X$  at any element  $z$  of a commutative  $\mathbb{Q}$ -algebra commutes both with each  $P_r^{\text{end}}$  and with the complete formal power series.

*Formal proof.* The definitions are `endpointTransferExponentCoefficient`, `endpointTransferPolynomial`, and `endpointTransferSeries` in `EndpointTransferPolynomials.lean`. Equations (546)–(548) are respectively `endpointTransferPolynomial_succ`, `endpointTransferSeries_eq_exp_subst`, and `endpointTransferPolynomial_eq_partitionExpSum`. The finite partition engine and its equality with the recursive exponential coefficients are `weightedPartitions`, `partitionExpSum`, and `partitionExpSum_eq_expCoeff` in `ExponentialPartition.lean` and `ExponentialBell.lean`. Evaluation and series naturality are `aeval_endpointTransferPolynomial` and `map_endpointTransferSeries`; the four identities in (549) are `endpointTransferPolynomial_zero`, `endpointTransferPolynomial_one`, `endpointTransferPolynomial_two`, and `endpointTransferPolynomial_three`.

All statements in this subsection are finite or formal. They do not identify the series analytically with  $\exp(X)(1 - Xt)^{1/t}$ , construct a tilted law, or prove a Taylor-window, tail, or endpoint-remainder estimate.  $\square$

## 12.10 Recursive elementary re-expansion of the phase

The exact Lambert phase can itself be expanded to arbitrary order. With

$$T(x) = -\log_2 x,$$

write  $\mathcal{L}_N(x)$  for `fabiusLambertLogarithmicApproximation(N, x)`. This approximation is built from the recursively defined polynomials `dyadicLambertDisplacementPolynomial`. For every  $N \in \mathbb{N}$ ,

$$\lambda(x) - \mathcal{L}_N(x) = \mathcal{O}(T(x)^{-N}) \quad (x \downarrow 0). \quad (550)$$



This is `fabiusLambertPhase_sub_logarithmicApproximation_isBigO`; the equivalent dyadic-scale form is `dyadicLambertAllOrderRemainder_isBigO`. Explicit coefficient polynomials are proved through index two, while higher orders are represented by the formal recursion.

The periodic term cannot simply be discarded from a sharp estimate. If  $\mathcal{M}_{\text{Wiki}}$  denotes the uncorrected elementary expression `fabiusWikipediaElementaryMain`, then Lean proves that  $x \mapsto \log F(x) - \mathcal{M}_{\text{Wiki}}(x)$  is not  $\mathcal{O}((-\log x)^{-1})$  as  $x \downarrow 0$ . This is `log_fabius_sub_wikipediaElementaryMain_not_isBigO` in `FabiusSharpAsymptotic.lean`. The obstruction is therefore a theorem, not a numerical diagnosis.

### 12.11 Three endpoint consequences already formalized

On the dyadic logarithmic scale, let  $g(t) = -\log F(2^{-t})$ . Then

$$\lim_{t \rightarrow \infty} \frac{g(t)}{t^2} = \frac{L}{2}. \quad (551)$$

This is `fabiusLogProfile_normalized_tendsto` in `FabiusLogSquaredAsymptotic.lean`. On the original small-argument scale, for every  $c > 0$ ,

$$\exp(-c/x) = o(F(x)) \quad (x \downarrow 0). \quad (552)$$

This is `exp_neg_div_isLittleO_fabius` in `FabiusDecayComparison.lean`. At the opposite endpoint, the exact global symmetry already proved in (15) gives only the identity

$$1 - F(1 - x) = F(x). \quad (553)$$

No additional right-endpoint asymptotic-transfer theorem is asserted here.

## 13 The Fourier–Legendre expansion of up

Let  $P_n$  be the Legendre polynomial normalized by  $P_n(1) = 1$ , and define

$$u_n = \frac{4n+1}{2} \int_{-1}^1 \text{up}(x) P_{2n}(x) \, dx. \quad (554)$$

Odd Legendre coefficients vanish by `rvachevFullLegendreCoefficient_odd_eq_zero`. The even coefficients have the exact finite formula

$$\begin{aligned} u_n &= 4^{-n} (4n+1) \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} \\ &\quad \times (2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \end{aligned} \quad (555)$$

This is `rvachevLegendreCoefficient_eq_fabius_sum` in `FabiusLegendreCoefficients.lean`.

The even series

$$\text{up}(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x) \quad (-1 \leq x \leq 1) \quad (556)$$

converges absolutely and uniformly in the formal development. The exact declarations are `summable_abs_rvachevLegendreCoefficient`, `hasSum_rvachevLegendreSeries`, and `hasSum_rvachevLegendreSeries_uniform` in `FabiusLegendreSeries.lean`. Its  $N$ -th polynomial partial sum minimizes squared  $L^2([-1, 1])$  error among all polynomials of degree at most  $2N+1$ , and every distinct competitor has strictly larger error. These are `rvachevLegendrePar`

`tialSum_least_squares` and `rvachevLegendrePartialSum_strict_least_squares` in `FabiusLegendreLeastSquares.lean`.

At compiled source checkpoint `ca66a1247`, the polynomial-form Legendre blocks  $B_n(x) = u_n P_{2n}(x)$  satisfy the complete orthogonality formula

$$\int_{-1}^1 B_m(x) B_n(x) dx = \delta_{mn} \frac{2u_n^2}{4n+1}. \quad (557)$$

This is `intervalIntegral_rvachevLegendreBlock_mul`, for the definition `rvachevLegendreBlock`. The coefficient-energy form of Parseval is also exact. Put  $S_N(x) = \sum_{n=0}^N u_n P_{2n}(x)$ . Then

$$\int_{-1}^1 \text{up}(x)^2 dx = \sum_{n=0}^{\infty} \frac{2u_n^2}{4n+1}, \quad (558)$$

$$\int_{-1}^1 (\text{up}(x) - S_N(x))^2 dx = \sum_{n=N+1}^{\infty} \frac{2u_n^2}{4n+1}. \quad (559)$$

The full-series and shifted-tail `HasSum` declarations are `hasSum_rvachevLegendreCoefficient_energy` and `hasSum_rvachevLegendreCoefficient_energy_tail`; the latter's `tsum` form is `rvachevLegendreSquaredError_partialSum_eq_tsum_tail`. The finite partial-sum norm is `integral_sq_eval_rvachevLegendrePartialSumPolynomial`. Finally, writing

$$A_2 := \int_0^1 F(t)^2 dt, \quad A_2 = \sum_{n=0}^{\infty} \frac{u_n^2}{4n+1}, \quad (560)$$

so that  $A_2 = \text{fabiusSquareEnergy } F$ , the bridge and series declarations are `integral_sq_rvachevUp_eq_two_mul_fabiusSquareEnergy`, `hasSum_fabiusSquareEnergy_legendre`, and `fabiusSquareEnergy_eq_tsum_legendre` in `FabiusLegendreEnergy.lean`. The block is defined directly in its polynomial form: these declarations do not formalize its finite up-translate realization, a Fourier-product or infinite-sinc integral for  $A_2$ , or rationality of its partial sums.

On  $[0, 2]$ , the signed global extension has the corresponding translated Legendre series `globalFabius_eq_tsum_translatedLegendre_formula`.

## 14 Additional proved summation identities

On the support interval  $-1 \leq t \leq 1$ , Poisson summation yields the exact cosine series

$$\text{up}(t) = \frac{1}{2} + \sum_{k=0}^{\infty} \widehat{\text{up}}\left(\frac{2k+1}{2}\right) \cos((2k+1)\pi t). \quad (561)$$

This is `rvachev_cosine_series`. The two-term support specialization

$$\text{up}(t) + \text{up}(t-1) = 1 \quad (0 \leq t \leq 1) \quad (562)$$

is `rvachev_add_shift_eq_one`. More general scaled Poisson support identities are sharp for either orientation of the lattice: whenever  $1/2 \leq |a|$ ,

$$1 + 2 \text{up}(a) = \sum_{m \in \mathbb{Z}} |a|^{-1} \widehat{\text{up}}(m/a), \quad (563)$$

$$|a| + 2|a| \text{up}(a) = \sum_{m \in \mathbb{Z}} \widehat{\text{up}}(m/a). \quad (564)$$

These are `rvachev_poisson_support_specialization_unscaled_of_one_half_le_abs` and `rvachev_poisson_support_specialization_of_one_half_le_abs` in `PoissonSummation.lean`; the oriented frequency  $m/a$  is retained while the covolume is  $|a|$ . They supplement the partition formulas (183)–(184); no unformalized derivative partition, probabilistic convolution interpretation, or moment statistic is inferred.

## 15 Effective computability

The centered spline construction gives a primitive-recursive rational approximation procedure. In the exact coding used by Lean, for every dyadic input code  $c$  and precision  $p$ ,

$$|F(c.\text{value}(p+3)) - \text{Approx}(c, p).\text{value}(p)| \leq 5 \cdot 2^{-(p+3)}. \quad (565)$$

This is `fabiusSplineApproxPR_error`; the procedure itself is proved primitive recursive by `fabiusSplineApproxPR_primrec`. Together with the sharp Lipschitz bound it proves that every bounded Fabius solution is sequentially computable and effectively uniformly continuous, hence a computable real function in the formal Grzegorzczuk-style definition. The declarations are `fabiusReal_sequentiallyComputable`, `fabiusReal_effectivelyUniformContinuous`, and `fabiusReal_isComputableRealFunction` in `FabiusComputableSpline.lean` and `FabiusComputability.lean`. The same effective continuity holds for every signed global extension: `extendedFabius_effectivelyUniformContinuous`. For the canonical extension  $\mathcal{F} = \text{globalFabius}$ , the explicit modulus is  $d(n) = 2n$ : whenever  $n > 0$  and  $|x - y| < 1/(2n)$ ,

$$|\mathcal{F}(x) - \mathcal{F}(y)| < \frac{1}{n}. \quad (566)$$

This is `globalFabius_effectively_uniformly_continuous`, together with the definition `fabiusEffectiveUniformModulus`, in `FabiusComputability.lean`.

The same finite spline code also gives a signed-global algorithm. Instead of clamping the positive numerator to the unit grid, it retains the unrestricted centered spline and rounds the signed rational numerator pair. Negative dyadic inputs collapse to the zero grid point, which is exact because  $\mathcal{F}(x) = 0$  for  $x \leq 0$ . The resulting primitive-recursive procedure satisfies

$$|\mathcal{F}(c.\text{value}(p+3)) - \text{Approx}_{\text{glob}}(c, p).\text{value}(p)| \leq 5 \cdot 2^{-(p+3)}. \quad (567)$$

These are `extendedFabiusSplineApproxPR_primrec` and `extendedFabiusSplineApproxPR_error`. Hence, for every  $F$  satisfying `IsFabiusF`, its signed extension `extendedFabiusF` is a computable real function, by `extendedFabius_isComputableRealFunction`; the canonical specialization is `globalFabius_isComputableRealFunction`. This is a primitive-recursive correctness statement, not a practical running-time estimate: the unrestricted positive grid numerator sets the length of the finite fold.

The quantitative analytic estimate is stable under moving the evaluation point:

$$|S_p(x) - \mathcal{F}(y)| \leq 2^{-p} + 2|x - y|. \quad (568)$$

This is `abs_fabiusUniformSpline_sub_extendedFabius_le_add`. In particular, if  $x_p \rightarrow x$ , then  $S_p(x_p) \rightarrow \mathcal{F}(x)$ , as packaged by `fabiusUniformSpline_tendsto_extendedFabius_of_tendsto`.

## 16 Formalization boundary and audit conventions

This exposition follows a fail-closed rule. A mathematical assertion remains in the main document only when an exact proved declaration can be identified in the Lean development. A nearby theorem, an executable computation, a numerical check, or an argument that appears straightforward is not treated as a substitute. When the formal theorem has stricter hypotheses, narrower domains, or different endpoint conventions, the prose uses those exact conditions.

That rule explains several choices made above:

- The finite probability approximation uses atomic convolution measures and the exact weak-convergence API, not an unformalized continuous box-convolution density.

- Arbitrary norm-summable vector weights have formal probability, head–tail, reflection, continuous-linear naturality, and operator-affine recursion laws; their laws have topological support exactly the series range, and arbitrary signed real weights have exact min/max interval ranges and supports. A nontrivial norm-summable real-weight law is absolutely continuous with respect to Lebesgue measure and has null singletons. The pushforward-law definition itself is total even without measurability; its probability and map identities state the summability, completeness, and Borel hypotheses explicitly. Scalar pointwise naturality alone needs neither summability nor completeness. The generic affine recurrence is conditional on a displayed shift relation. Separately, `GeometricUniformLaw.lean` defines the named weights  $(1 - q)q^n$ , their series and probability law, and proves self-similarity, reflection, absolute continuity, null singletons, and support equal to the series range under  $|q| < 1$ , with exact support  $[0, 1]$  under  $0 \leq q < 1$ . `ContinuousCDF.lean` and `GeometricUniformCDF.lean` add the named CDF, continuity and reflection for  $|q| < 1$ , the exterior tails for  $0 \leq q < 1$ , and, for  $0 < q < 1$ , the conditioning equations, everywhere derivative, explicit compactly supported Radon–Nikodym density, and real-space  $C^\infty$  package. `ProbabilityRepresentation.lean` supplies the exact half-base CDF and density bridges. `AffineIndependentCopy.lean` proves probability preservation, product-map and characteristic-function factorization, finite characteristic-function iteration, and uniqueness of probability fixed points for  $|q| < 1$  in every complete second-countable real inner-product target. Its `GeometricUniformUniqueness.lean` specialization characterizes the named law, including  $q = 0$  and negative  $q$ , without support, density, or moment assumptions. The selected density has no proved interpretation here at  $q = 0$  or negative  $q$ . `GeometricSincFactorization.lean` gives the exact finite residual sinc factorization and pointwise phase-bearing prefix limit for every  $|q| < 1$ , with dyadic wrappers; `geometricSincProduct` already packages the same rescaled sinc factors as a `tprod`. A named characteristic-function-to-`geometricSincProduct` bridge (or explicit `HasProd` theorem), compact-uniform convergence, general- $q$  rapid decay and Fourier inversion, centered packaging, closed higher-derivative and endpoint-flatness wrappers, shape results, further transforms, and explicit Bernoulli-cumulant/Bell-moment formulas and asymptotics remain outside this interface. The exact-support oriented transform DDE and adjacent Cauchy-power recurrence for every nonzero  $|q| < 1$ , including negative ratios, are direct theorems of `GeometricUniformCauchy.lean`; these are distinct from the still-open characteristic-function-to-infinite-product and Fourier-inversion formulas.
- Quantile transport is formal at the measure level first for arbitrary real probability laws, with no support or range restriction and an almost-everywhere sublevel characterization; a compact pointwise specialization feeds the clamped Fabius inverse. This proves strongly measurable Banach-valued substitution with Lean’s totalized Bochner integral. A separate weighted-subgraph development proves Banach-valued survival layer cake and the full-interval Fabius inverse identity for an explicit interval-integrable primitive or a pointwise- $C^1$  primitive. Its strict-order hypothesis actually makes the two compact-interval clock restrictions a Galois connection, so the generic explicit and pointwise- $C^1$  forms now derive clock measurability rather than assuming it. For a real-valued absolutely continuous primitive, a further generic subtraction theorem and its normalized and Fabius forms accept every real or complex  $L^1$  weight and require no separate complementary-integrability hypothesis. Finally, `InversePairIntegral.lean` proves the full and variable-upper two-function identities for arbitrary absolutely-continuous outer functions on an ordered compact rectangle, derives monotonicity on that rectangle and measurable canonical clamped extensions from the strict inverse-order relation, and supplies the two Fabius wrappers, the full area sum, and the partial inverse-area identity. These declarations do not package reversed-endpoint, higher-order, improper-integral, fractional, complex-valued arbitrary-AC-primitive, or complex-Mellin identities.
- Step cells use the formal half-endpoint indicator, including its point value at a shared boundary.

- The Fourier chapter distinguishes the real punctured-limit statement from the stronger proved local theory: every nonzero signed integer has an exact complex analytic order, all lower complex derivatives vanish, and the first surviving derivative is evaluated explicitly in [Section 5.3](#).
- Summable-scale infinite products are formal under the exact norm-summability, continuity, local-compactness, completeness, and big- $O$  hypotheses stated in [Section 5.2](#). For complex  $q$  with  $\|q\| < 1$ , this gives the entire reciprocal-Gamma product, its zero orbit, Mahler and reflection equations, and the dyadic zero and meromorphic pole orders. The Gamma-side functions are totalized pointwise inverses: raw products of Gamma factors away from their poles, arithmetic canonical regroupings, logarithmic, digamma, Malmsten, heat-trace, integral, D-finite, reciprocal-Gamma derivative-tower, and general-base order statements are not claimed.
- Flatness uses natural powers, and the analytic-locus claims use precisely the domains proved in `NowhereAnalytic.lean`.
- The  $q$ -calculus distinguishes the half-base Fabius identities from the base-generic finite Gaussian/Richardson algebra: recursive Gaussian coefficients, functoriality, both Pascal orientations, reflection symmetry, product concatenation, the full denominator-free factorial identity, principal specializations, denominator-free geometric residual moments, forward closed finite-node numerators, and the proved quarter-base specialization are formal. Generic finite formal-series filters act diagonally coefficientwise, and exact geometric rows have the all-degree Gaussian residual multiplier. Their analytic companion interchanges finite filters with unconditionally summable weighted coefficient series, splits exact heads from infinite tails, and gives the exact Gaussian tsum tail for geometric and Lagrange rows. Zero-weight nodes impose no convergence condition. Named even Fourier modes now instantiate that tail for the actual infinite Rvachev sinc product: arbitrary complex  $c, z$  need only injectivity of the finite powers of  $c^2$ , and  $c = 1/2, q = 1/4$  is assumption-free. The two  $q$ -Vandermonde orientations, the central convolution, the total positive shift, and both  $k \leq N$  negative-shift forms are also denominator-free commutative-semiring theorems. Conditionally-only convergent boundary series, formal-to-analytic evaluation outside the named entire series, the report’s finite-prefix quotient and Bell package, analytic sign/error bounds, uniform or derivative convergence, and asymptotic quarter-sinc acceleration remain outside the formal corpus.
- Integer-order Volterra integration is formal at arbitrary base point and endpoint in a real Banach space, including strict FTC, order composition, polynomial weights, Taylor reconstruction, and the exact Fabius ladder. A separate total real-order operator over arbitrary real normed spaces has the classical continuous-input kernel and additivity theorems when  $0 < \alpha$  and  $a \leq x$ , and agrees with the integer operator at every positive natural order. The shifted real beta kernel is evaluated exactly when both orders are positive and the endpoint inequality is strict; its raw kernel is interval-integrable also at a coincident endpoint. In a complete target, shifted powers and constants have exact Gamma-quotient values, and two positive orders compose for a continuous input on an ordered compact interval. The atom-preserving lower-CDF layer cake also gives the positive-real Fabius stopped-power identity for every real endpoint and its unit-endpoint moment. Positive affine covariance holds at arbitrary real order for an ordered endpoint and positive scale, and Gamma-normalized order raising needs only a continuous primitive, an interval-integrable displayed derivative, and that derivative from the right on the open interval. The causal Rvachev primitive, its compact-support cutoff, its positive-natural bridge, and its positive-order semigroup are formalized on  $x \geq -1$ . The first exact fractional shifts are proved for the signed extension on  $x \geq 0$ , the bounded function on  $[0, 1]$ , and the up-to- $F$  bridge based at  $-1$  on  $x \geq -1$ . The modules prove no corresponding semigroup law for merely interval-integrable inputs or noncomplete targets, no Caputo or Riemann–Liouville derivative operator, no complex-order result, and no iterated or negative-branch shift, shifted lattice, or endpoint-up wrapper. Separately, the weighted inverse-layer-cake API permits real or

complex scalar weights and a Banach-valued explicit or pointwise- $C^1$  primitive on the full unit interval; its arbitrary-absolutely-continuous form retains a real-valued primitive. Variable or reversed endpoints, complex-valued arbitrary-AC primitives, higher inverse primitives, fractional inverse identities, improper-integral limits, and complex Mellin continuation remain outside these results.

- Endpoint asymptotics retain exact phases and the proved remainder scopes; numerical fits and unproved coefficient formulas are absent.

The source drafts have been discharged into two destinations. Proved material was rewritten into the theorem-backed narrative here. The remaining research material is preserved under

`Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/`.

In particular, inverse-function repeated integration and saddle expansions, improper inverse integrals, complex-power and complex-order integration, conditionally convergent boundary transforms, sinc-tail instantiations and estimates, small-argument analysis, and other deferred results remain available there with an explicit warning that they are not part of the formal corpus. Positive-real order raising, the first fractional Fabius–Rvachev shifts, and unconditional generic- $q$  series transforms have moved into the theorem-backed narrative. Natural-order Volterra integration and finite polynomial weights are formalized in [Section 6.2](#); the compact-rectangle absolutely-continuous inverse calculus is [Theorem 4.8](#); and the finite base-generic Gaussian/Richardson algebra and its formal and unconditionally summable series filters are formalized in [Propositions 10.7 to 10.9](#).

## 17 Conclusion

The formal development links three global presentations without confusing their domains. The bounded  $F$  supplies the probability and arithmetic viewpoints; the compactly supported up supplies refinement, Fourier, spline, and Legendre viewpoints; and the signed  $\mathcal{F}$  supplies the global dilation and binary-translation formulas. Exact bridges connect them to finite atomic laws, weak convergence, centered splines, moments, generic weighted distributions, geometrically weighted uniform laws, exact quantile transport, Banach-valued survival layer cake, weighted inverse layer cake, arbitrary absolutely-continuous inverse-pair calculus, reciprocal-power values, dyadic algorithms, half-base  $q$ -binomial identities, prefix sums, denominator-free arbitrary-base geometric interpolation, finite formal-series and unconditionally summable analytic filters with their Gaussian residuals, normalized integer-order and positive-real fractional Volterra foundations, exact integer Fabius primitives and first fractional Fabius–Rvachev shifts, polynomial weights, endpoint Laplace analysis, and computable approximation.

The most important methodological conclusion is equally concrete: the primary document is now a map of proved mathematics. Research claims can move back from the frontier only after a matching Lean theorem exists, with the same hypotheses, domain, normalization, endpoint convention, and conclusion.

## 18 Affine Cauchy–Stieltjes calculus

The current library packages the report-oriented kernel without tying it to a particular probability law. For a finite real measure  $\mu$ , write

$$\mathcal{C}_\mu(z) = \int_{\mathbb{R}} \frac{d\mu(x)}{z - x}, \quad \mathcal{P}_{\mu,n}(z) = \int_{\mathbb{R}} (z - x)^{-n} d\mu(x).$$

These are `measureCauchyTransform` and `measureCauchyPower`, on the natural open domain `measureCauchyDomain`. Their integral, index-one, affine-pushforward, open-domain, derivative,

and holomorphy declarations are, respectively, `measureCauchyTransform_apply`, `measureCauchyPower_one`, `measureCauchyPower_map_affine`, `measureCauchyTransform_map_affine`, `isOpen_measureCauchyDomain`, `hasDerivAt_measureCauchyTransform`, and `analyticOn_measureCauchyTransform`. The affine identities are total under Lean’s Bochner-integral convention and allow every nonzero real scale, including a negative one.

The reusable fixed-point interface assumes only that  $\mu$  is finite, its support lies in a carrier  $K$ , the map  $(u, x) \mapsto c + au + bx$  preserves  $K$ , both  $a$  and  $b$  are nonzero, and

$$\mu = (\text{Unif}[0, 1] \times \mu)_*(u, x \mapsto c + au + bx).$$

It requires no probability normalization and no closedness, compactness, measurability, or nonemptiness assumption on  $K$ . Off the complexification of that carrier, the direct named theorems `hasDerivAt_measureCauchyTransform_of_uniformAffineFixedPoint` and `measureCauchyPower_succ_of_uniformAffineFixedPoint` give

$$\mathcal{C}'_{\mu}(z) = \frac{1}{ab} \left( \mathcal{C}_{\mu} \left( \frac{z-c}{b} \right) - \mathcal{C}_{\mu} \left( \frac{z-c-a}{b} \right) \right)$$

and, for  $n \geq 1$ ,

$$\mathcal{P}_{\mu, n+1}(z) = \frac{1}{nab^n} \left( \mathcal{P}_{\mu, n} \left( \frac{z-c-a}{b} \right) - \mathcal{P}_{\mu, n} \left( \frac{z-c}{b} \right) \right).$$

For the geometric uniform law the definitions `geometricUniformStieltjesDomain`, `geometricUniformStieltjesTransform`, and `geometricUniformStieltjesPower` are total in real  $q$ . Their remaining direct public declarations are `geometricUniformStieltjesTransform_apply`, `geometricUniformStieltjesPower_one`, `isOpen_geometricUniformStieltjesDomain`, `analyticOn_geometricUniformStieltjesTransform`, `hasDerivAt_geometricUniformStieltjesTransform_refinement`, and `geometricUniformStieltjesPower_succ`. Holomorphy assumes  $|q| < 1$ ; the divided differential and adjacent-power equations also assume  $q \neq 0$ . Their proofs use the exact series range as invariant carrier, so negative ratios are included. The case  $q = 0$  is excluded only from the divided formulas.

At  $q = 1/2$ , for every bounded Fabius solution, the canonical unit and centered hierarchies are related by the direct named all-order bridge

$$S_n(z) = 2^n R_n(2z - 1).$$

$$\text{fabiusStieltjesPower\_eq\_two\_pow\_mul\_rvachevCauchyPower}$$

The module `CauchyTransform.lean` contains six public definitions: `rvachevCauchyDomain`, `fabiusStieltjesDomain`, `rvachevCauchyTransform`, `fabiusStieltjesTransform`, `rvachevCauchyPower`, and `fabiusStieltjesPower`. Its direct theorem surface is `rvachevCauchyTransform_apply`, `fabiusStieltjesTransform_apply`, `rvachevCauchyPower_one`, `fabiusStieltjesPower_one`, `rvachevCauchyTransform_eq_integral_rvachevUp`, `isOpen_rvachevCauchyDomain`, `isOpen_fabiusStieltjesDomain`, `hasDerivAt_rvachevCauchyTransform`, `hasDerivAt_fabiusStieltjesTransform`, `analyticOn_rvachevCauchyTransform`, `analyticOn_fabiusStieltjesTransform`, `fabiusStieltjesTransform_eq_two_mul_rvachevCauchyTransform`, `rvachevCauchyTransform_eq_inv_two_mul_fabiusStieltjesTransform`, the all-order bridge above, `hasDerivAt_fabiusStieltjesTransform_refinement`, `fabiusStieltjesPower_succ`, and `rvachevCauchyPower_succ`. In particular, the unit-interval equation  $S'(z) = 4(S(2z) - S(2z - 1))$  and both adjacent-order recurrences on their complex slit domains are now direct named results, not merely informal consequences of affine transport.

This closure does not supply a complex logarithmic fixed point, complex order, pointwise or nontangential Sokhotski–Plemelj limits, a principal-value Hilbert-transform identity, or the full

Jacobi/Padé theory. Separate current modules do prove the real logarithmic identity for  $z > 1$ , real order lowering for  $\alpha > 1$ , exterior Laurent series, finite-height Herglotz–Poisson identities, integrated interval Stieltjes–Perron inversion, and initial exact Jacobi data; none of those statements is read as one of the stronger boundary or continuation claims just listed.

## A Lean module map

Every retained result above points to one or more exact declaration names. The principal source groups are:

| Topic                                                                                                                                                                                                                                                                                                                                                                             | Modules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definitions, existence, uniqueness                                                                                                                                                                                                                                                                                                                                                | <code>Basic.lean</code> , <code>Existence.lean</code> , <code>OriginalCharacterization.lean</code> , <code>OriginalUniqueness.lean</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Shape and regularity                                                                                                                                                                                                                                                                                                                                                              | <code>Differential.lean</code> , <code>Monotonicity.lean</code> , <code>Convexity.lean</code> , <code>Regularity.lean</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Probability, weighted-law naturality, absolute continuity, exact support and affine recursion, contractive affine-law uniqueness, geometric sinc-prefix factorization, quantile transport, survival/CDF layer cake, fractional stopped powers, affine Cauchy–Stieltjes calculus, renormalization, Laurent/Herglotz/inversion layers, initial Jacobi data, and finite approximants | <code>WeightedUniformSeries.lean</code> , <code>WeightedUniformDistribution.lean</code> , <code>WeightedUniformSupport.lean</code> , <code>GeometricUniformLaw.lean</code> , <code>AffineIndependentCopy.lean</code> , <code>GeometricUniformUniqueness.lean</code> , <code>GeometricSincFactorization.lean</code> , <code>ContinuousCDF.lean</code> , <code>GeometricUniformCDF.lean</code> , <code>ProbabilityRepresentation.lean</code> , <code>QuantileTransport.lean</code> , <code>SubgraphFubini.lean</code> , <code>SurvivalLayerCake.lean</code> , <code>CDFLayerCake.lean</code> , <code>FractionalCDFLayerCake.lean</code> , <code>FabijsFractionalIntegral.lean</code> , <code>InverseLayerCake.lean</code> , <code>InversePairIntegral.lean</code> , <code>MeasureCauchyTransform.lean</code> , <code>GeometricUniformCauchy.lean</code> , <code>CauchyTransform.lean</code> , <code>CauchyCDF.lean</code> , <code>CauchySurvival.lean</code> , <code>CauchyHigherPowers.lean</code> , <code>CauchyRenormalization.lean</code> , <code>StieltjesLogFixedPoint.lean</code> , <code>StieltjesResolventHierarchy.lean</code> , <code>StieltjesGeneralizedOrder.lean</code> , <code>StieltjesMomentLaurent.lean</code> , <code>StieltjesCauchyTransform.lean</code> , <code>StieltjesHerglotz.lean</code> , <code>StieltjesInversion.lean</code> , <code>StieltjesPerron.lean</code> , <code>OrthogonalPolynomialJacobi.lean</code> , <code>EarlyApproximants.lean</code> , <code>EarlyMeasureBridge.lean</code> , <code>WeakConvergence.lean</code> , <code>ProbabilityLaplaceMoments.lean</code> , <code>LaplaceMoments.lean</code> , <code>StepMeasureBridge.lean</code> , <code>StepApproximationLimit.lean</code> , <code>FabijsUniformSpline.lean</code> , <code>FabijsComputability.lean</code> |



| Topic                                                                                                                                                                                                                                                                                                                         | Modules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fourier analysis                                                                                                                                                                                                                                                                                                              | FourierProduct.lean, FourierAnalytic.lean, RvachevQBinomialFilter.lean, OriginalPaperSupplement.lean, PoissonSummation.lean, SummableCyclicAlias.lean, ThueMorseFourier.lean, ThueMorseSineProduct.lean, ThueMorseComplexProductBridge.lean, LacunaryProductToSum.lean, RenormalizationIdentity.lean, SincProductShells.lean, SincPrefixBound.lean, SincCanonicalProduct.lean, PhiZeroOrder.lean, IntegerZeroLocalFactorization.lean, IntegerZeroAnalyticOrder.lean, ThueMorseLobeSign.lean, RvachevProductContinuity.lean, PeakRayEnvelope.lean, GlobalDecayEnvelope.lean, RvachevFourierDecay.lean, VarianceBookkeeping.lean, BirkhoffVariance.lean, BirkhoffProductBridge.lean |
| Summable-scale products and geometric reciprocal-Gamma factorization                                                                                                                                                                                                                                                          | ScaledInfiniteProducts.lean, GeometricReciprocalGamma.lean, ReciprocalIntegerGammaZeros.lean, DyadicGammaOrder.lean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Reciprocal-Gamma first jets and the Thue–Morse Gamma tower                                                                                                                                                                                                                                                                    | ReciprocalGammaJets.lean, ThueMorseGammaTower.lean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Local all-order jet calculus                                                                                                                                                                                                                                                                                                  | LeadingJet.lean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Normalized integer- and positive-real-order Volterra calculus, affine covariance, shifted-power and constant evaluations, positive affine covariance and right-derivative order raising, base-point shifts, exact integer Fabius primitives, causal Rvachev fractional primitives, and first fractional Fabius–Rvachev shifts | NormalizedVolterra.lean, FractionalVolterra.lean, FractionalVolterraSemigroup.lean, FractionalVolterraCalculus.lean, FabiusAntiderivatives.lean, FabiusFractionalVolterra.lean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Global derivatives, value distributions, and flatness                                                                                                                                                                                                                                                                         | GlobalExtension.lean, GlobalBounds.lean, BoundedDerivatives.lean, RvachevDerivativeDistribution.lean, EffectiveFlatness.lean, SharpFlatness.lean, FabiusFlatness.lean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Analyticity, inverse, elementary obstruction                                                                                                                                                                                                                                                                                  | NowhereAnalytic.lean, FabiusInverse.lean, ElementaryFunction.lean, NotElementary.lean, InverseBranch.lean, InverseNotElementary.lean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Moments and exact arithmetic                                                                                                                                                                                                                                                                                                  | Arithmetic.lean, AnalyticMoments.lean, MomentPowerSeries.lean, UnitLaplaceMomentBounds.lean, NormalizedEvenMoments.lean, BernoulliRecurrences.lean, MomentCumulantAlgebra.lean, PolynomialExpectationCumulant.lean, WeightedScaleMultiplicity.lean, ExactInversePower.lean, Parity.lean, TwoAdic.lean, CentralBinomialValuation.lean                                                                                                                                                                                                                                                                                                                                              |
| Universal exponential, Bell, and endpoint-transfer algebra                                                                                                                                                                                                                                                                    | SaddleExpansionAlgebra.lean, ExponentialPartition.lean, ExponentialBell.lean, EndpointTransferPolynomials.lean, LogSeriesMultiplicative.lean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Topic                                                                                                                               | Modules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Formal implicit roots and non-monic Hensel lifting                                                                                  | <code>ImplicitPowerSeries.lean</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Finite base-layer arithmetic                                                                                                        | <code>BaseLayerRegrouping.lean</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Finite polynomial, formal-series, and unconditionally summable analytic-series filters, and geometric Lagrange/Richardson exactness | <code>FinitePolynomialFunctional.lean</code> , <code>CompleteHomogeneous.lean</code> , <code>CompleteHomogeneousGenerating.lean</code> , <code>FinitePowerSeriesFilter.lean</code> , <code>GeometricPowerSeriesFilter.lean</code> , <code>AnalyticSeriesFilter.lean</code> , <code>GeometricLagrange.lean</code> , <code>LagrangeResidualMoments.lean</code> , <code>GeometricCompleteHomogeneous.lean</code> , <code>GeometricResidualMoments.lean</code> , <code>GeometricQBinomialLagrange.lean</code> , <code>GeometricRichardson.lean</code> , <code>GeometricLagrangeWeights.lean</code> , <code>GeometricLagrangeQBinomial.lean</code> , <code>GeometricLagrangeQMoments.lean</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Phase-locked reciprocal-grid Richardson algebra                                                                                     | <code>LambertPhaseLockedRichardson.lean</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Dyadic and $q$ -binomial formulas                                                                                                   | <code>DyadicClosedForm.lean</code> , <code>DyadicAnalytic.lean</code> , <code>GlobalDyadic.lean</code> , <code>DyadicCorrectness.lean</code> , <code>FiniteQBinomialCore.lean</code> , <code>QBinomialVandermonde.lean</code> , <code>HalfQBinomial.lean</code> , <code>FabijsDyadicQBinomialScalar.lean</code> , <code>FabijsQBinomialFormula.lean</code> , <code>FabijsRawQBinomialScalar.lean</code> , <code>FabijsBinaryReductionSeries.lean</code> , <code>FabijsParityPowerSeries.lean</code> , <code>FabijsGlobalQBinomialSeries.lean</code> , <code>FabijsDiscreteLimitToeplitz.lean</code> , <code>FabijsDiscreteLimitIntegration.lean</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Thue–Morse algebra, prefixes, disc series, and ring-valued Prouhet identities                                                       | <code>ThueMorseBitSupport.lean</code> , <code>ThueMorsePrefix.lean</code> , <code>ThueMorseGenerating.lean</code> , <code>ThueMorseBlockAlgebra.lean</code> , <code>ThueMorseInfiniteProduct.lean</code> , <code>ThueMorseDiscSeries.lean</code> , <code>ThueMorseNaturalBoundary.lean</code> , <code>ThueMorseExponential.lean</code> , <code>ThueMorseApproximation.lean</code> , <code>ThueMorseSparseProuhet.lean</code> , <code>PaperKFoldThueMorse.lean</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Endpoint asymptotics                                                                                                                | <code>NegativeLaplace.lean</code> , <code>NegativeLaplaceVertical.lean</code> , <code>PeriodicCorrection.lean</code> , <code>PeriodicFourier.lean</code> , <code>MellinBose.lean</code> , <code>MellinFinitePart.lean</code> , <code>BoseFinitePartIntegral.lean</code> , <code>PrincipalLambertW.lean</code> , <code>LowerLambertW.lean</code> , <code>PowerExponentialLambert.lean</code> , <code>PowerExponentialLambertCalculus.lean</code> , <code>PowerExponentialLambertInverse.lean</code> , <code>PowerExponentialLambertFabijs.lean</code> , <code>FabijsLambertPhase.lean</code> , <code>FabijsLambertSaddle.lean</code> , <code>FabijsSharpLambertMain.lean</code> , <code>FabijsWikipediaMain.lean</code> , <code>FabijsDecayComparison.lean</code> , <code>FabijsSharpAsymptotic.lean</code> , <code>FabijsFullAsymptoticExpansion.lean</code> , <code>FabijsSharpLambertTransfer.lean</code> , <code>LaplaceMomentBounds.lean</code> , <code>FabijsDyadicSharpCumulant.lean</code> , <code>SaddleLogExpansionAlgebra.lean</code> , <code>SaddleLogExpansionPowerSeries.lean</code> , <code>FabijsSecondSaddleCorrection.lean</code> , <code>FabijsSecondSaddleExpansion.lean</code> |

| Topic                                | Modules                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Legendre expansion and computability | FabiusLegendreCoefficients.lean, FabiusLegendreSeries.lean, FabiusLegendreLeastSquares.lean, FabiusLegendreEnergy.lean, FabiusTranslatedLegendreSeries.lean, FabiusComputableSpline.lean |

## B Notation crosswalk

| Article                                                                  | Lean                                                                                       | Meaning                                                                                               |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| $F$                                                                      | fabiusReal fabius                                                                          | bounded canonical Fabius function                                                                     |
| up                                                                       | rvachevUp fabius                                                                           | compactly supported fold                                                                              |
| $\mathcal{F}$                                                            | extendedFabius fabius                                                                      | signed global extension                                                                               |
| $S_w$                                                                    | ProbabilityRepresentation.weightedUniformSeries w                                          | weighted coordinate series                                                                            |
| $\mu_w$                                                                  | ProbabilityRepresentation.weightedUniformDistribution w                                    | its pushforward law                                                                                   |
| $G_q$                                                                    | geometricReciprocalGamma q                                                                 | entire reciprocal-Gamma product for $\ q\  < 1$                                                       |
| $\Gamma_{\langle q \rangle}$                                             | geometricGamma q                                                                           | totalized meromorphic inverse of $G_q$                                                                |
| $G_{\text{dy}}, \Gamma_{\text{dy}}$                                      | dyadicReciprocalGamma dyadicGamma                                                          | dyadic reciprocal product and totalized meromorphic inverse                                           |
| $V_a f$                                                                  | volterraPrimitive a f                                                                      | first oriented primitive at base $a$                                                                  |
| $L_a^n f$                                                                | iteratedPrimitive n a f                                                                    | literal $n$ -fold primitive                                                                           |
| $J_a^n f$                                                                | normalizedVolterra n a f                                                                   | normalized Cauchy-kernel primitive                                                                    |
| $I_{a+}^\alpha f$                                                        | fractionalVolterra                                                                         | total real-order Gamma-normalized Volterra operator                                                   |
| $I_{a+}^{\alpha+\beta} f = I_{a+}^\alpha I_{a+}^\beta f$                 | fractionalVolterra_add                                                                     | positive orders, ordered endpoint, continuous Banach-valued input                                     |
| $U_\beta(x)$                                                             | rvachevFractionalPrimitive                                                                 | causal Rvachev fractional primitive based at $-1$                                                     |
| $I_{0+}^\alpha F(x) = \mathbb{E}[(x - X)_+^\alpha] / \Gamma(\alpha + 1)$ | ProbabilityRepresentation.fractionalVolterra_fabiusReal_eq_uniformProduct_integral_posPart | positive real order, every real endpoint                                                              |
| affine $I_{a+}^\alpha$                                                   | fractionalVolterra_affine                                                                  | arbitrary real order, positive scale, ordered endpoint                                                |
| $I_{a+}^{\alpha+1} g'$                                                   | fractionalVolterra_add_one_deriv                                                           | positive order, continuous primitive, integrable interior right derivative, exact Gamma boundary term |
| fractional $\mathcal{F}$                                                 | fractionalVolterra_add_one_extendedFabius_of_nonneg                                        | $I_{0+}^{\alpha+1} \mathcal{F}(x) = 2^\alpha I_{0+}^\alpha \mathcal{F}(x/2)$ , $\alpha > 0, x \geq 0$ |

| Article                     | Lean                                                                | Meaning                                                               |
|-----------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------|
| $\mathcal{A}_{S,x,w}(a; z)$ | <code>finiteAnalyticSeriesFilter</code>                             | finite filter of unconditionally summable weighted coefficient series |
| $I$                         | <code>fabiusInv</code> <code>fabius</code> <code>fabius_spec</code> | totalized inverse                                                     |
| $c_n$                       | <code>moment n</code>                                               | even moment                                                           |
| $d_n$                       | <code>halfMoment n</code>                                           | half moment                                                           |
| $P_w(n)$                    | <code>inclusivePrefixSum w n</code>                                 | inclusive weighted prefix                                             |
| $W_{b,w}(n)$                | <code>weightedScaleMultiplicity b w n</code>                        | weighted scale multiplicity                                           |
| $\nu_b(n)$                  | <code>padicValNat b n</code>                                        | maximal whole-base power exponent                                     |
| $\varepsilon_n$             | <code>thueMorseSign n</code>                                        | signed Thue–Morse value                                               |
| $\text{wt}_2(n)$            | <code>binaryWeight n</code>                                         | binary digit sum                                                      |

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The generic affine independent-copy probability operator, its characteristic-function factorization and finite iteration, its contractive probability-fixed-point uniqueness theorem, and the geometric-uniform uniqueness specialization are included in compiler-validated checkpoint d312c0603. <https://github.com/VladimirReshetnikov/ProveIt/tree/d312c0603/Analysis/FabiusFunction>.